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JEE MAIN · PHYSICS · CHAPTER-WISE PYQS

Semiconductor

JEE Main Physics · Previous Year Questions with Answer Key

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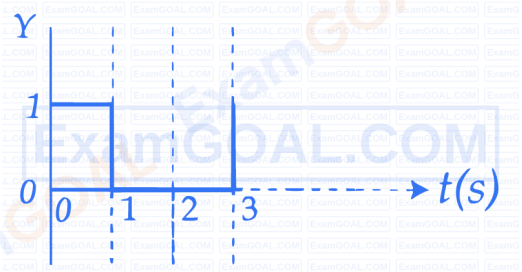
MCQ (Single Correct Answer) 191 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

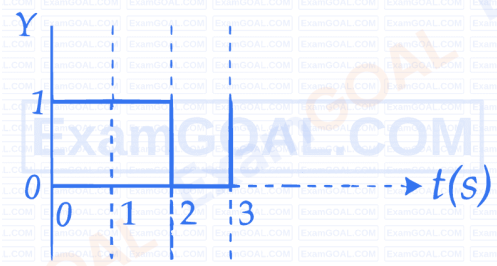
Open solution

The output Y for the given inputs A and B to the circuit is :

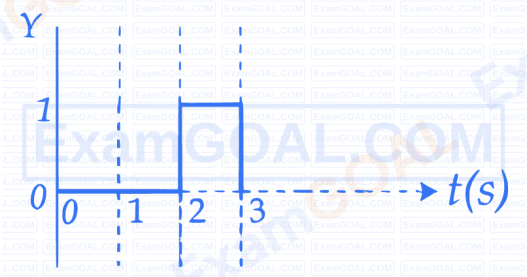
(A)



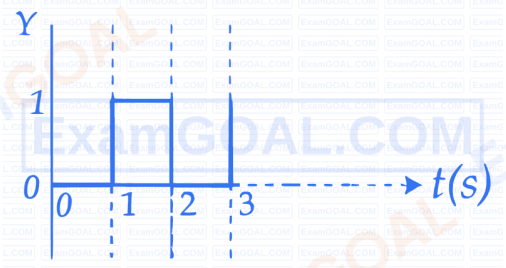
(B)



(C)



(D)



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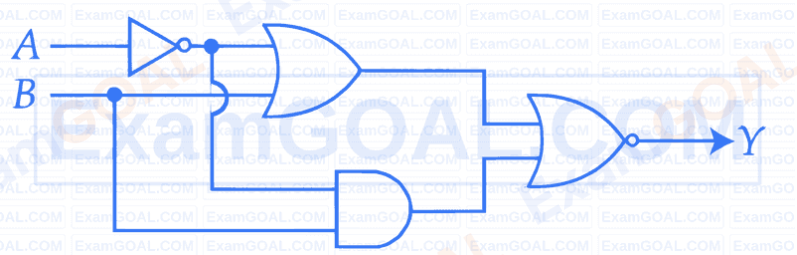
Q2. JEE Main 2026 (Online) 6th April Evening Shift

Open solution

Refer to the logic circuit given below. For two inputs ($A = 1, B = 1$) and ($A = 0, B = 1$), output (Y) will be ____ .



(A) 1, 0 respectively

(B) 0, 1 respectively

(C) 0, 0 respectively

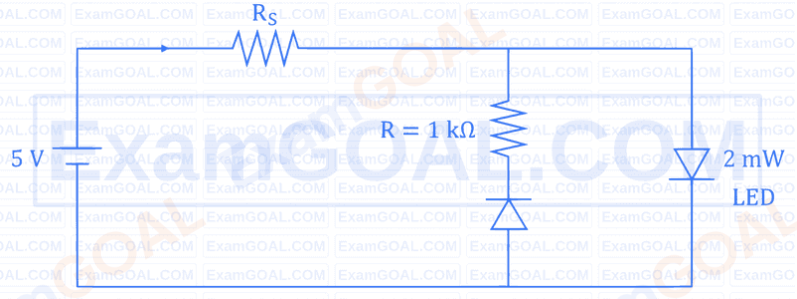
(D) 1, 1 respectively

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Open solution

The maximum rated power of the LED is 2 mW and it is used in the circuit with input voltage of 5 V as shown in the figure below. The current through resistance R_S is 0.5 mA .

The minimum value of the resistance of R_S, to ensure that the LED is not damaged is ____ $\mathrm{k\Omega}$.



(A) 6

(B) 2

(C) 4

(D) 5

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

In a semiconductor p-n diode, the doping concentrations on p-side and n-side are 10^{15} atoms $/\mathrm{cm}^3$ and 10^{18} atoms $/\mathrm{cm}^3$, respectively. Which one of the following statements is true?

(A) Widths of depletion region on either side of the interface are equal

(B) The depletion region width is more on p -side compared to that in n -side

(C) The depletion region width is more on n-side compared to that in p-side

(D) No depletion region forms because of unequal doping concentrations on p and n -sides

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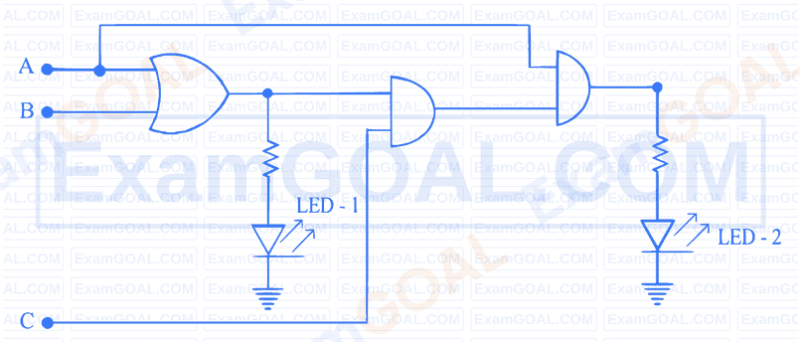
Q5. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow?



(A) $\mathrm{A}=0,\ \mathrm{B}=1, \mathrm{C}=1$

(B) $\mathrm{A}=1,\ \mathrm{B}=0, \mathrm{C}=0$

(C) $\mathrm{A}=1,\ \mathrm{B}=0, \mathrm{C}=1$

(D) $\mathrm{A}=1,\ \mathrm{B}=1, \mathrm{C}=0$

Q6. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason $\mathbf{R}$

Assertion A : A diode under reverse-biased condition provides very small current which is nearly independent of voltage until a critical limit at which the current increases drastically.

Reason R : Below the critical voltage limit, only majority charge carriers flow which increases drastically above critical voltage.

choose the correct answer from the options given below

(A) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

(C) A is true but $\mathbf{R}$ is false

(D) A is false but $\mathbf{R}$ is true

Q7. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Two 4 bits binary numbers, $A = 1101$ and $B = 1010$ are given in the inputs of a logic circuit shown in figure below. The output (Y) will be :

(A) $Y = 1101$

(B) $Y = 0010$

(C) $Y = 0111$

(D) $Y = 1000$

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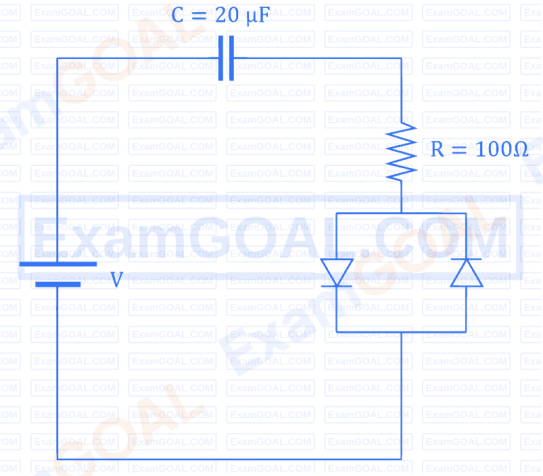
Q8. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Consider a circuit consisting of a capacitor $(20\mu\,\mathrm{F})$, resistor (100Ω) and two identical diodes as shown in figure. The resistance of diode under forward biasing condition is 10Ω. The time constant of the circuit is $\alpha \times 10^{-3}$ s. The value of α is ____

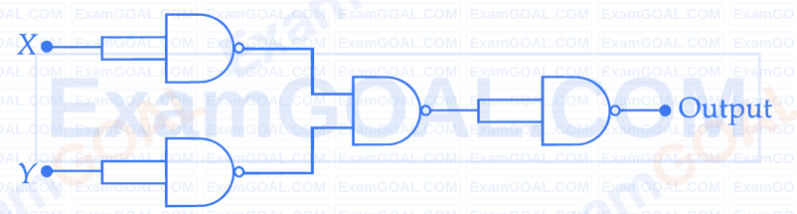


(A) 2.2 (B) 2.0 (C) 2.1 (D) 2.4

Q9. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

If X and Y are the inputs, the given circuit works as _____.



(A) OR gate (B) AND gate (C) NAND gate (D) NOR gate

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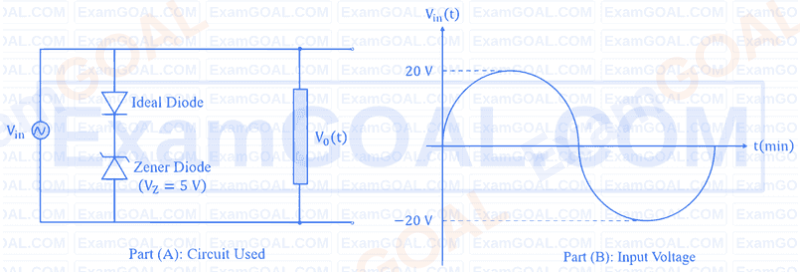
Q10. JEE Main 2026 (Online) 2nd April Morning Shift

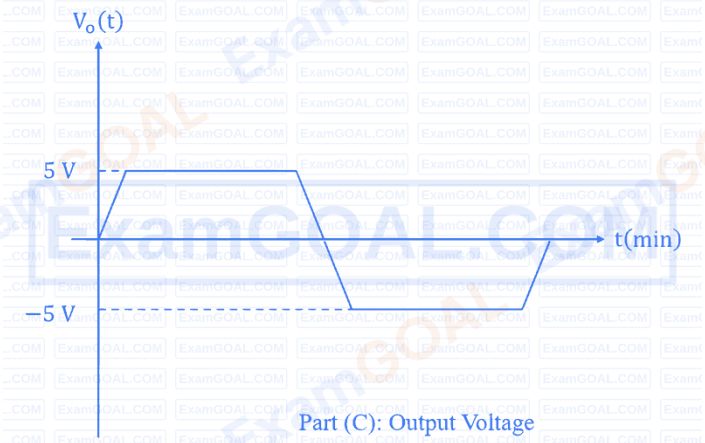
Open solution

For the given circuit (shown in part (A)) the time dependent input voltage $v_{in}(t)$ and corresponding output $v_o(t)$ are shown in part (B) and part (C), respectively. Identify the components that are used in the circuit between points X and Y.



Part (A): Circuit Used

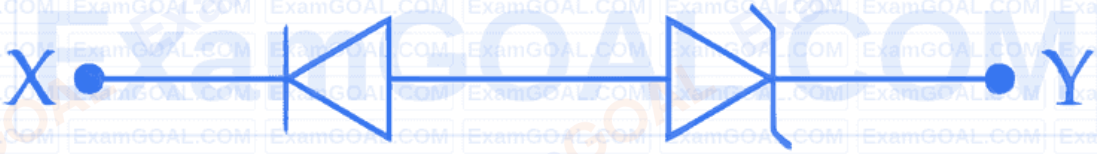
Part (B): Input Voltage



Part (C): Output Voltage

(A)

(B)



(C)

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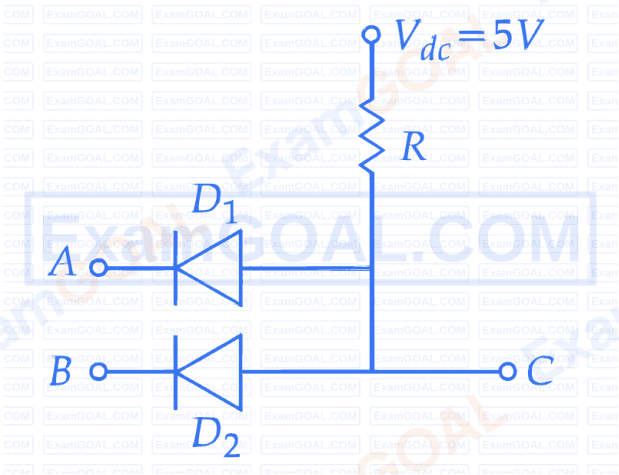
(D)

Q11. JEE Main 2026 (Online) 28th January Evening Shift

Two p-n junction diodes D_1 and D_2 are connected as shown in figure. A and B are input signals and C is the output. The given circuit will function as a _____.

Open solution



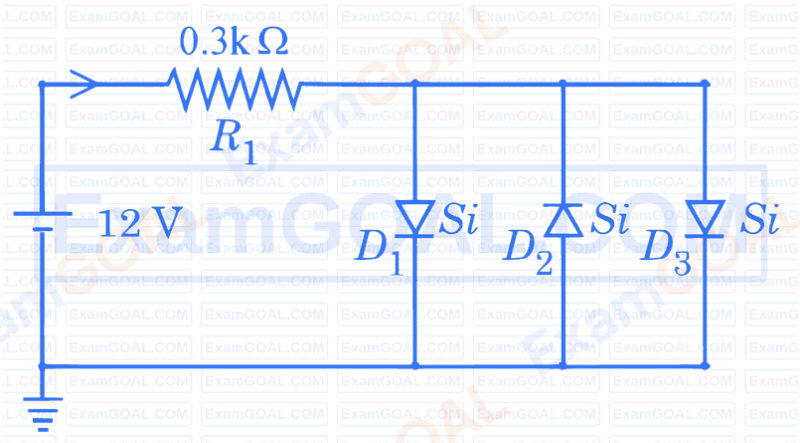
(A) AND Gate (B) OR Gate (C) NOR Gate (D) NAND Gate

Q12. JEE Main 2026 (Online) 28th January Morning Shift

Assuming in forward bias condition there is a voltage drop of 0.7 V across a silicon diode, the current through diode D_1 in the circuit is ____ mA .

(Assume all diodes in the given circuit are identical)

Open solution



(A) 17.6 (B) 18.8 (C) 20.15 (D) 11.7

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Q13. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Identify the correct truth table of the given logic circuit.

(A)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

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Q14. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

For the given logic gate circuit, which of the following is the correct truth table?

(A)

n	m	z
0	0	1
0	1	1
1	1	0
1	0	0

(B)

n	m	z
0	0	1
0	1	0
1	1	1
1	0	0

(C)

n	m	z
0	0	0
0	1	1
1	1	0
1	0	1

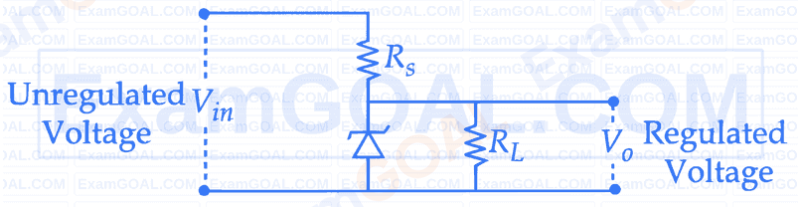
(D)

n	m	z
0	0	1
0	1	0
1	1	0
1	0	0

Q15. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

The following diagram shows a Zener diode as a voltage regulator. The Zener diode is rated at $V_z = 5$ V and the desired current in load is 5 mA . The unregulated voltage source can supply upto 25 V . Considering the Zener diode can withstand four times of the load current, the value of resistor R_s (shown in circuit) should be ____ Ω.



(A) 100 (B) 10 (C) 1000 (D) 4000

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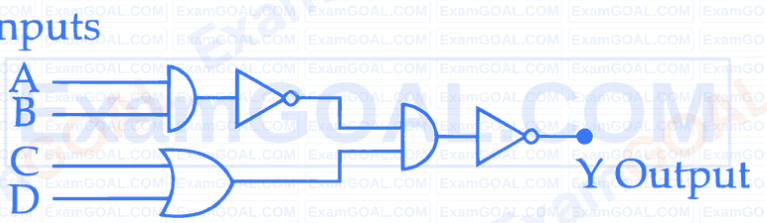
Q16. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

The correct truth table for the given input data of the following logic gate is :



(A)

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	0
1	1	1	1	1

(B)

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	1
1	0	1	0	1
1	1	1	1	1

(C)

Inputs				Output
A	B	C	D	Y
1	1	0	1	1
0	0	1	1	0
1	0	1	0	1
1	1	1	1	0

(D)

Inputs				Output
A	B	C	D	Y
1	1	0	1	0
0	0	1	1	0
1	0	1	0	1
1	1	1	1	1

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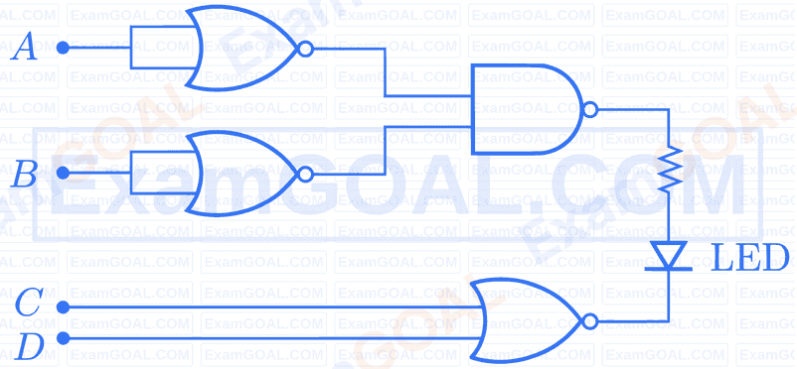
Q17. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Find the correct combination of A, B, C and D inputs which can cause the LED to glow.

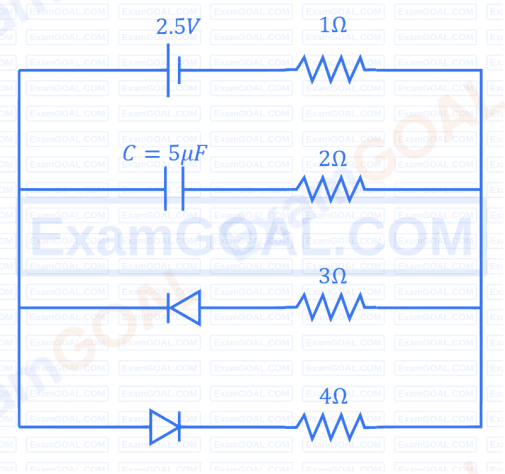


(A) 1000 (B) 0011 (C) 0100 (D) 1101

Q18. JEE Main 2026 (Online) 21st January Evening Shift

Open solution

The charge stored by the capacitor C in the given circuit in the steady state is _____ μC.



(A) 7.5 (B) 5 (C) 10 (D) 12.5

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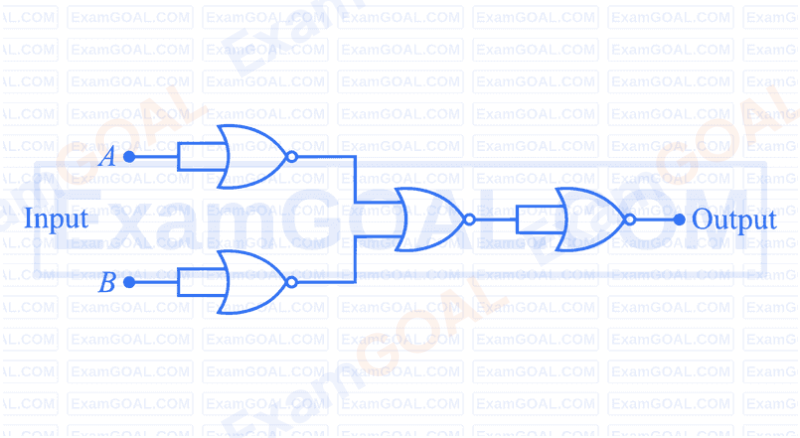
Q19. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

The given circuit works as :

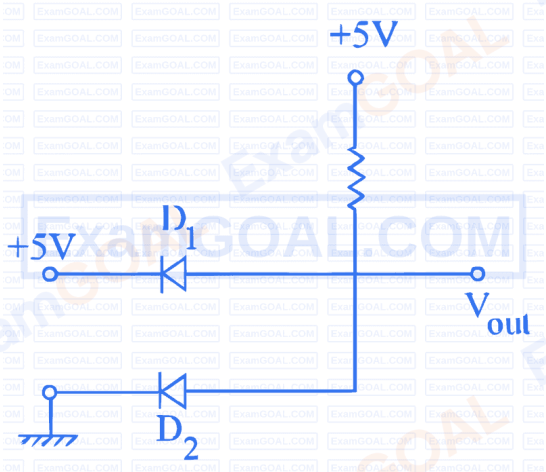


(A) NOR gate (B) OR gate (C) AND gate (D) NAND gate

Q20. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

The output voltage in the following circuit is (Consider ideal diode case)



(A) 0 V (B) +5 V (C) −5 V (D) 10 V

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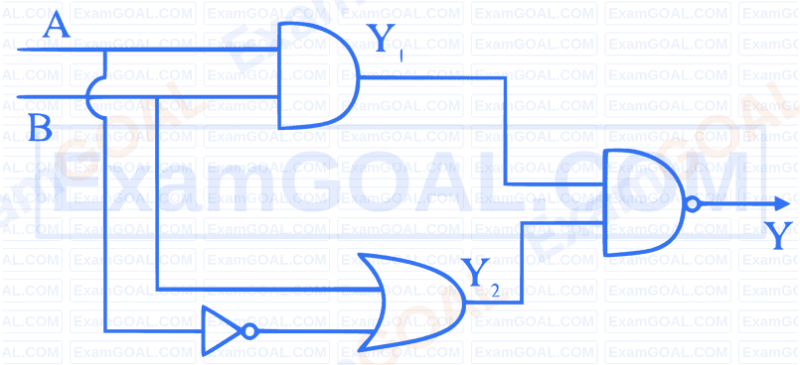
Q21. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

Consider the following logic circuit.



The output is Y = 0 when :

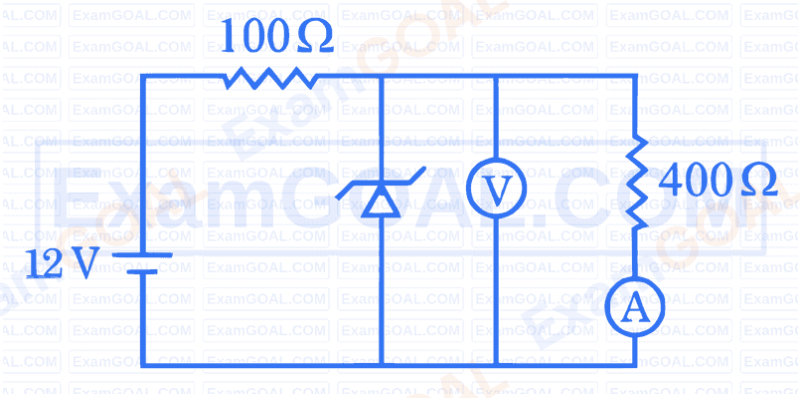
(A) A = 1 and B = 0 (B) A = 0 and B = 0 (C) A = 1 and B = 1 (D) A = 0 and B = 1

Q22. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

In the following circuit, the reading of the ammeter will be

(Take Zener breakdown voltage $= 4\ \mathrm{V}$)



(A) 60 mA (B) 80 mA (C) 10 mA (D) 24 mA

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Q23. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Consider a n-type semiconductor in which n_e and n_h are number of electrons and holes, respectively.

(A) Holes are minority carriers

(B) The dopant is a pentavalent atom

(C) $n_e n_h \neq n_i^2$

(where n_i is number of electrons or holes in semiconductor when it is intrinsic form)

(D) $n_e n_h \geqslant n_i^2$

(E) The holes are not generated due to the donors

Choose the correct answer from the options given below :

(A) (A), (B), (E) only

(B) (A), (C), (E) only

(C) (A), (B), (C) only

(D) (A), (C), (D) only

Q24. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

The Boolean expression $Y = A\bar{B}C + \bar{A}\bar{C}$ can be realised with which of the following gate configurations.

A. One 3-input AND gate, 3 NOT gates and one 2-input OR gate, one 2-input AND gate

B. One 3 -input AND gate, 1 NOT gate, One 2 -input NOR gate and one 2 -input OR gate

C. 3 -input OR gate, 3 NOT gates and one 2 -input AND gate

Choose the correct answer from the options given below:

(A) B, C Only

(B) A, B, C Only

(C) A, B Only

(D) A, C Only

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Q25. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

The truth table corresponding to the circuit given below is:

(A)

A	B	C
0	0	0
0	1	0
1	0	1
1	1	1

(B)

A	B	C
0	0	1
1	0	0
0	1	0
1	1	0

(C)

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

(D)

A	B	C
0	0	0
1	0	0
0	1	0
1	1	1

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Q26. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

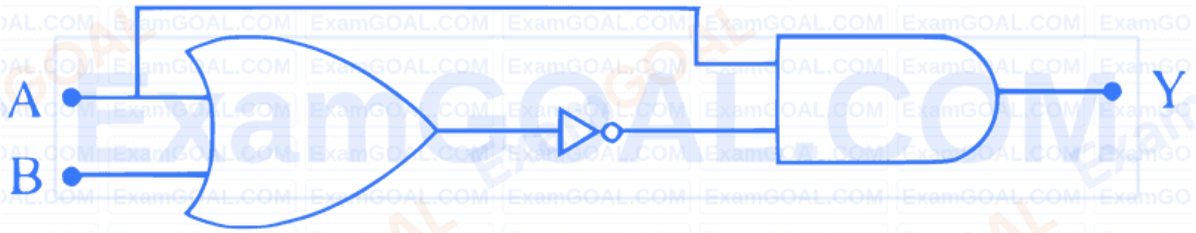
Choose the correct logic circuit for the given truth table having inputs A and B.

Inputs		Output
A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(A)

(B)

(C)



(D)

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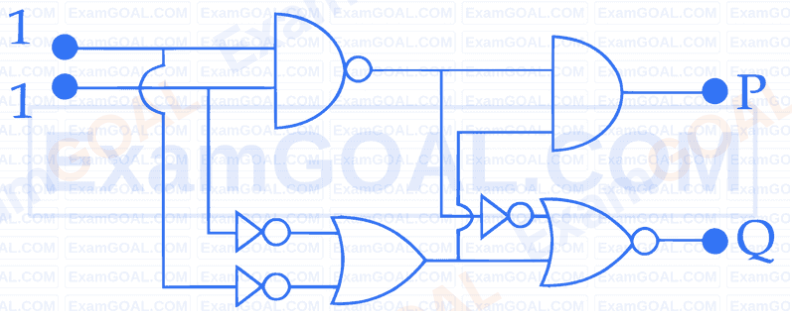
Q27. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

In the digital circuit shown in the figure, for the given inputs the P and Q values are :



(A) $P = 0, Q = 1$ (B) $P = 1, Q = 0$ (C) $\mathrm{P} = 0, \mathrm{Q} = 0$ (D) $P = 1, Q = 1$

Q28. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

A zener diode with 5 V zener voltage is used to regulate an unregulated dc voltage input of 25 V . For a 400Ω resistor connected in series, the zener current is found to be 4 times load current. The load current (I_L) and load resistance (R_L) are :

(A) $\mathrm{I_L = 0.02\,mA; R_L = 250\Omega}$ (B) $\mathrm{I_L = 10\,A; R_L = 0.5\Omega}$

(C) $\mathrm{I_L = 10\,mA; R_L = 500\Omega}$ (D) $\mathrm{I_L = 20\,mA; R_L = 250\Omega}$

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Q29. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

The truth table for the circuit given below is:

(A)

A	B	Y
0	0	0
1	1	1
1	0	1
0	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	0
1	1	0

(C)

A	B	Y
0	0	0
1	0	0
1	1	0
0	1	1

(D)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

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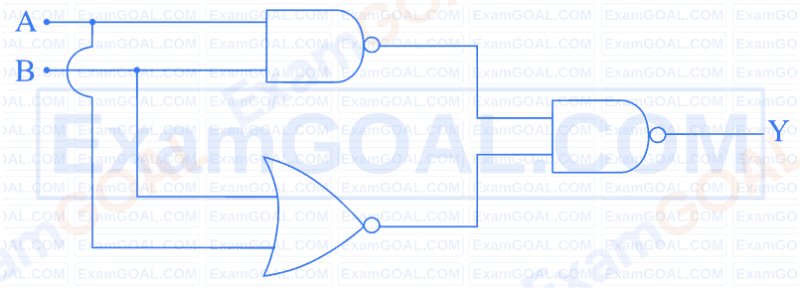
Q30. JEE Main 2025 (Online) 29th January Morning Shift



Open solution

For the circuit shown above, equivalent GATE is :

(A) NOT gate (B) NAND gate (C) AND gate (D) OR gate

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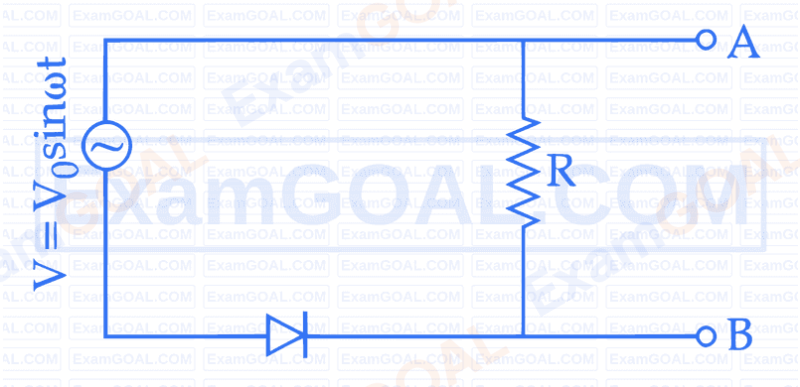
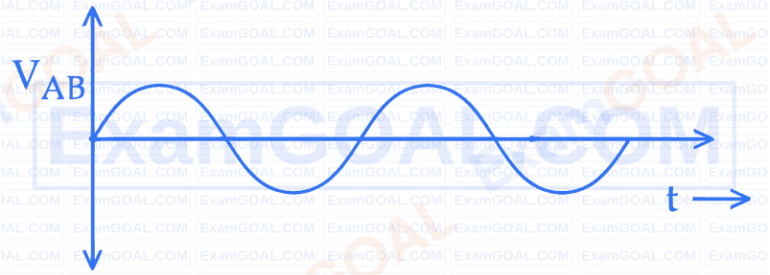
Q31. JEE Main 2025 (Online) 28th January Evening Shift

Open solution



In the circuit shown here, assuming threshold voltage of diode is negligibly small, then voltage V_{AB} is correctly represented by:

(A)



(B)

(C) V_{AB} would be zero at all times.

(D)

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Q32. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Which of the following circuits has the same output as that of the given circuit?

(A)

(B)

(C)

(D)

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Q33. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

The output of the circuit is low (zero) for :

(A) $X = 0, Y = 0$

(B) $X = 0, Y = 1$

(C) $X = 1, Y = 0$

(D) $X = 1, Y = 1$

Choose the correct answer from the options given below :

(A) (A), (B) and (C) only

(B) (A), (C) and (D) only

(C) (B), (C) and (D) only

(D) (A), (B) and (D) only

Q34. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

Consider the following statements :

A. The junction area of solar cell is made very narrow compared to a photo diode.

B. Solar cells are not connected with any external bias.

C. LED is made of lightly doped p-n junction.

D. Increase of forward current results in continuous increase of LED light intensity.

E. LEDs have to be connected in forward bias for emission of light.

Choose the correct answer from the options given below :

(A) A, C, E Only

(B) B, E Only

(C) A, C Only

(D) B, D, E Only

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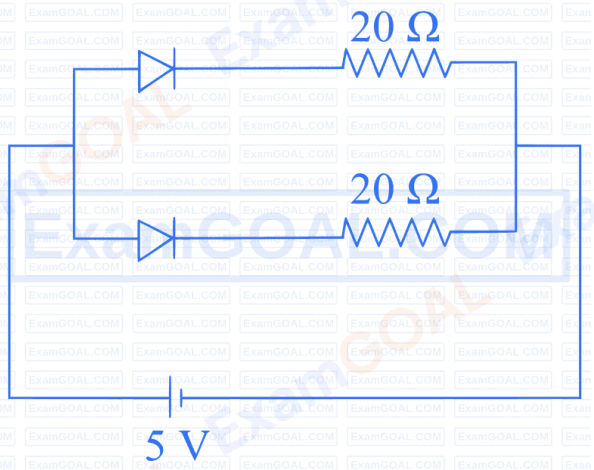
Q35. JEE Main 2025 (Online) 23rd January Evening Shift

What is the current through the battery in the circuit shown below?

Open solution



(A) 1.5 A (B) 0.25 A (C) 1.0 A (D) 0.5 A

Q36. JEE Main 2025 (Online) 23rd January Morning Shift

Refer to the circuit diagram given in the figure. which of the following observations are correct?

A. Total resistance of circuit is 6Ω

B. Current in Ammeter is 1 A

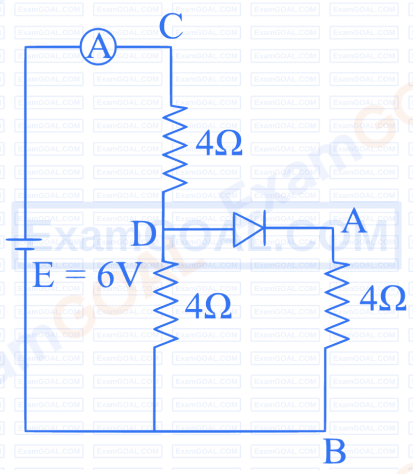
C. Potential across AB is 4 Volts.

D. Potential across CD is 4 Volts

E. Total resistance of the circuit is 8Ω.

Choose the correct answer from the options given below:

Open solution



(A) B, C and E Only (B) A, C and D Only (C) A, B and C Only (D) A, B and D Only

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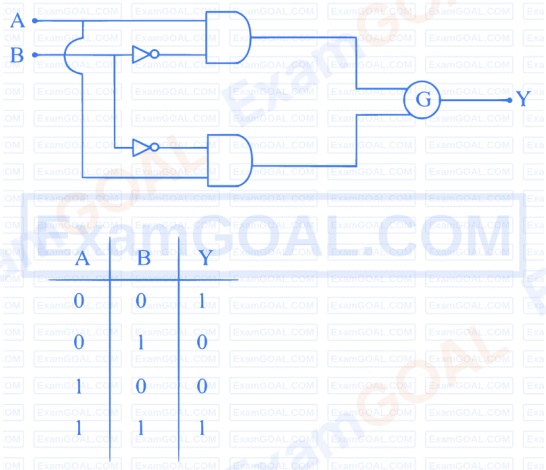
Q37. JEE Main 2025 (Online) 22nd January Evening Shift



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

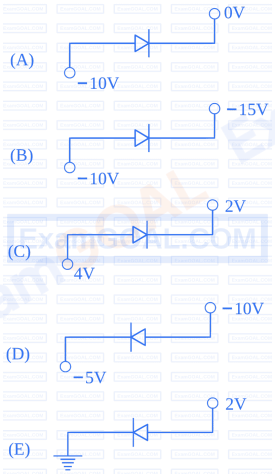
Open solution

To obtain the given truth table, following logic gate should be placed at G :

(A) NOR Gate **(B)** OR Gate **(C)** AND Gate **(D)** NAND Gate

Q38. JEE Main 2025 (Online) 22nd January Morning Shift

Which of the following circuits represents a forward biased diode?



Open solution

Choose the correct answer from the options given below:

(A) (C) and (E) only **(B)** (A) and (D) only

(C) (B), (D) and (E) only **(D)** (B), (C) and (E) only

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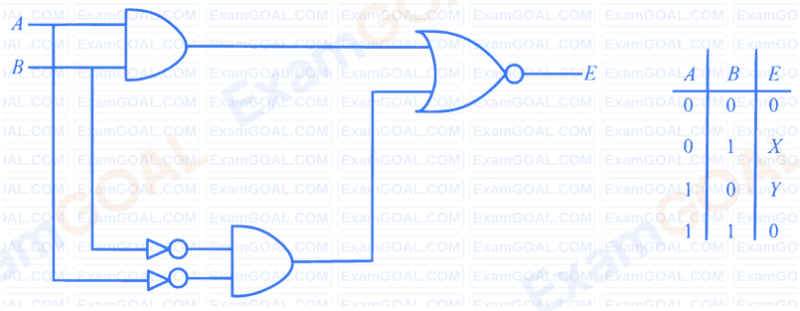
Q39. JEE Main 2024 (Online) 9th April Evening Shift



A	B	E
0	0	0
0	1	X
1	0	Y
1	1	0

Open solution

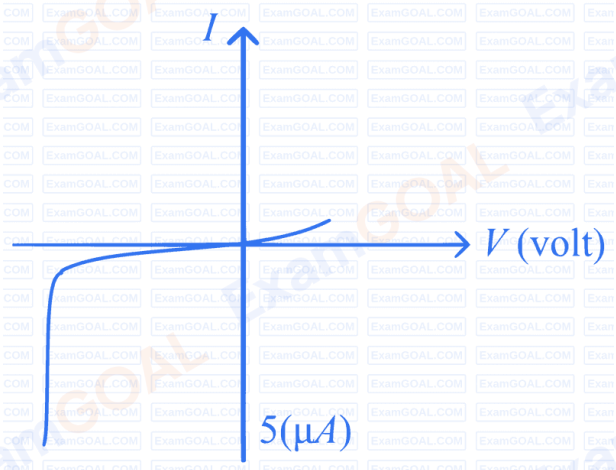
In the truth table of the above circuit the value of X and Y are :

(A) 1, 1 (B) 0, 1 (C) 1, 0 (D) 0, 0

Q40. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The I-V characteristics of an electronic device are shown in the figure. The device is :



(A) a solar cell

(B) a transistor which can be used as an amplifier

(C) a diode which can be used as a rectifier

(D) a zener diode which can be used as a voltage regulator

Q41. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

A light emitting diode (LED) is fabricated using GaAs semiconducting material whose band gap is $1.42\ \text{eV}$. The wavelength of light emitted from the LED is :

(A) 1243 nm (B) 875 nm (C) 650 nm (D) 1400 nm

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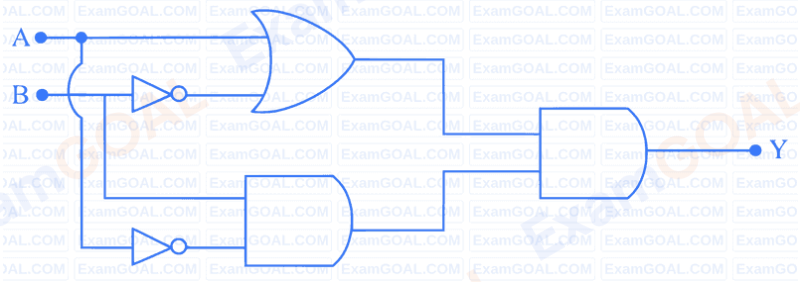
Q42. JEE Main 2024 (Online) 8th April Morning Shift

The output Y of following circuit for given inputs is :



Open solution

(A) $A \cdot B$ (B) $\bar{A} \cdot B$ (C) $A \cdot B(A + B)$ (D) 0

Q43. JEE Main 2024 (Online) 6th April Evening Shift

The acceptor level of a p-type semiconductor is $6\ \text{eV}$. The maximum wavelength of light which can create a hole would be : Given $\text{hc} = 1242\ \text{eV nm}$.

(A) 407 nm (B) 103.5 nm (C) 414 nm (D) 207 nm

Open solution

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Q44. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

The correct truth table for the following logic circuit is :

(A)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(B)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(D)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

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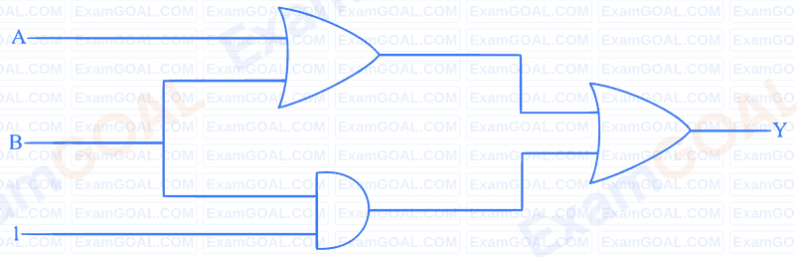
Q45. JEE Main 2024 (Online) 5th April Evening Shift

The output (Y) of logic circuit given below is 0 only when :

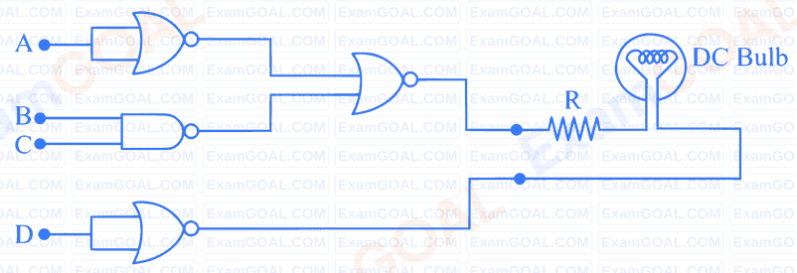


Open solution

(A) $A = 0, B = 0$ (B) $A = 0, B = 1$ (C) $A = 1, B = 0$ (D) $A = 1, B = 1$

Q46. JEE Main 2024 (Online) 5th April Morning Shift

Following gates section is connected in a complete suitable circuit.



Open solution

For which of the following combination, bulb will glow (ON) :

(A) $A = 1, B = 1, C = 1, D = 0$

(B) $A = 1, B = 0, C = 0, D = 0$

(C) $A = 0, B = 0, C = 0, D = 1$

(D) $A = 0, B = 1, C = 1, D = 1$

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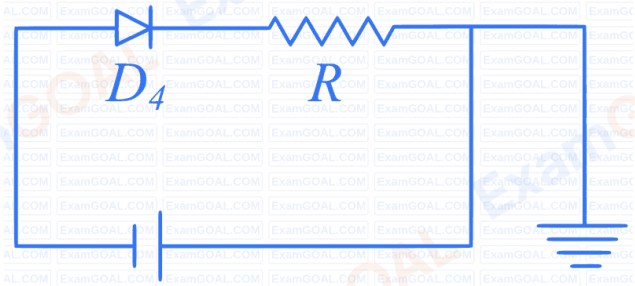
Q47. JEE Main 2024 (Online) 4th April Evening Shift

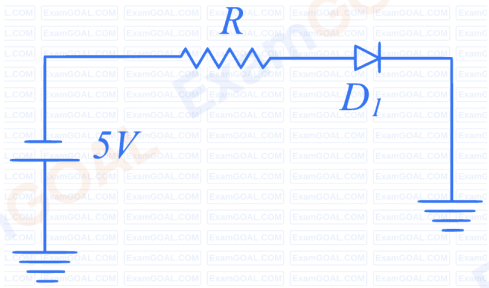
Which of the diode circuits shows correct biasing used for the measurement of dynamic resistance of p-n junction diode :

Open solution

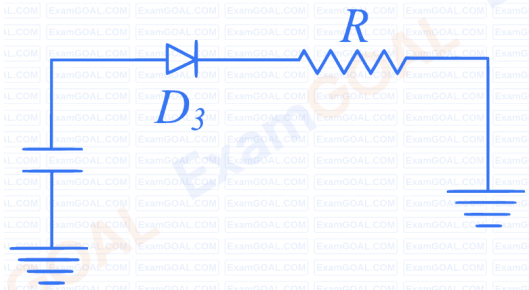
(A)



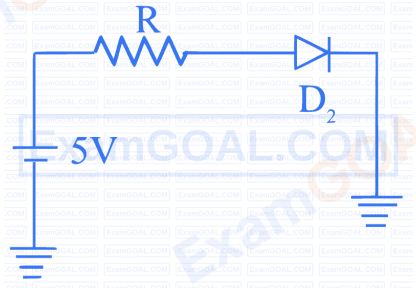
(B)



(C)



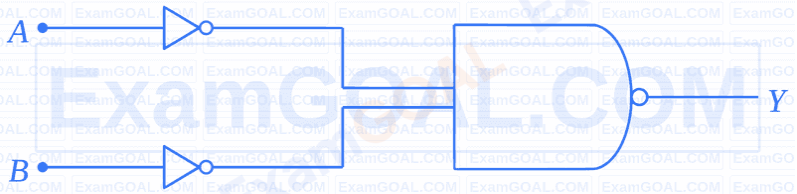
(D)



Q48. JEE Main 2024 (Online) 4th April Evening Shift

Identify the logic gate given in the circuit :

Open solution



(A) OR gate (B) NAND gate (C) NOR gate (D) AND gate

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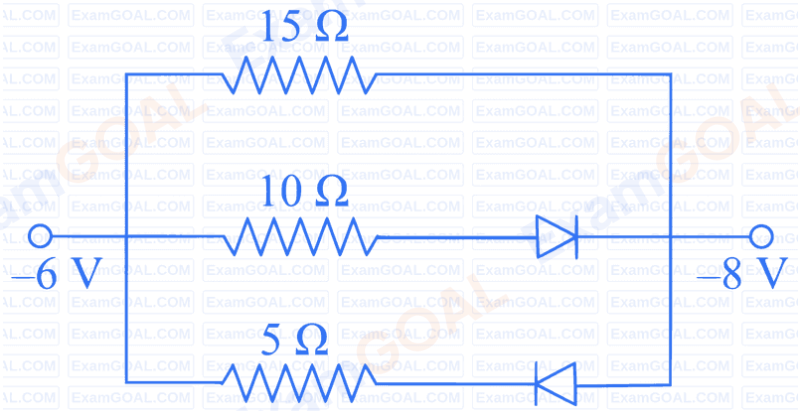
Q49. JEE Main 2024 (Online) 4th April Morning Shift

The value of net resistance of the network as shown in the given figure is :



Open solution

(A) $(30/11)\Omega$ (B) 6Ω (C) $(5/2)\Omega$ (D) $(15/4)\Omega$

Q50. JEE Main 2024 (Online) 1st February Evening Shift

Conductivity of a photodiode starts changing only if the wavelength of incident light is less than $660\ \mathrm{nm}$. The band gap of photodiode is found to be $\left(\frac{\mathrm{X}}{8}\right)\mathrm{eV}$. The value of X is :

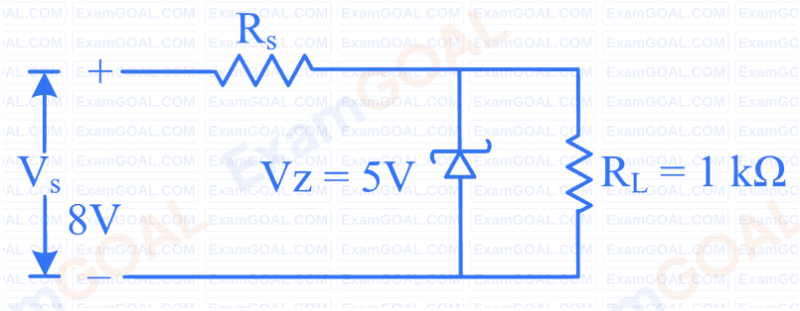
(Given, $\mathrm{h} = 6.6 \times 10^{-34}\mathrm{Js}, \mathrm{e} = 1.6 \times 10^{-19}\mathrm{C}$)

Open solution

(A) 11 (B) 13 (C) 15 (D) 21

Q51. JEE Main 2024 (Online) 1st February Morning Shift

In the given circuit if the power rating of Zener diode is $10\ \mathrm{mW}$, the value of series resistance R_s to regulate the input unregulated supply is :



Open solution

(A) $10\mathrm{k}\Omega$

(B) 10Ω

(C) $1\mathrm{k}\Omega$

(D) $\frac{3}{7}\mathrm{k}\Omega < \mathrm{R_s} < \frac{3}{5}\mathrm{k}\Omega$

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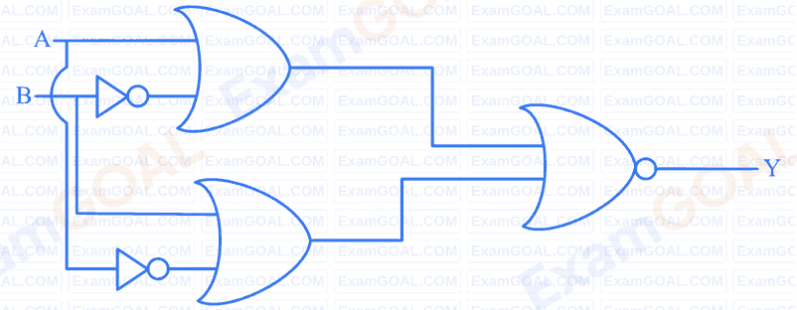
Q52. JEE Main 2024 (Online) 31st January Evening Shift



Open solution

The output of the given circuit diagram is -

(A)

A	B	Y
0	0	0
1	0	0
0	1	1
1	1	0

(B)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(C)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0

(D)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

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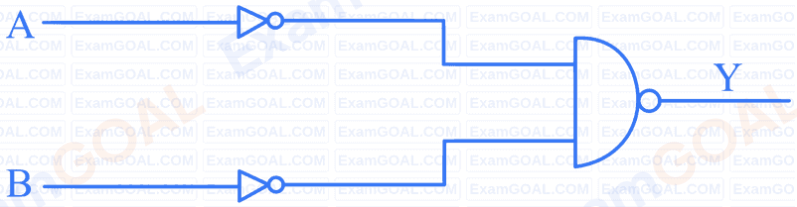
Q53. JEE Main 2024 (Online) 31st January Morning Shift

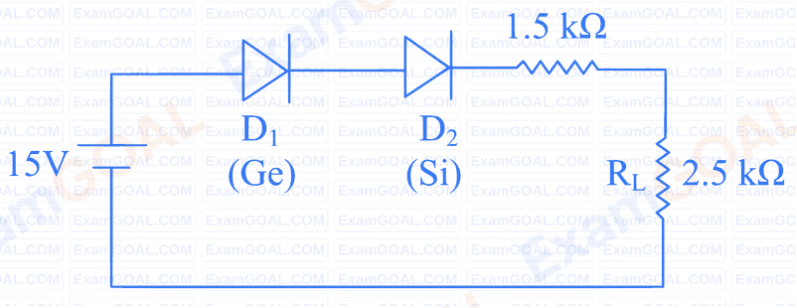
Identify the logic operation performed by the given circuit.



Open solution

(A) AND (B) NOR (C) OR (D) NAND

Q54. JEE Main 2024 (Online) 30th January Evening Shift



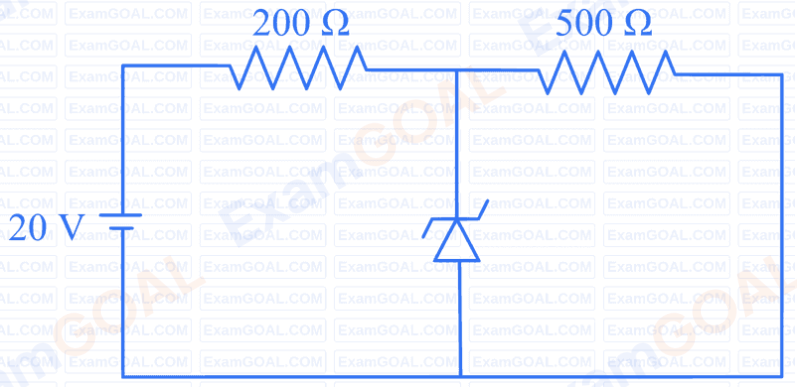
Open solution

In the given circuit, the voltage across load resistance (R_L) is :

(A) 8.75 V (B) 9.00 V (C) 8.50 V (D) 14.00 V

Q55. JEE Main 2024 (Online) 30th January Morning Shift

A Zener diode of breakdown voltage $10\ V$ is used as a voltage regulator as shown in the figure. The current through the Zener diode is :



Open solution

(A) 0 (B) 30 mA (C) 20 mA (D) 50 mA

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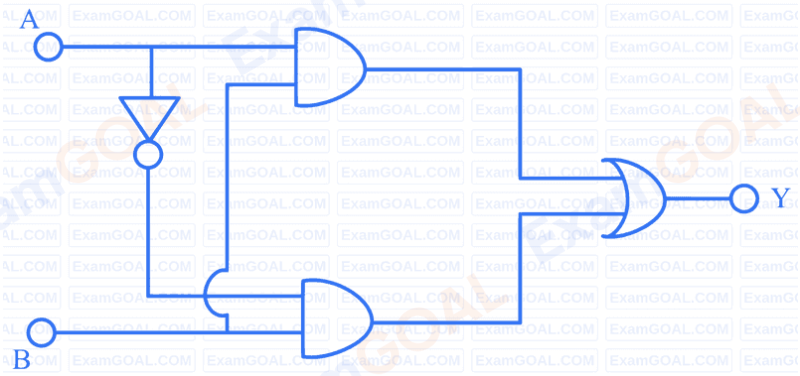
Q56. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

The truth table for this given circuit is :



(A)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(B)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(C)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

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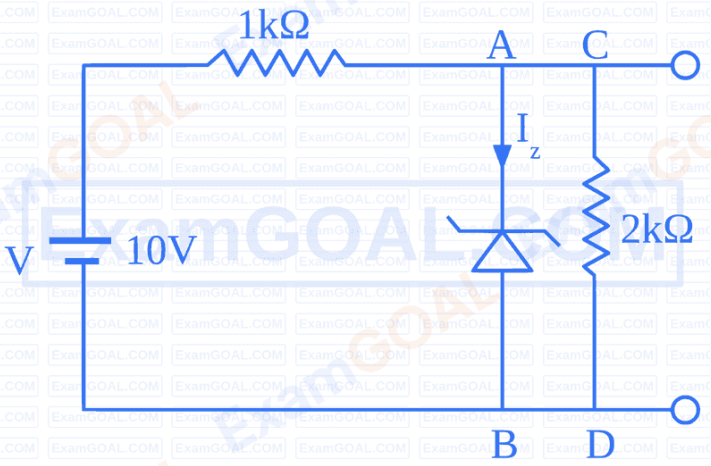
Q57. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

In the given circuit, the breakdown voltage of the Zener diode is $3.0\ \mathrm{V}$. What is the value of $\mathrm{I_z}$?



(A) 3.3 mA (B) 10 mA (C) 5.5 mA (D) 7 mA

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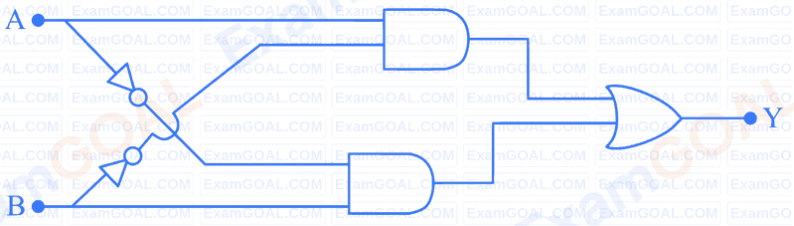
Q58. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

The truth table of the given circuit diagram is :



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(C)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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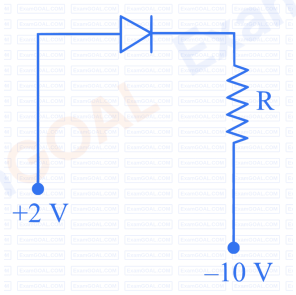
Q59. JEE Main 2024 (Online) 27th January Morning Shift

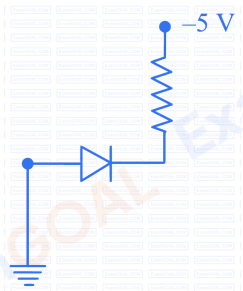
Open solution

Which of the following circuits is reverse - biased?

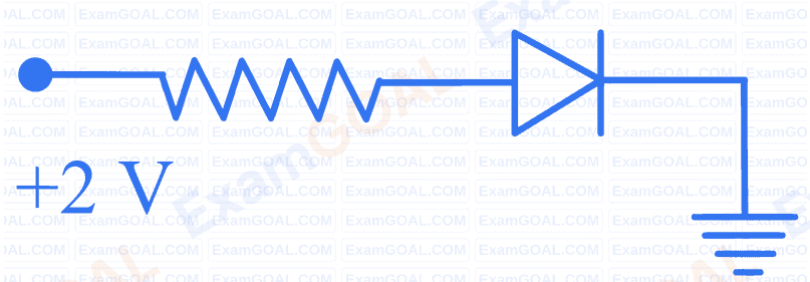
(A)



(B)



(C)



(D)

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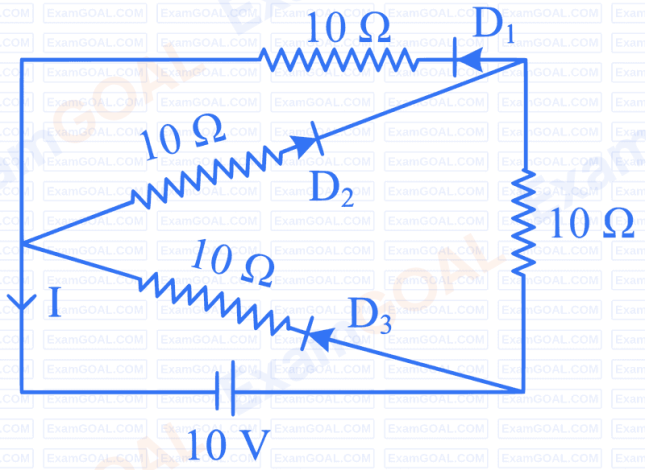
Q60. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

In the given circuit, the current (I) through the battery will be



(A) 1A (B) 2.5 A (C) 2 A (D) 1.5 A

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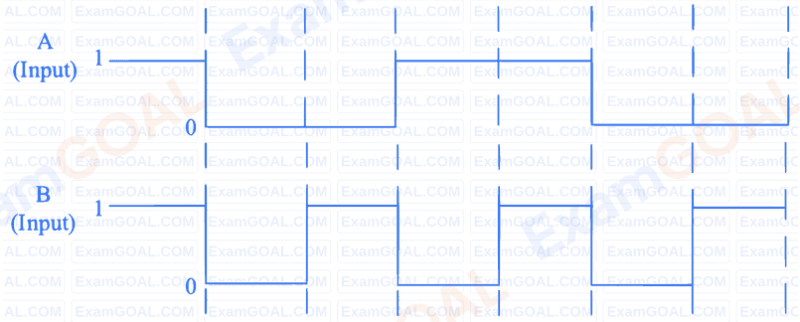
Q61. JEE Main 2023 (Online) 13th April Evening Shift

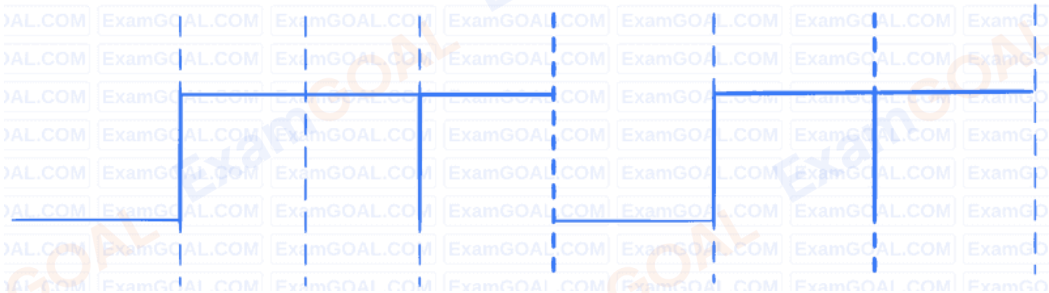
Open solution

The output from NAND gate having inputs A and B given below will be,

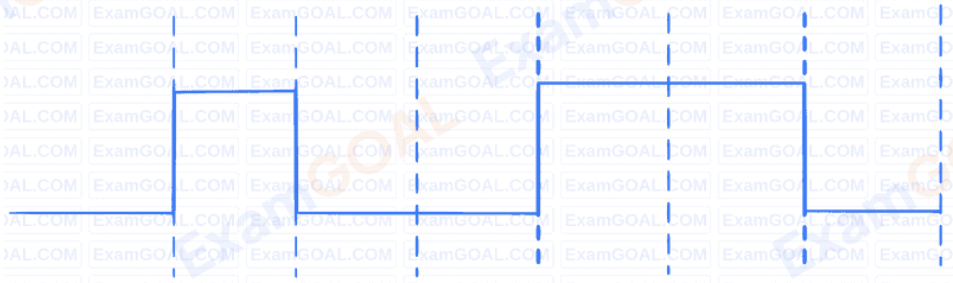


(A)

(B)



(C)



(D)

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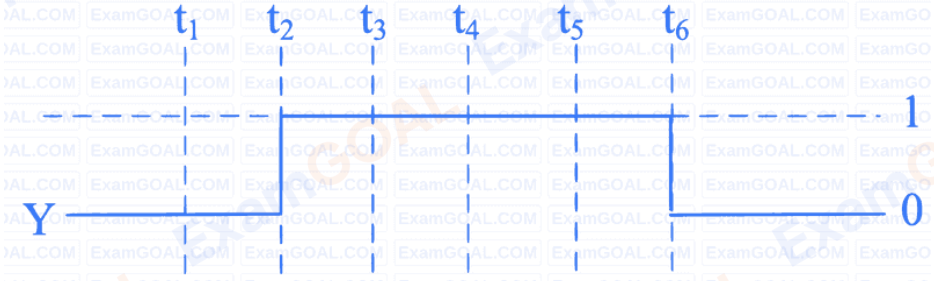
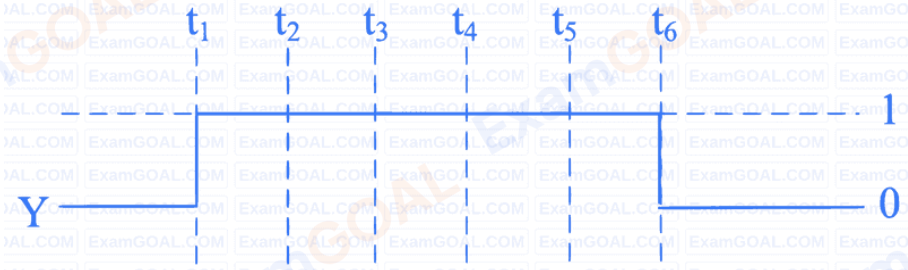
Q62. JEE Main 2023 (Online) 13th April Morning Shift

For the following circuit and given inputs A and B, choose the correct option for output 'Y'

(A)



(B)



(C)

(D)

Open solution

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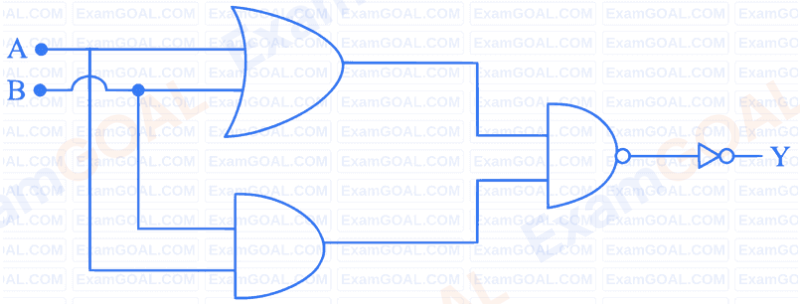
Q63. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

The logic operations performed by the given digital circuit is equivalent to:

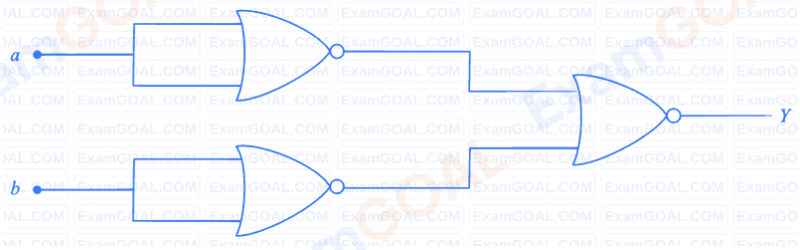


(A) NOR (B) AND (C) NAND (D) OR

Q64. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

The logic performed by the circuit shown in figure is equivalent to :

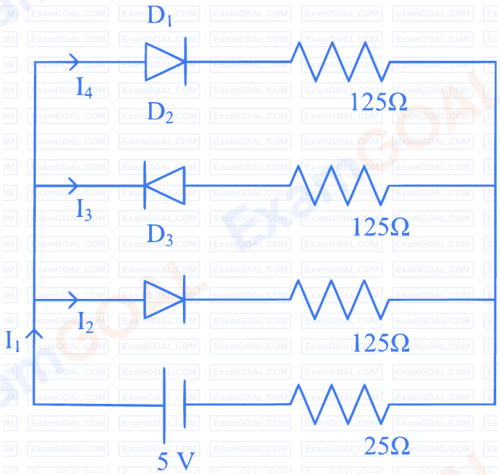


(A) NAND (B) AND (C) OR (D) NOR

Q65. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

If each diode has a forward bias resistance of $25\ \Omega$ in the below circuit,



Which of the following options is correct :

(A) $\frac{I_3}{I_4} = 1$ (B) $\frac{I_1}{I_2} = 2$

(C) $\frac{I_2}{I_3} = 1$ (D) $\frac{I_1}{I_2} = 1$

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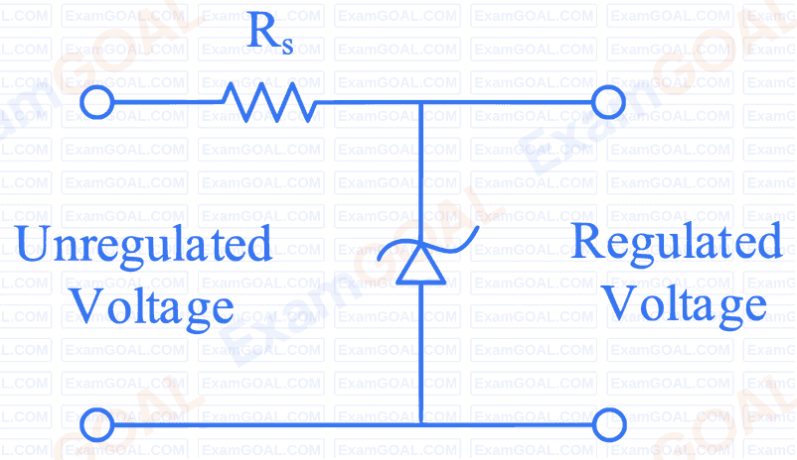
Q66. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

A zener diode of power rating 1.6 W is to be used as voltage regulator. If the zener diode has a breakdown of 8V and it has to regulate voltage fluctuating between 3 V and 10 V. The value of resistance R_s for safe operation of diode will be



(A) 13 Ω (B) 13.3 Ω (C) 10 Ω (D) 12 Ω

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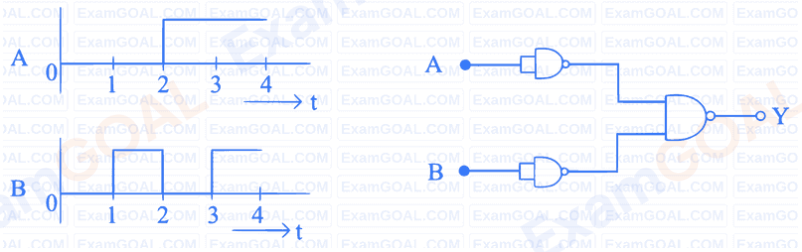
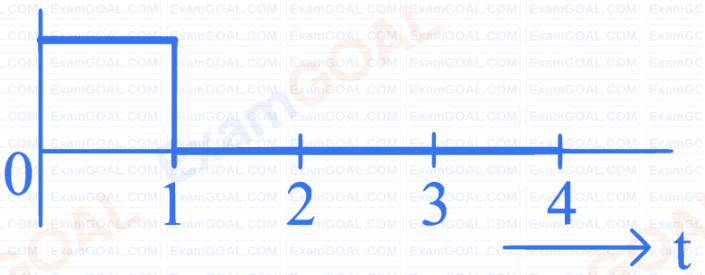
Q67. JEE Main 2023 (Online) 8th April Morning Shift

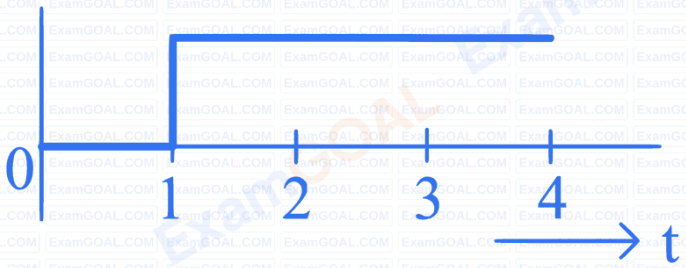
For the logic circuit shown, the output waveform at Y is:



(A)

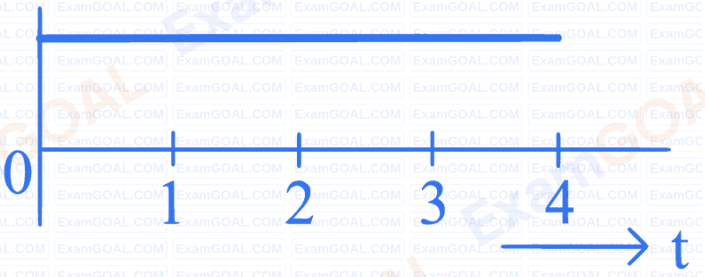


(B)



(C)

(D)



Open solution

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Q68. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R

Assertion A: Diffusion current in a p-n junction is greater than the drift current in magnitude if the junction is forward biased.

Reason R: Diffusion current in a p-n junction is from the n-side to the p-side if the junction is forward biased.

In the light of the above statements, choose the most appropriate answer from the options given below

(A) Both A and R are correct but R is NOT the correct explanation of A

(B) A is correct but R is not correct

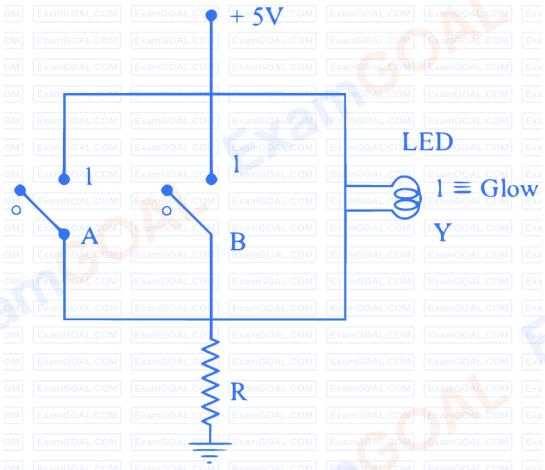
(C) A is not correct but R is correct

(D) Both A and R are correct and R is the correct explanation of A

Q69. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Name the logic gate equivalent to the diagram attached



(A) NOR (B) NAND (C) AND (D) OR

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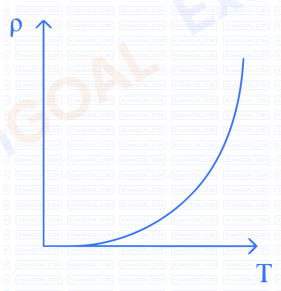
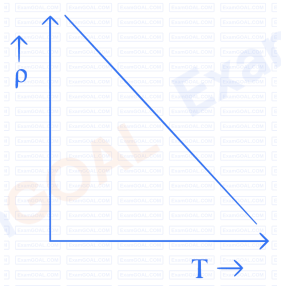
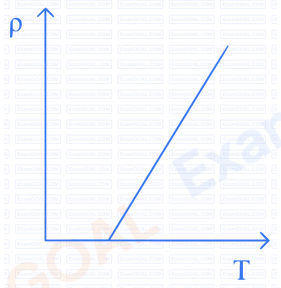
Q70. JEE Main 2023 (Online) 6th April Morning Shift

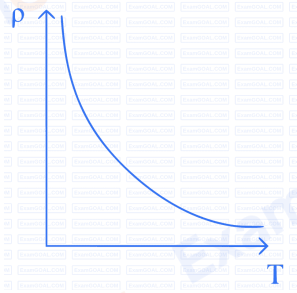
Open solution

The resistivity (ρ) of semiconductor varies with temperature. Which of the following curve represents the correct behaviour :

(A) 

(B) 

(C) 

(D) 

Q71. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

Choose the correct statement about Zener diode :

(A) It works as a voltage regulator in reverse bias and behaves like simple pn junction diode in forward bias.

(B) It works as a voltage regulator in both forward and reverse bias.

(C) It works as a voltage regulator only in forward bias.

(D) It works as a voltage regulator in forward bias and behaves like simple pn junction diode in reverse bias.

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Q72. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Match List I with List II:

	List I		List II
A.	Intrinsic semiconductor	I.	Fermi-level near the valence band
B.	n-type semiconductor	II.	Fermi-level in the middle of valence and conduction band.
C.	p-type semiconductor	III.	Fermi-level near the conduction band
D.	Metals	IV.	Fermi-level inside the conduction band

Choose the correct answer from the options given below :

(A) A-III, B-I, C-II, D-IV

(B) A-II, B-I, C-III, D-IV

(C) A-I, B-II, C-III, D-IV

(D) A-II, B-III, C-I, D-IV

Q73. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

The effect of increase in temperature on the number of electrons in conduction band (n_e) and resistance of a semiconductor will be as:

(A) n_e decreases, resistance increases

(B) Both n_e and resistance increase

(C) n_e increases, resistance decreases

(D) Both n_e and resistance decrease

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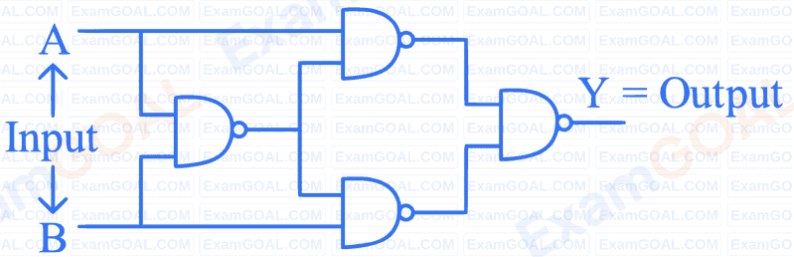
Q74. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

The output Y for the inputs A and B of circuit is given by



Truth table of the shown circuit is:

(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(D)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

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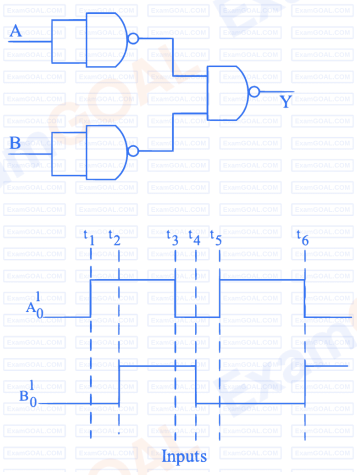
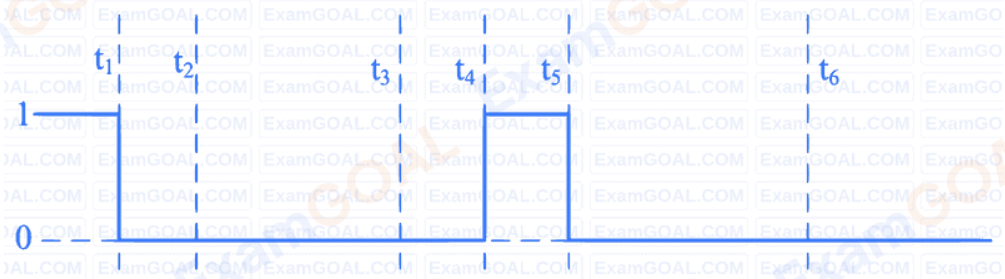
Q75. JEE Main 2023 (Online) 30th January Morning Shift

The output waveform of the given logical circuit for the following inputs A and B as shown below, is :

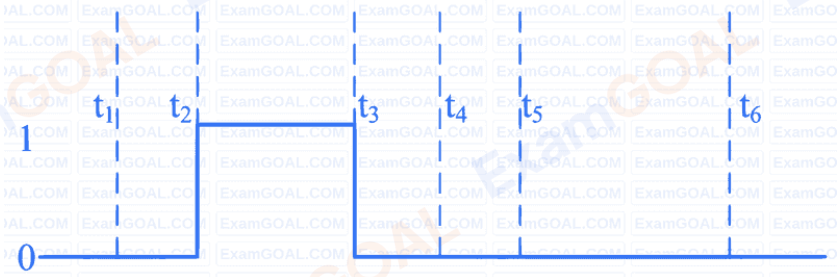
Open solution



(A)

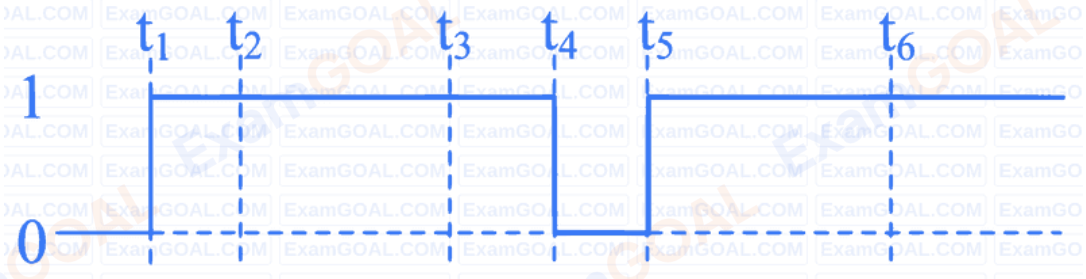


(B)



(C)

(D)



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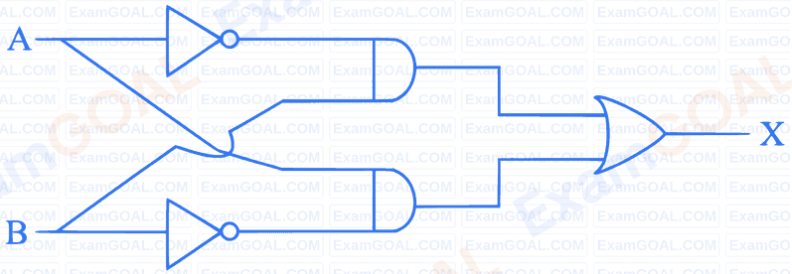
Q76. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

For the given logic gates combination, the correct truth table will be



(A)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(B)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(C)

A	B	X
0	0	1
0	1	0
1	0	1
1	1	0

(D)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

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Q77. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Which one of the following statement is **not** correct in the case of light emitting diodes?

A. It is a heavily doped p-n junction.

B. It emits light only when it is forward biased.

C. It emits light only when it is reverse biased.

D. The energy of the light emitted is equal to or slightly less than the energy gap of the semiconductor used.

Choose the correct answer from the options given below:

(A) A **(B)** B **(C)** C and D **(D)** C

Q78. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Statement I : When a Si sample is doped with Boron, it becomes P type and when doped by Arsenic it becomes N-type semi conductor such that P-type has excess holes and N-type has excess electrons.

Statement II : When such P-type and N-type semi-conductors, are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter.

In the light of above statements, choose the most appropriate answer from the options given below

(A) Statement I is incorrect but statement II is correct

(B) Both Statement I and statement II are correct

(C) Statement I is correct but statement II is incorrect

(D) Both Statement I and Statement II are incorrect

Q79. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Photodiodes are used in forward bias usually for measuring the light intensity.

Reason R : For a p-n junction diode, at applied voltage V the current in the forward bias is more than the current in the reverse bias for $|V_z| > \pm V \geq |V_0|$ where V_0 is the threshold voltage and V_z is the breakdown voltage.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true but R is NOT the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true

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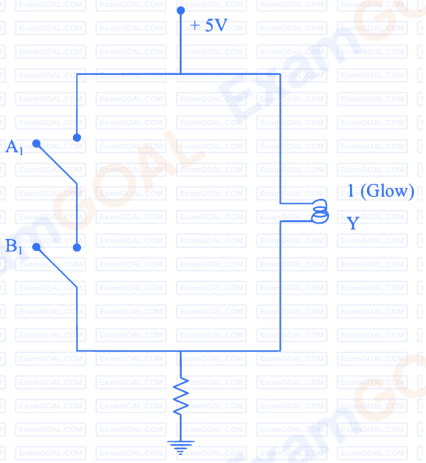
Q80. JEE Main 2023 (Online) 24th January Evening Shift

Open solution



The logic gate equivalent to the given circuit diagram is :

(A) NOR **(B)** OR **(C)** NAND **(D)** AND

Q81. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Photodiodes are preferably operated in reverse bias condition for light intensity measurement.

Reason R : The current in the forward bias is more than the current in the reverse bias for a $p-n$ junction diode.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true and R is the correct explanation of A

(B) A is false but R is true

(C) A is true but R is false

(D) Both A and R are true but R is NOT the correct explanation of A

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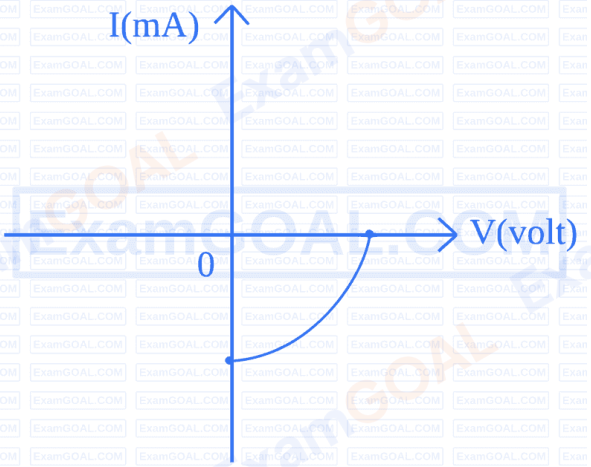
Q82. JEE Main 2022 (Online) 30th July Evening Shift

Open solution



The I-V characteristics given in above figure is related to:

(A) Photodiode in reverse bias

(B) LED in forward bias

(C) Zener diode

(D) Solar cell

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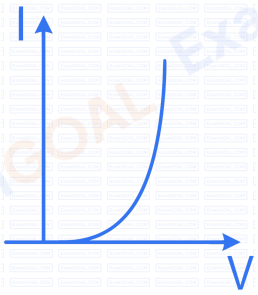
Q83. JEE Main 2022 (Online) 28th July Morning Shift

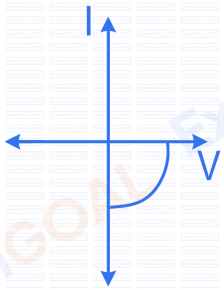
Open solution

Identify the solar cell characteristics from the following options :

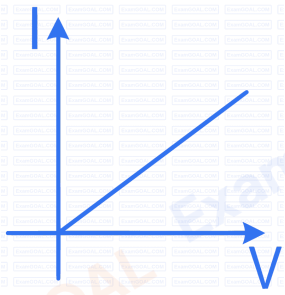
(A)



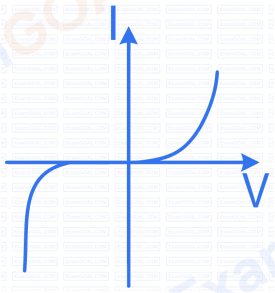
(B)



(C)



(D)



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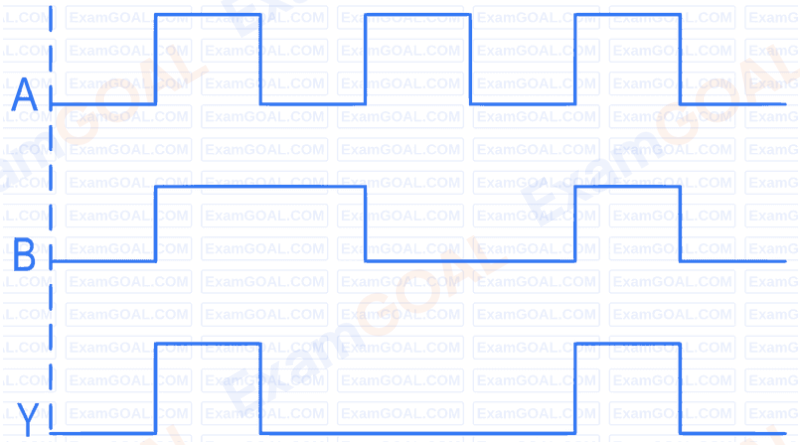
Q84. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

A logic gate circuit has two inputs A and B and output Y. The voltage waveforms of A, B and Y are shown below.



The logic gate circuit is :

(A) AND gate (B) OR gate (C) NOR gate (D) NAND gate

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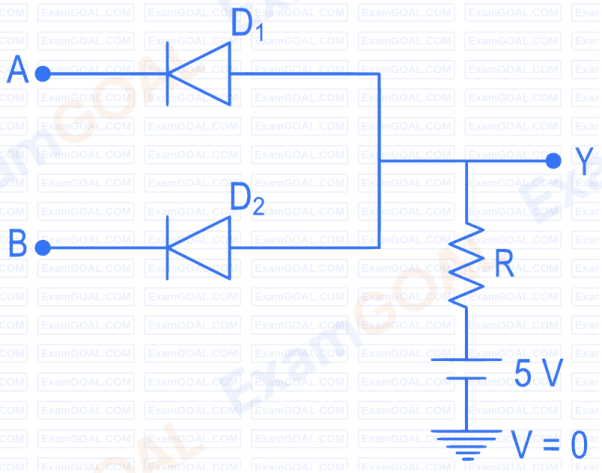
Q85. JEE Main 2022 (Online) 25th July Morning Shift

Open solution

In the circuit, the logical value of $A = 1$ or $B = 1$ when potential at A or B is $5\ \text{V}$ and the logical value of $A = 0$ or $B = 0$ when potential at A or B is $0\ \text{V}$.



The truth table of the given circuit will be :

(A)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

(B)

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

(C)

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

(D)

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

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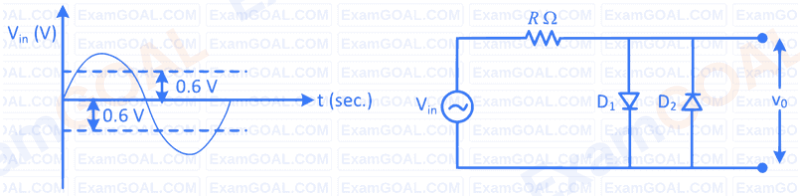
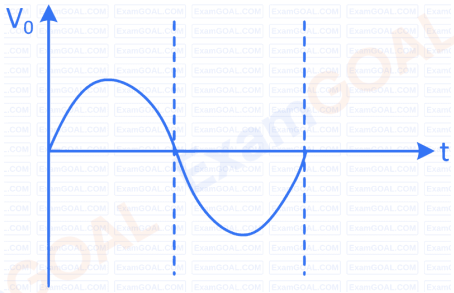
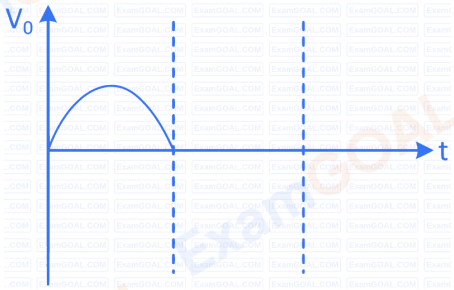
Q86. JEE Main 2022 (Online) 28th June Evening Shift

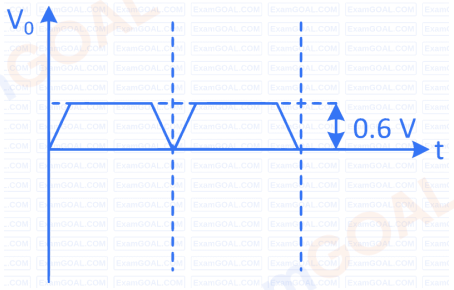
Open solution

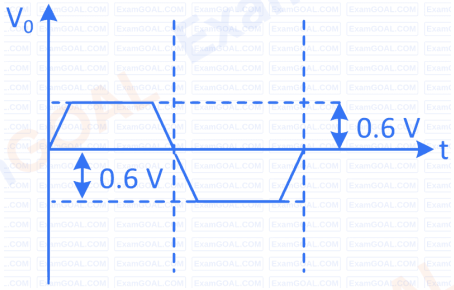
In the given circuit the input voltage V_{in} is shown in figure. The cut-in voltage of p-n junction diode (D_1 or D_2) is 0.6 V. Which of the following output voltage (V_0) waveform across the diode is correct?



(A)


(B)


(C)


(D)


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Q87. JEE Main 2022 (Online) 28th June Morning Shift

For using a multimeter to identify diode from electrical components, choose the correct statement out of the following about the diode :

(A) It is two terminal device which conducts current in both directions.

(B) It is two terminal device which conducts current in one direction only.

(C) It does not conduct current gives an initial deflection which decays to zero.

(D) It is three terminal device which conducts current in one direction only between central terminal and either of the remaining two terminals

Open solution

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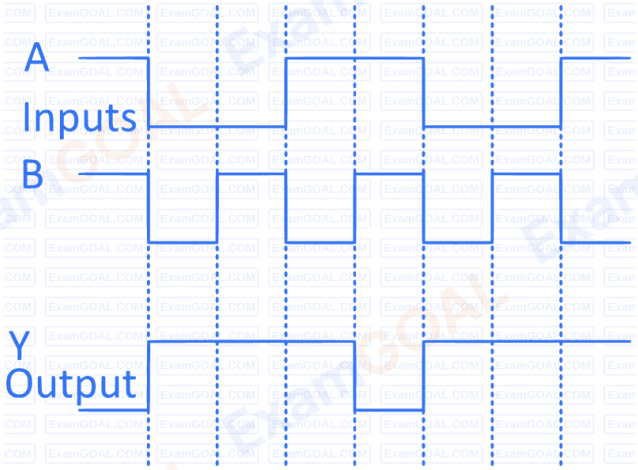
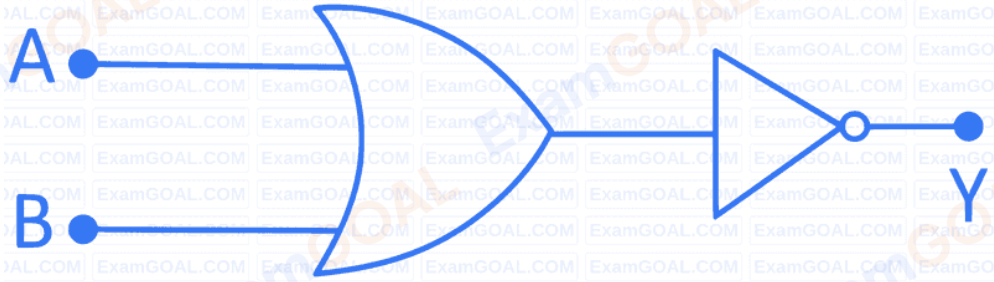
Q88. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Identify the correct Logic Gate for the following output (Y) of two inputs A and B.

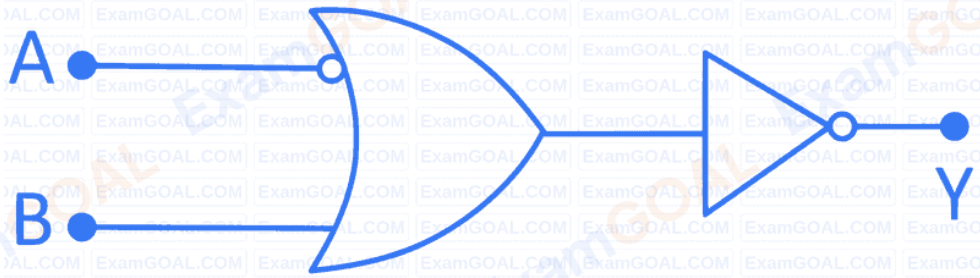


(A)

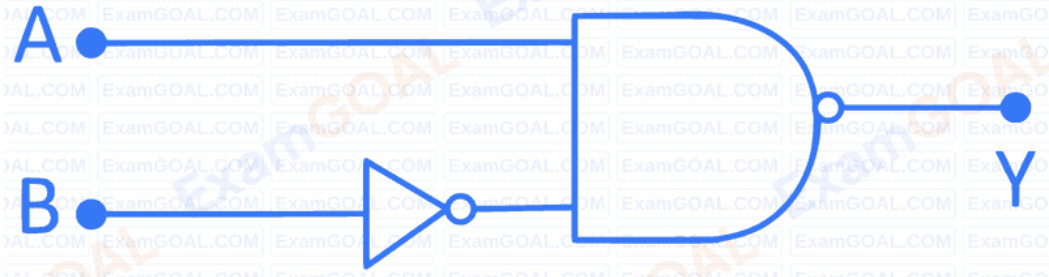


(B)

(C)



(D)



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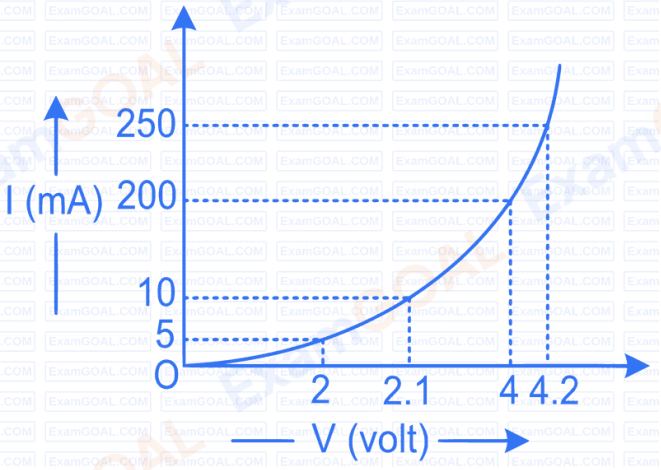
Q89. JEE Main 2022 (Online) 26th June Morning Shift

The I-V characteristics of a p-n junction diode in forward bias is shown in the figure. The ratio of dynamic resistance, corresponding to forward bias voltage of 2 V and 4 V respectively, is :

Open solution

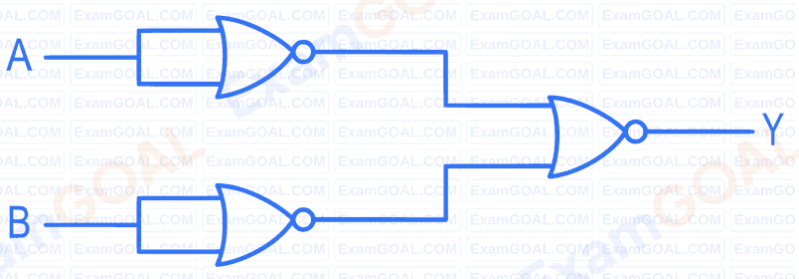


(A) 1 : 2 (B) 5 : 1 (C) 1 : 40 (D) 20 : 1

Q90. JEE Main 2022 (Online) 25th June Evening Shift

Identify the logic operation performed by the given circuit:

Open solution



(A) AND gate (B) OR gate (C) NOR gate (D) NAND gate

Q91. JEE Main 2022 (Online) 25th June Morning Shift

The photodiode is used to detect the optical signals. These diodes are preferably operated in reverse biased mode because :

Open solution

(A) fractional change in majority carriers produce higher forward bias current

(B) fractional change in majority carriers produce higher reverse bias current

(C) fractional change in minority carriers produce higher forward bias current

(D) fractional change in minority carriers produce higher reverse bias current

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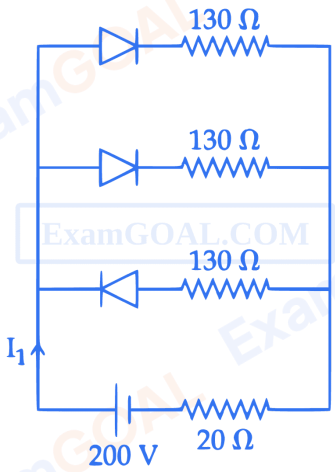
Q92. JEE Main 2021 (Online) 1st September Evening Shift

Open solution

In the given figure, each diode has a forward bias resistance of 30Ω and infinite resistance in reverse bias. The current I_1 will be :



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(A) 3.75 A (B) 2.35 A (C) 2 A (D) 2.73 A

Q93. JEE Main 2021 (Online) 31st August Evening Shift

Open solution

Statement - I :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect a capacitor across the output parallel to the load R_L.

Statement - II :

To get a steady dc output from the pulsating voltage received from a full wave rectifier we can connect an inductor in series with R_L.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

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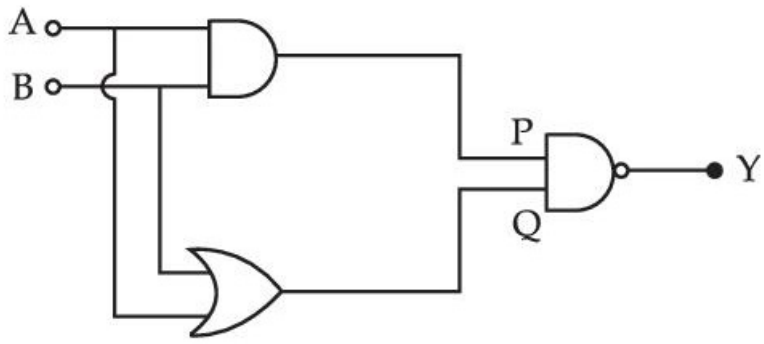
Q94. JEE Main 2021 (Online) 31st August Morning Shift

In the following logic circuit the sequence of the inputs A, B are (0, 0), (0, 1), (1, 0) and (1, 1). The output Y for this sequence will be :

Open solution



(A) 1, 0, 1, 0 (B) 0, 1, 0, 1 (C) 1, 1, 1, 0 (D) 0, 0, 1, 1

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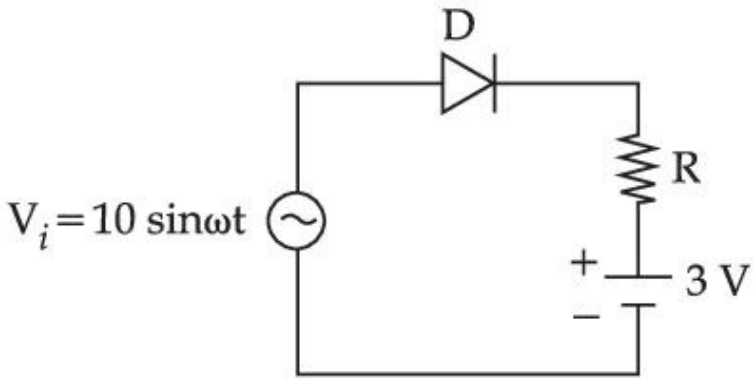
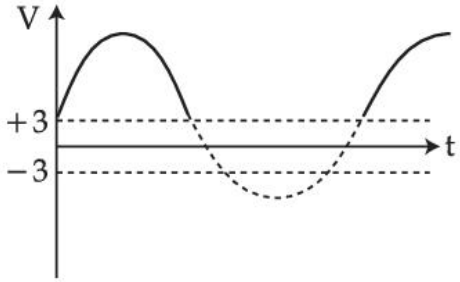
Q95. JEE Main 2021 (Online) 31st August Morning Shift

Choose the correct waveform that can represent the voltage across R of the following circuit, assuming the diode is ideal one :

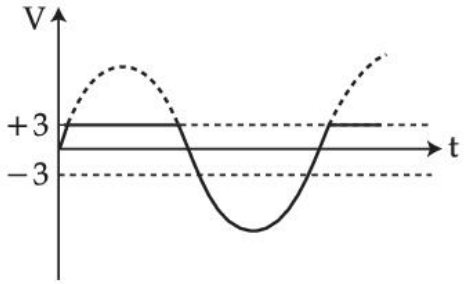
Open solution



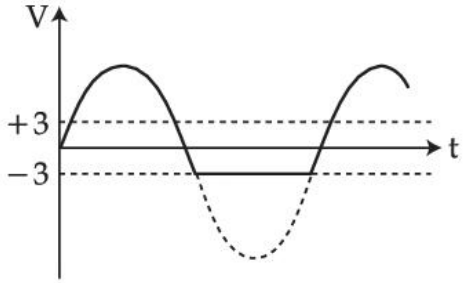
(A)



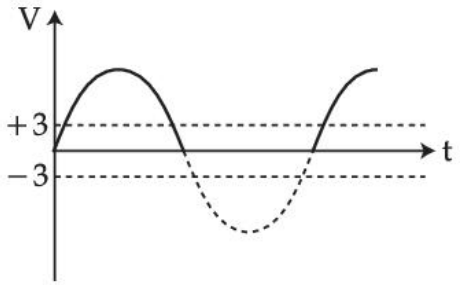
(B)



(C)



(D)



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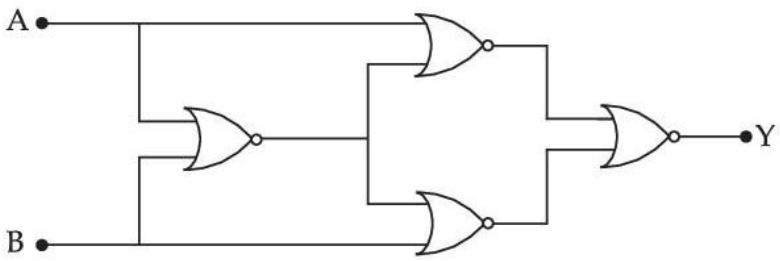
Q96. JEE Main 2021 (Online) 26th August Evening Shift

Four NOR gates are connected as shown in figure. The truth table for the given figure is :

Open solution



(A)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(C)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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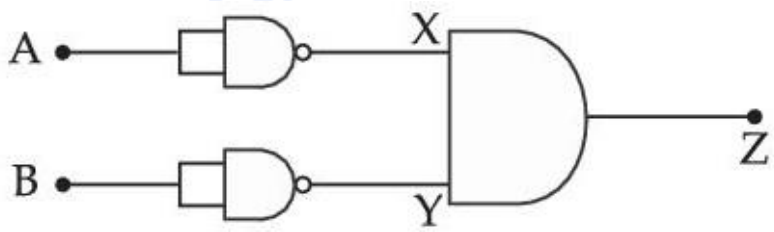
Q97. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Identify the logic operation carried out by the given circuit :-



(A) OR (B) AND (C) NOR (D) NAND

Q98. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Statement I : By doping silicon semiconductor with pentavalent material, the electrons density increases.

Statement II : The n-type semiconductor has net negative charge.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement - I is true but Statement - II is false.

(B) Statement - I is false but Statement - II is true.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are false.

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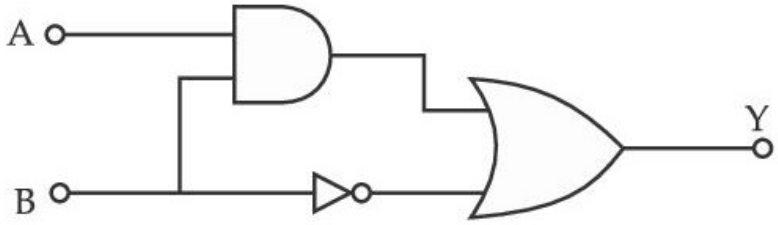
Q99. JEE Main 2021 (Online) 27th July Evening Shift

Open solution

Find the truth table for the function Y of A and B represented in the following figure.



(A)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(D)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

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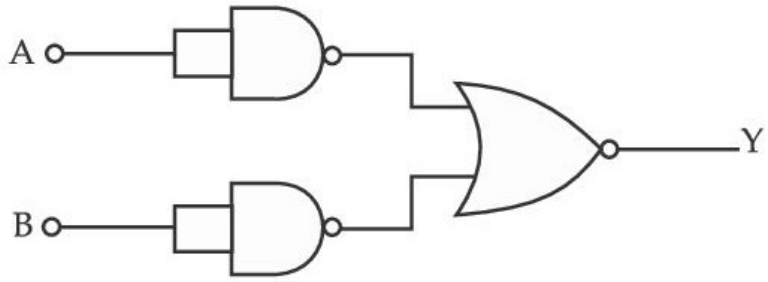
Q100. JEE Main 2021 (Online) 25th July Morning Shift

Identify the logic operation carried out.

Open solution



(A) OR (B) AND (C) NOR (D) NAND

Q101. JEE Main 2021 (Online) 22th July Evening Shift

Consider a situation in which reverse biased current of a particular P-N junction increases when it is exposed to a light of wavelength $\leq$ 621 nm. During this process, enhancement in carrier concentration takes place due to generation of hole-electron pairs. The value of band gap is nearly.

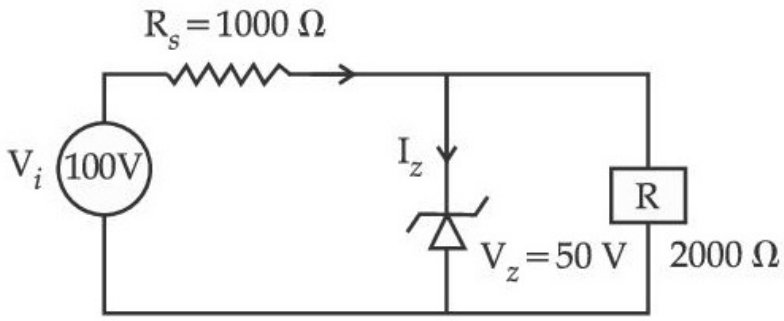
Open solution

(A) 1 eV (B) 4 eV (C) 0.5 eV (D) 2 eV

Q102. JEE Main 2021 (Online) 20th July Morning Shift

For the circuit shown below, calculate the value of I_z :

Open solution



(A) 25 mA (B) 0.1 A (C) 0.15 A (D) 0.05 A

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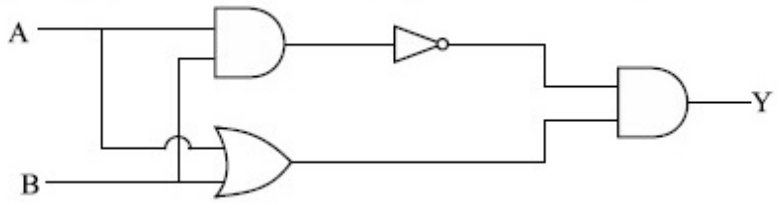
Q103. JEE Main 2021 (Online) 17th March Evening Shift

Which one of the following will be the output of the given circuit?

Open solution

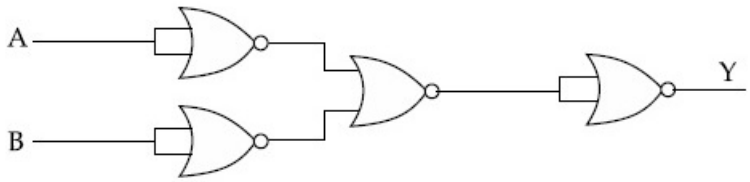


(A) XOR Gate (B) NOR Gate (C) NAND Gate (D) AND Gate

Q104. JEE Main 2021 (Online) 17th March Morning Shift

The output of the given combination gates represents :

Open solution

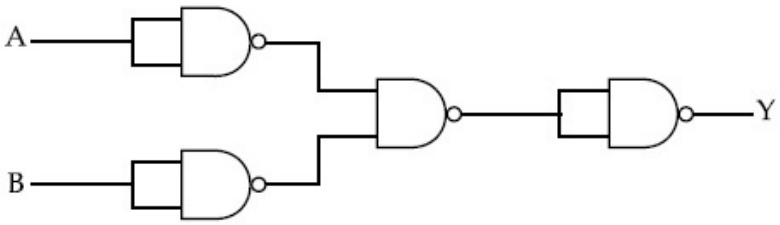


(A) AND Gate (B) NOR Gate (C) NAND Gate (D) XOR Gate

Q105. JEE Main 2021 (Online) 16th March Evening Shift

The following logic gate is equivalent to :

Open solution



(A) NOR Gate (B) NAND Gate (C) OR Gate (D) AND Gate

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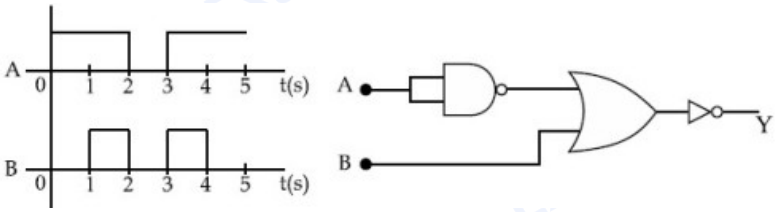
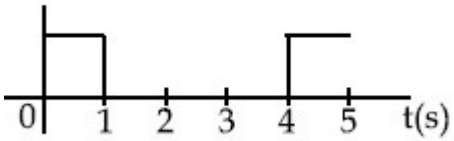
Q106. JEE Main 2021 (Online) 26th February Evening Shift

Open solution

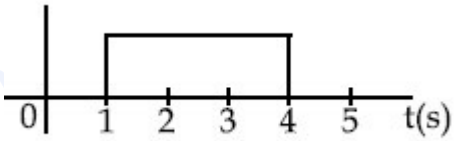
Draw the output signal Y in the given combination of gates.



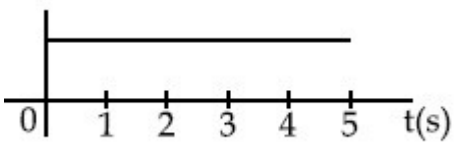
(A)



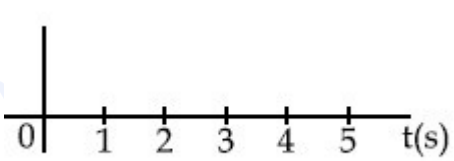
(B)



(C)



(D)



Q107. JEE Main 2021 (Online) 26th February Morning Shift

Open solution

LED is constructed from Ga-As-P semiconducting material. The energy gap of this LED is 1.9 eV. Calculate the wavelength of light emitted and its colour.

[h = 6.63 $\times$ 10^{-34} Js and c = 3 $\times$ 10^{8} ms^{-1}]

(A) 654 nm and orange colour

(B) 654 nm and red colour

(C) 1046 nm and red colour

(D) 1046 nm and blue colour

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Q108. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

Match List I with List II.

	List I		List II
(a)	Rectifier	(i)	Used either for stepping up or stepping down the a.c. voltage
(b)	Stabilizer	(ii)	Used to convert a.c. voltage into d.c. voltage
(c)	Transformer	(iii)	Used to remove any ripple in the rectified output voltage
(d)	Filter	(iv)	Used for constant output voltage even when the input voltage or load current change

Choose the correct answer from the options given below :

(A) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

(D) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Q109. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

For extrinsic semiconductors; when doping level is increased;

(A) Fermi-level of both p-type and n-type semiconductors will go upward for $T > T_F K$ and downward for $T < T_F K$, where T_F is Fermi temperature.

(B) Fermi-level of p-type semiconductor will go upward and Fermi-level of n-type semiconductors will go downward

(C) Fermi-level of p and n-type semiconductors will not be affected.

(D) Fermi-level of p-type semiconductors will go downward and Fermi-level of n-type semiconductor will go upward.

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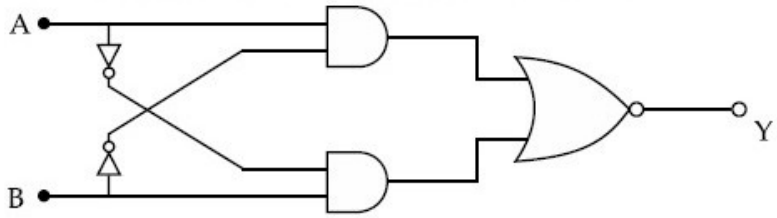
Q110. JEE Main 2021 (Online) 25th February Evening Shift

Open solution

The truth table for the following logic circuit is :



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(C)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

(D)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

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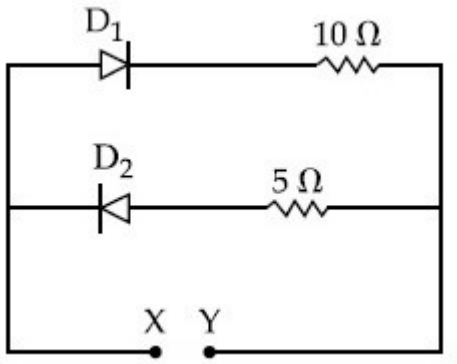
Q111. JEE Main 2021 (Online) 25th February Morning Shift

A 5V battery is connected across the points X and Y. Assume D_1 and D_2 to be normal silicon diodes. Find the current supplied by the battery if the +ve terminal of the battery is connected to point X.

Open solution



(A) $\sim$ 0.5 A (B) $\sim$ 1.5 A (C) $\sim$ 0.43 A (D) $\sim$ 0.86 A

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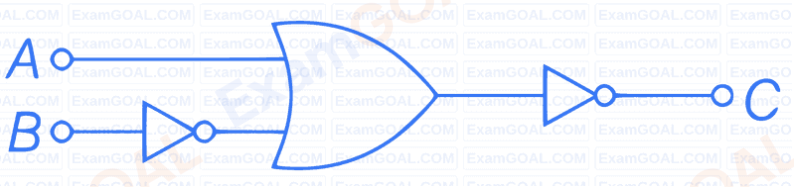
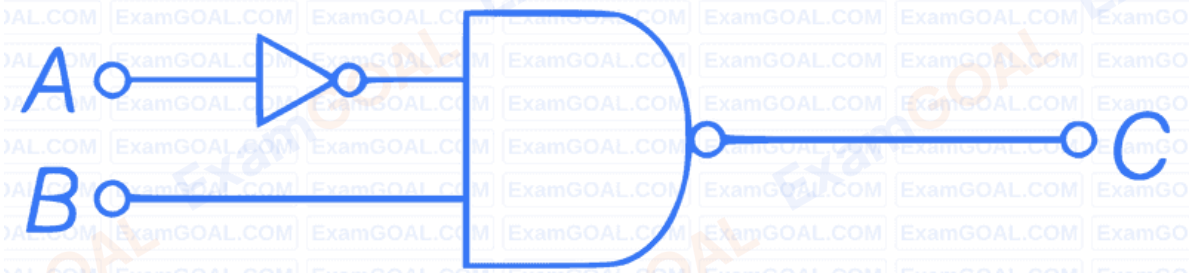
Q112. JEE Main 2021 (Online) 24th February Evening Shift



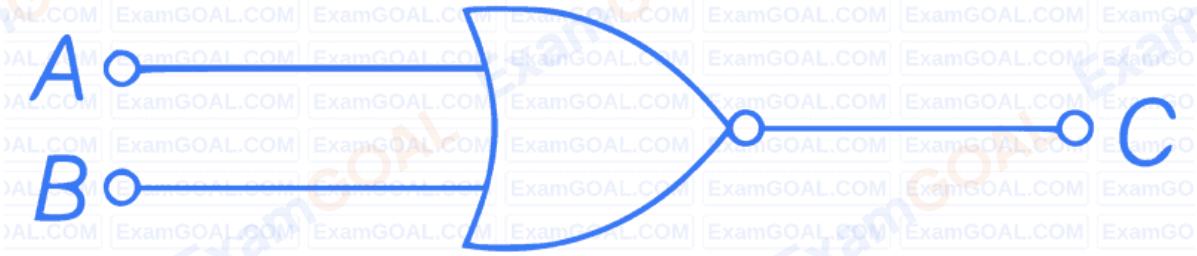
Open solution

The logic circuit shown above is equivalent to :

(A)



(B)



(C)

(D)

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Q113. JEE Main 2021 (Online) 24th February Evening Shift

Zener breakdown occurs in a p – n junction having p and n both :

(A) heavily doped and have wide depletion layer.

(B) lightly doped and have narrow depletion layer.

(C) heavily doped and have narrow depletion layer.

(D) lightly doped and have wide depletion layer.

Open solution

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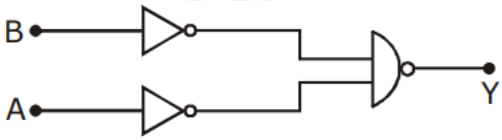
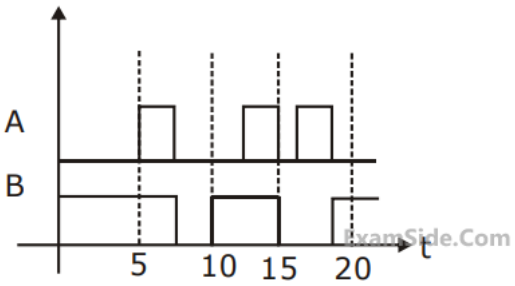
Q114. JEE Main 2020 (Online) 6th September Morning Slot

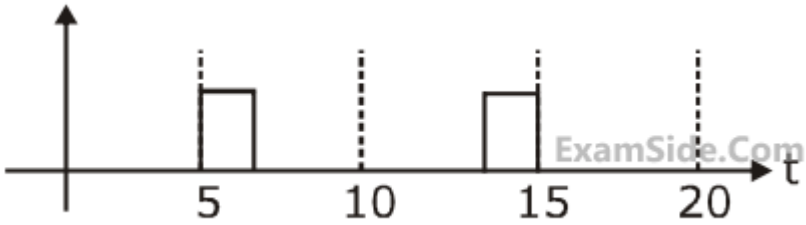
Open solution

Identify the correct output signal Y in the given combination of gates (as shown) for the given inputs A and B.

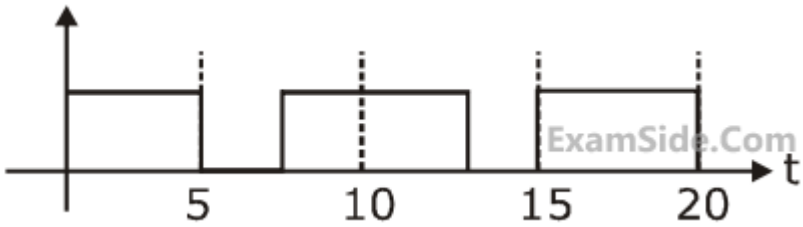




(A)

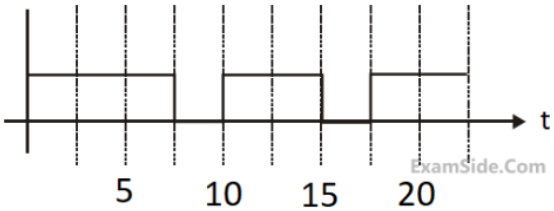


(B)



(C)

(D)



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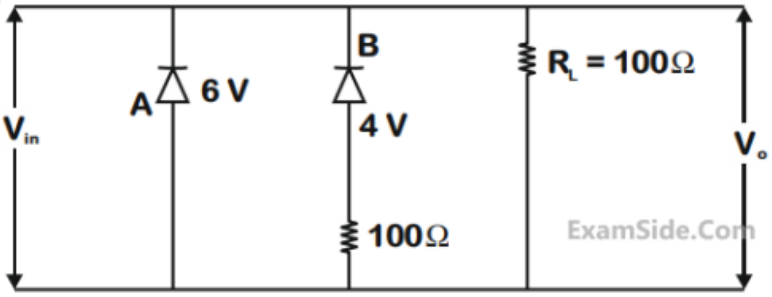
Q115. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

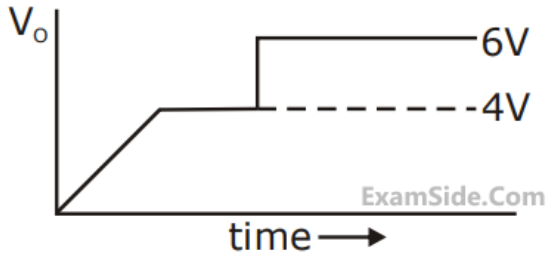
Two Zener diodes (A and B) having breakdown voltages of 6 V and 4 V respectively, are connected as shown in the circuit below. The output voltage V_0 variation with input voltage linearly increasing with time, is given by :

(V_{input} = 0 V at t = 0)

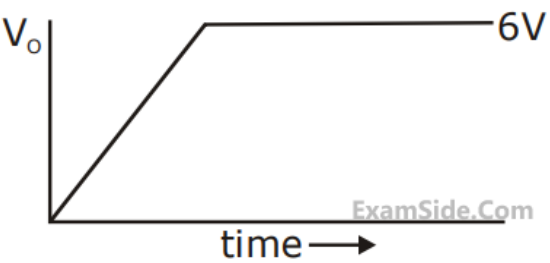
(figures are qualitative)



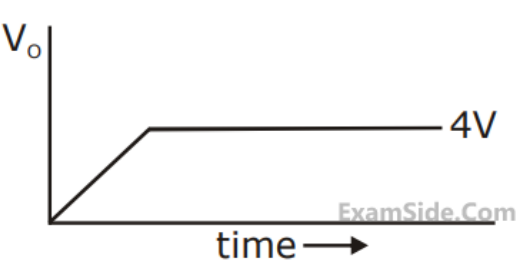
(A)



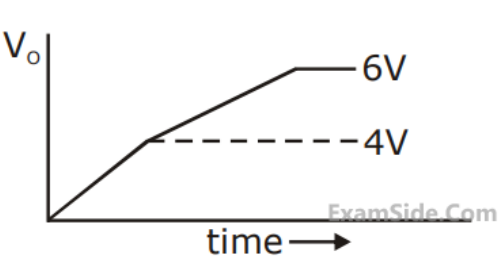
(B)



(C)



(D)



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Q116. JEE Main 2020 (Online) 5th September Morning Slot

With increasing biasing voltage of a photodiode, the photocurrent magnitude :

(A) Increases initially and after attaining certain value, it decreases

(B) Increases linearly

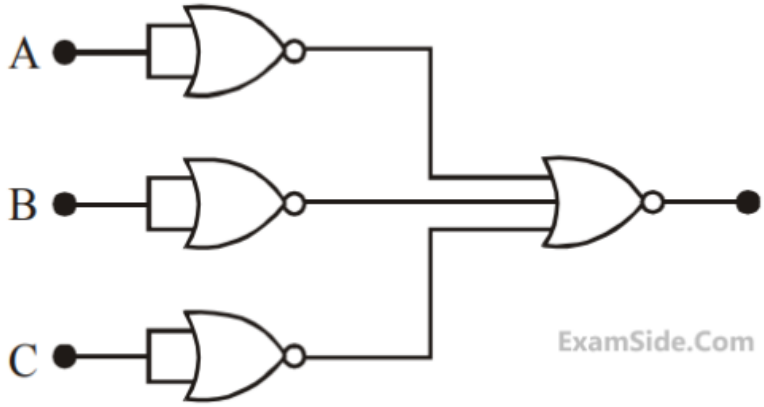
(C) Increases initially and saturates finally

(D) Remains constant

Open solution

Q117. JEE Main 2020 (Online) 4th September Evening Slot

Identify the operation performed by the circuit given below :



(A) AND (B) NAND (C) OR (D) NOT

Open solution

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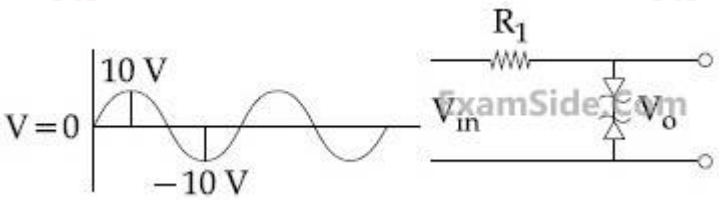
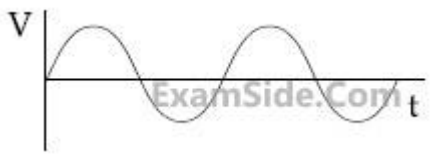
Q118. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

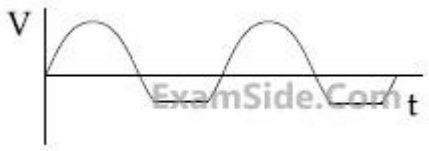
Take the breakdown voltage of the zener diode used in the given circuit as 6V. For the input voltage shown in figure below, the time variation of the output voltage is: (Graphs drawn are schematic and not to scale)



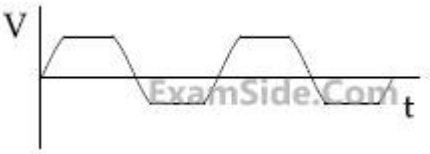
(A)



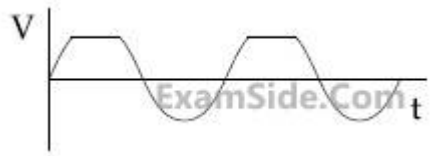
(B)



(C)



(D)



Q119. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

If a semiconductor photodiode can detect a photon with a maximum wavelength of 400 nm, then its band gap energy is :

Planck's constant h = 6.63×10^{-34} J.s. Speed of light c = 3×10^{8} m/s

(A) 1.5 eV (B) 2.0 eV (C) 3.1 eV (D) 1.1 eV

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Q120. JEE Main 2020 (Online) 3rd September Evening Slot

Which of the following will NOT be observed when a multimeter (operating in resistance measuring mode) probes connected across a component, are just reversed?

(A) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is metal wire.

(B) Multimeter shows a deflection, accompanied by a splash of light out of connected component in one direction and NO deflection on reversing the probes if the chosen component is LED.

(C) Multimeter shows an equal deflection in both cases i.e. before and after reversing the probes if the chosen component is resistor.

(D) Multimeter shows NO deflection in both cases i.e. before and after reversing the probes if the chosen component is capacitor.

Open solution

Q121. JEE Main 2020 (Online) 3rd September Morning Slot

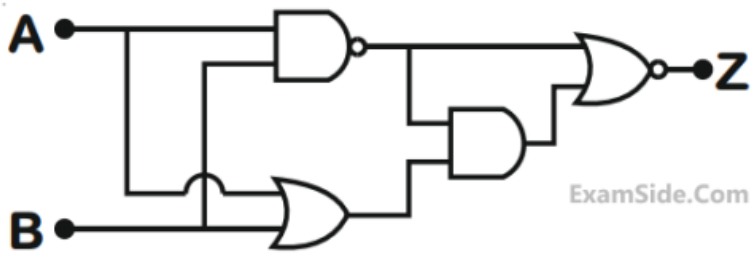
When a diode is forward biased, it has a voltage drop of 0.5 V. The safe limit of current through the diode is 10 mA. If a battery of emf 1.5 V is used in the circuit, the value of minimum resistance to be connected in series with the diode so that the current does not exceed the safe limit is

(A) 50 Ω (B) 200 Ω (C) 300 Ω (D) 100 Ω

Open solution

Q122. JEE Main 2020 (Online) 2nd September Evening Slot

In the following digital circuit, what will be the output at 'Z', when the input (A, B) are (1, 0), (0, 0), (1, 1,), (0, 1)



(A) 1, 1, 0, 1 (B) 0, 1, 0, 0 (C) 1, 0, 1, 1 (D) 0, 0, 1, 0

Open solution

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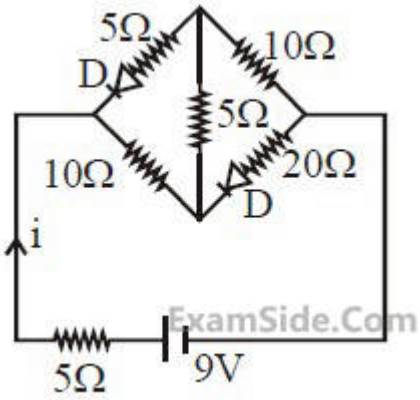
Q123. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

The current i in the network is :

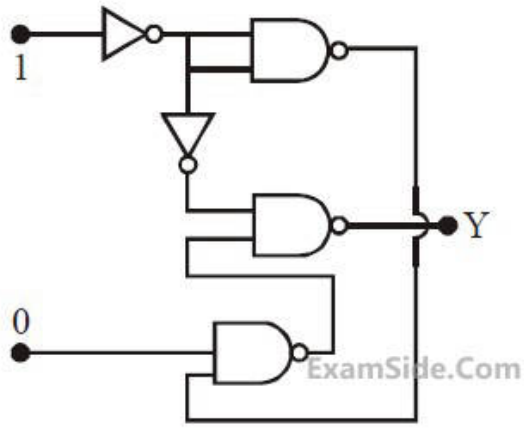


(A) 0.6 A (B) 0.3 A (C) 0 A (D) 0.2 A

Q124. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

In the given circuit, value of Y is :



(A) toggles between 0 and 1 (B) 1

(C) will not execute (D) 0

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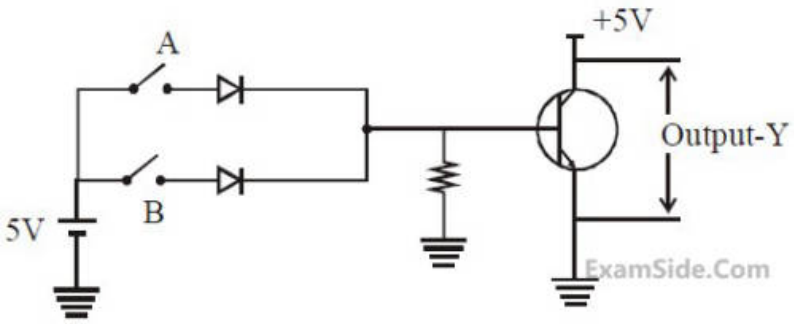
Q125. JEE Main 2020 (Online) 8th January Morning Slot

Boolean relation at the output stage-Y for the following circuit is :

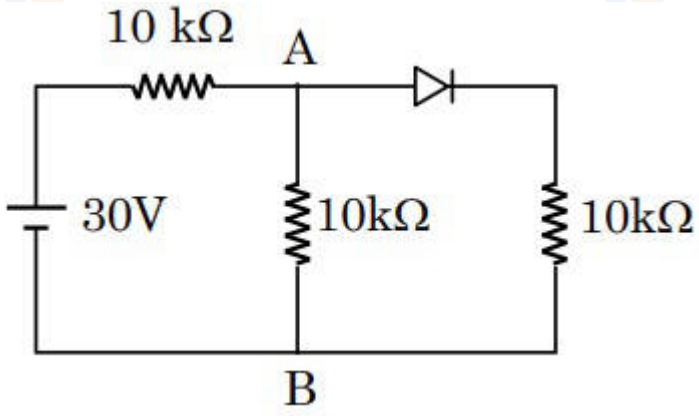


Open solution

(A) A.B (B) $\overline{A}+\overline{B}$ (C) $\overline{A}.\overline{B}$ (D) A + B

Q126. JEE Main 2020 (Online) 7th January Evening Slot

In the figure, potential difference between A and B is :



Open solution

(A) 10 V (B) 5 V (C) zero (D) 15 V

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Q127. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

Which of the following gives a reversible operation?

(A)

(B)

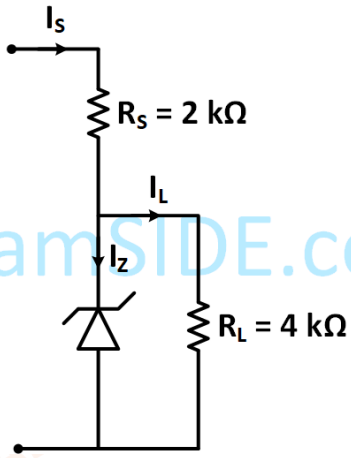
(C)

(D)

Q128. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

Figure shows a DC voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is maximum Zener current?



(A) 3.5 mA (B) 1.5 mA (C) 2.5 mA (D) 7.5 mA

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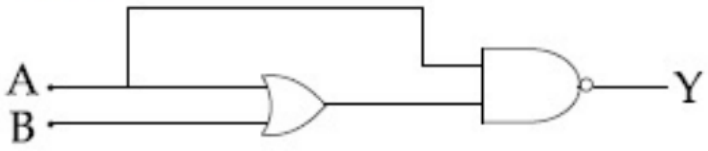
Q129. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

The truth table for the circuit given in the fig. is:



(A)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	1

(B)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(C)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

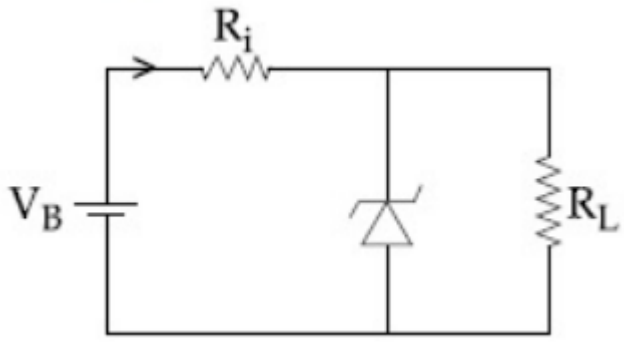
(D)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

Q130. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is, R_L = 4kΩ. The series resistance of the circuit is R_i = 1 kΩ. If the battery voltage V_B varies from 8 V to 16 V, what are the minimum and maximum values of the current through Zener diode?



(A) 0.5 mA; 8.5 mA (B) 1.5 mA; 8.5 mA (C) 1 mA; 8.5 mA (D) 0.5 mA; 6 mA

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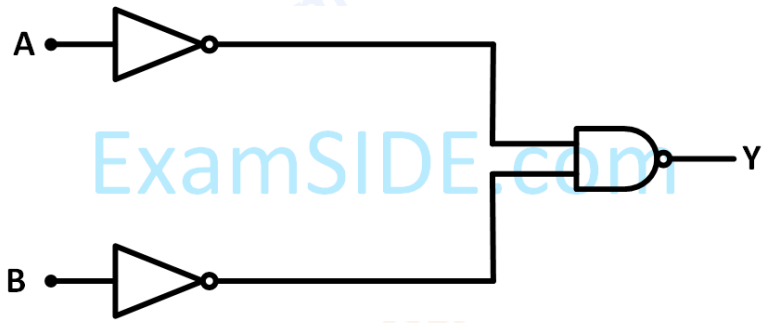
Q131. JEE Main 2019 (Online) 9th April Evening Slot

The logic gate equivalent to the given logic circuit is :-



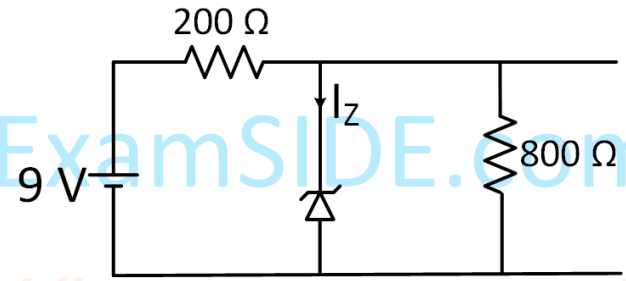
Open solution

(A) NAND (B) AND (C) NOR (D) OR

Q132. JEE Main 2019 (Online) 8th April Morning Slot

The reverse breakdown voltage of a Zener diode is 5.6 V in the given circuit.

The current IZ through the Zener is :

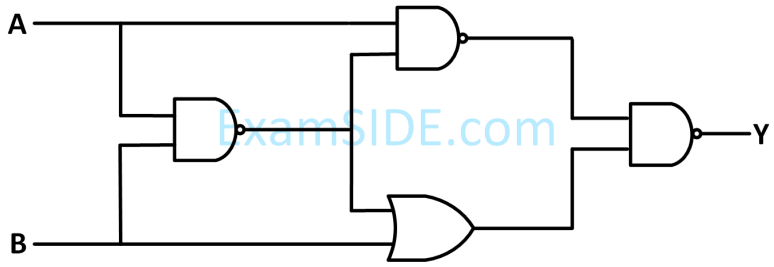


Open solution

(A) 7 mA (B) 17 mA (C) 15mA (D) 10 mA

Q133. JEE Main 2019 (Online) 12th January Morning Slot

The output of the given logic circuit is :



Open solution

(A) $\overline{A}B$ (B) $AB+\overline{A}\overline{B}$ (C) $A\overline{B}+\overline{A}B$ (D) $A\overline{B}$

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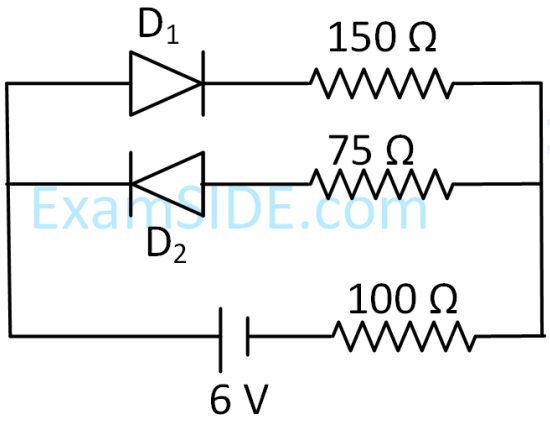
Q134. JEE Main 2019 (Online) 11th January Evening Slot

The circuit shown below contains two ideal diodes, each with a forward resistance of 50 Ω. If the battery voltage is 6 V, the current through the 100 Ω resistance (in Amperes) is :

Open solution

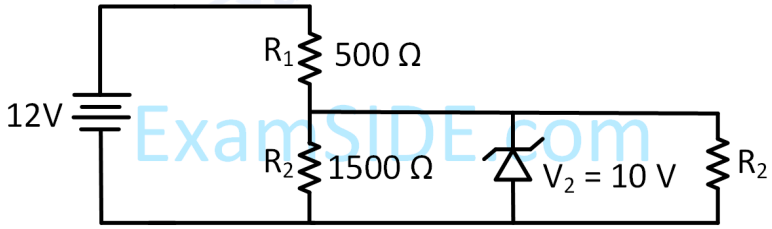


(A) 0.027 (B) 0.030 (C) 0.036 (D) 0.020

Q135. JEE Main 2019 (Online) 11th January Morning Slot

In the given circuit the current through Zener Diode is close to:

Open solution

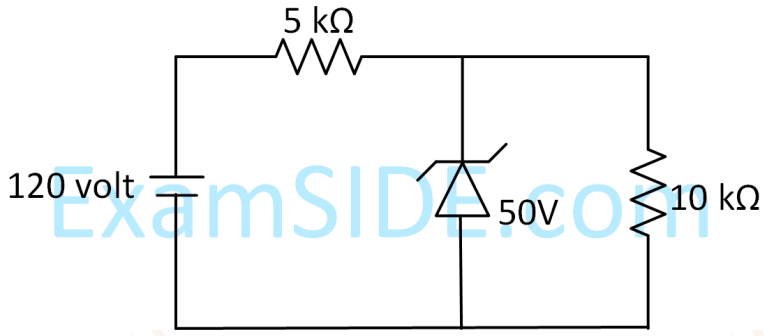


(A) 0.0 mA (B) 6.7 mA (C) 4.0 mA (D) 6.0 mA

Q136. JEE Main 2019 (Online) 10th January Evening Slot

For the circuit shown below, the current through the Zener diode is -

Open solution



(A) 5 mA (B) zero (C) 14 mA (D) 9 mA

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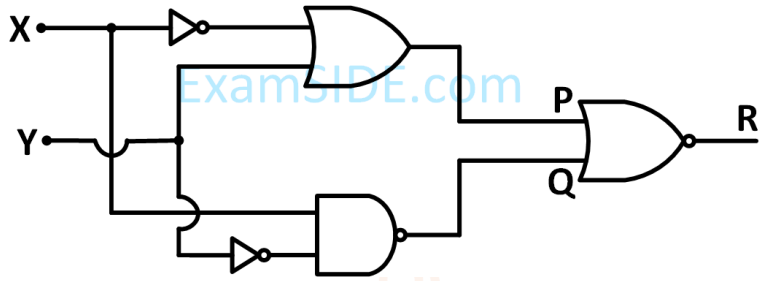
Q137. JEE Main 2019 (Online) 10th January Morning Slot

To get output 1 at R, for the given logic gate circuit the input values must be



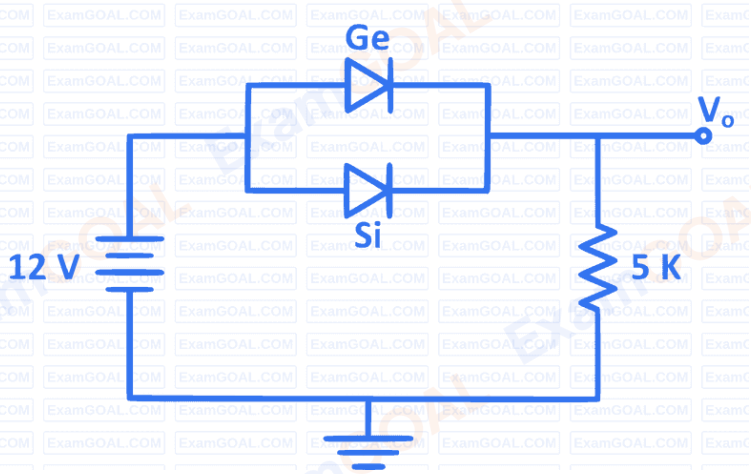
Open solution

(A) x = 0, y = 0 (B) x = 1, y = 0 (C) x = 0, y = 1 (D) x = 1, y = 1

Q138. JEE Main 2019 (Online) 9th January Evening Slot

Ge and Si diodes start conducting at 0.3 V and 0.7 V respectively. In the following figure if Ge diode connection are reversed, the value of V_0 changes by : (assume that the Ge diode has large breakdown voltage)

Open solution



(A) 0.8 V (B) 0.6 V (C) 0.2 V (D) 0.4 V

Q139. JEE Main 2019 (Online) 9th January Morning Slot

Mobility of electrons in a semiconductor is defined as the ratio of their drift velocity to the applied electric field. If, for an n-type semiconductor, the density of electrons is 10^{19} m^{-3} and their mobility is 1.6 $m^2/(V.s)$ then the resistivity of the semiconductor (since it is an n-type semiconductor contribution of holes is ignored) is close to :

Open solution

(A) $2\,\Omega m$ (B) $4\,\Omega m$ (C) $0.4\ \Omega m$ (D) $0.2\ \Omega m$

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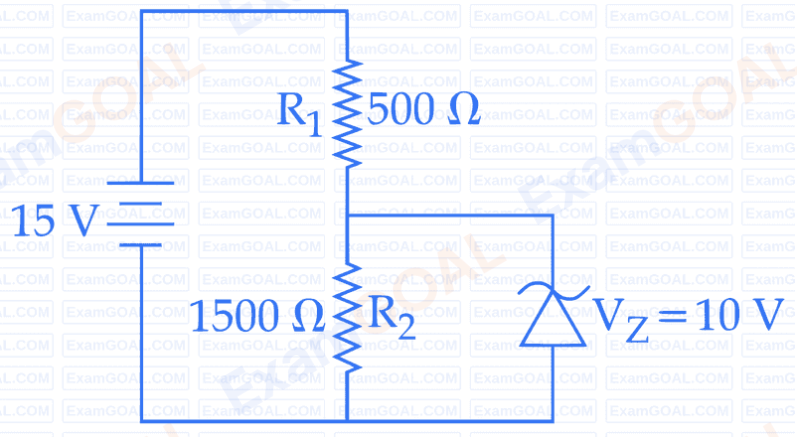
Q140. JEE Main 2018 (Online) 16th April Morning Slot

In the given circuit, the current through zener diode is :

Open solution

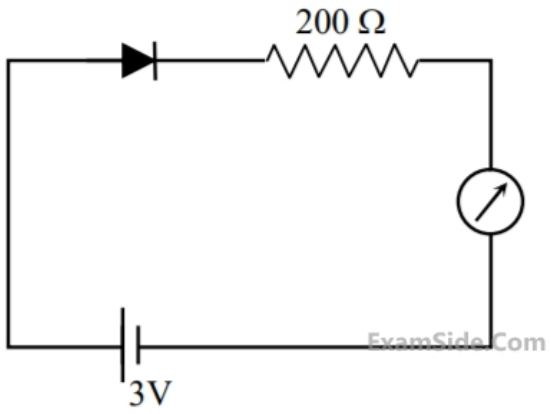


(A) 5.5 mA (B) 6.7 mA (C) 2.5 mA (D) 3.3 mA

Q141. JEE Main 2018 (Offline)

The reading of the ammeter for a silicon diode in the given circuit is :

Open solution



(A) 13.5 mA (B) 0 (C) 15 mA (D) 11.5 mA

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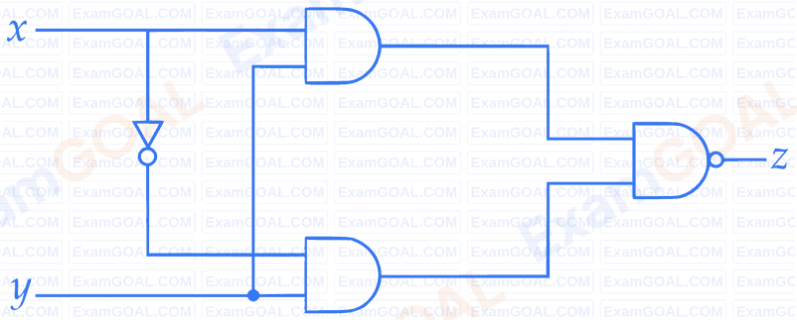
Q142. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

Truth table for the following digital circuit will be :



(A)

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

(B)

x	y	z
0	0	1
0	1	1
1	0	1
1	1	0

(C)

x	y	z
0	0	1
0	1	1
1	0	1
1	1	1

(D)

x	y	z
0	0	0
0	1	1
1	0	1
1	1	1

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Q143. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

What is the conductivity of a semiconductor sample having electron concentration of $5 \times 10^{18}\ m^{-3}$, hole concentration of $5 \times 10^{19}\ m^{-3}$, electron mobility of 2.0 $m^2\ V^{-1}\ s^{-1}$ and hole mobility of 0.01 $m^2\ V^{-1}\ s^{-1}$?

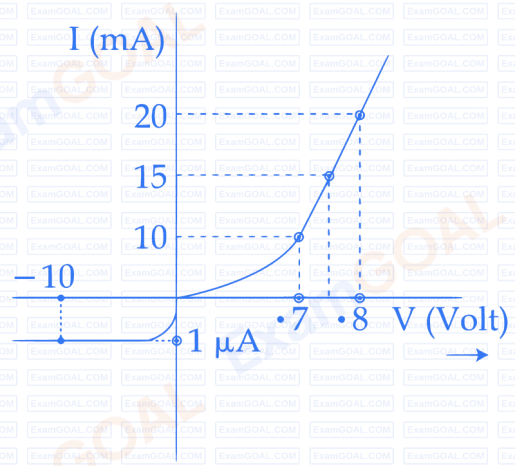
(Take charge of electron as 1.6×10^{-19} C)

(A) 1.68 $(\Omega\text{-m})^{-1}$ (B) 1.83 $(\Omega\text{-m})^{-1}$ (C) 0.59 $(\Omega\text{-m})^{-1}$ (D) 1.20 $(\Omega\text{-m})^{-1}$

Q144. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

The V-I characteristic of a diode is shown in the figure. The ratio of forward to reverse bias resistance is :

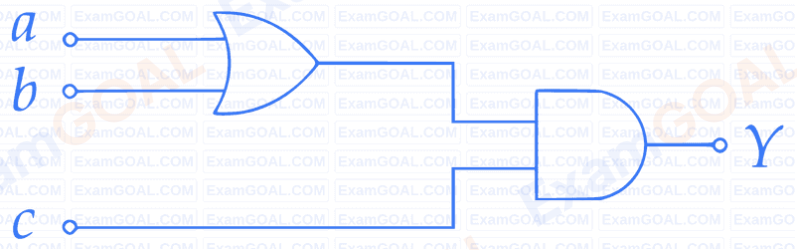


(A) 10 (B) 10^{-6} (C) 10^6 (D) 100

Q145. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

To get an output of 1 from the circuit shown in figure the input must be :



(A) a = 0, b = 1, c = 0 (B) a = 1, b = 0, c = 0

(C) a = 1, b = 0, c = 1 (D) a = 0, b = 0, c = 1

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Q146. JEE Main 2016 (Online) 9th April Morning Slot

The truth table given in fig. represents :

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Open solution

(A) AND - Gate (B) OR - Gate (C) NAND - Gate (D) NOR - Gate

Q147. JEE Main 2016 (Online) 9th April Morning Slot

An experiment is performed to determine the *I* - V characteristics of a Zener diode, which has a protective resistance of R = 100 Ω, and a maximum power of dissipation rating of 1 W. The minimum voltage range of the DC source in the circuit is :

Open solution

(A) $0 - 5$ V (B) $0 - 8$ V (C) $0 - 12$ V (D) $0 - 24$ V

Q148. JEE Main 2016 (Offline)

The temperature dependence of resistance of Cu and undoped Si in the temperature range $300 - 400\ K$, is best described by :

Open solution

(A) Linear increases for Cu, exponential decrease of Si.

(B) Linear decrease for Cu, linear decrease for Si

(C) Linear increase for Cu, linear increase for Si.

(D) Linear increase for Cu, exponential increase for Si

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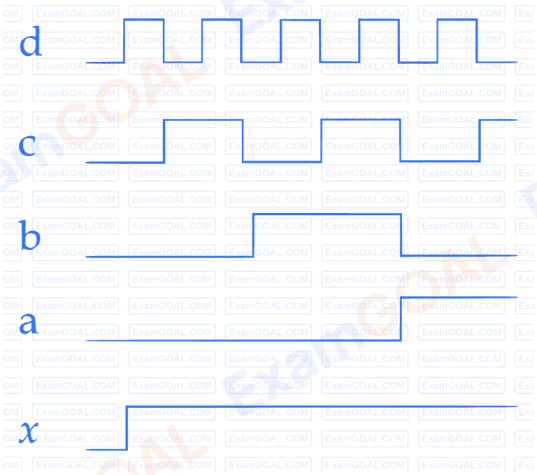
Q149. JEE Main 2016 (Offline)

Open solution

If a, b, c, d are inputs to a gate and x is its output, then, as per the following time graph, the gate is :

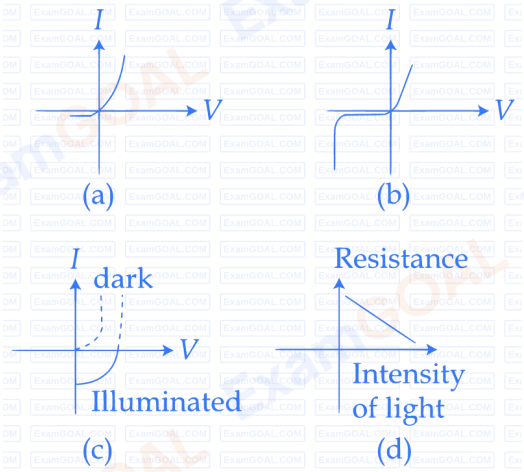


(A) OR (B) $NAND$ (C) NOT (D) AND

Q150. JEE Main 2016 (Offline)

Open solution

Identify the semiconductor devices whose characteristics are given below, in the order $(a), (b), (c), (d)$:



(A) Simple diode, Zener diode, Solar cell, Light dependent resistance

(B) Zener diode, Simple diode, Light dependent resistance, Solar cell

(C) Solar cell, Light dependent resistance, Zener diode, simple diode

(D) Zener diode, Solar cell, simple diode, Light dependent resistance

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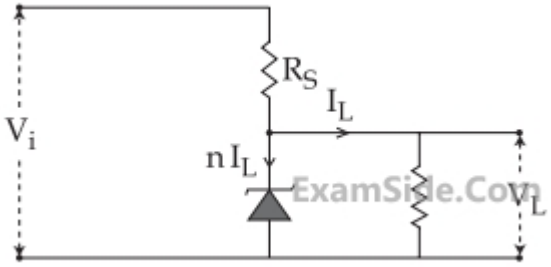
Q151. JEE Main 2015 (Online) 11th April Morning Slot

The value of the resistor, R_S, needed in the dc voltage regulator circuit shown here, equals :

Open solution



(A) $(V_i - V_L)/n\, I_L$

(B) $(V_i + V_L)/n\, I_L$

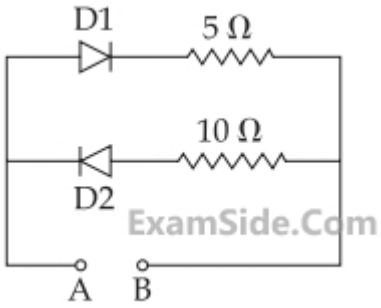
(C) $(V_i - V_L)/(n + 1)\, I_L$

(D) $(V_i + V_L)/(n + 1)\, I_L$

Q152. JEE Main 2015 (Online) 11th April Morning Slot

A 2V battery is connected across AB as shown in the figure. The value of the current supplied by the battery when in one case battery's positive terminal is connected to A and in other case when positive terminal of battery is connected to B will respectively be :

Open solution



(A) 0.2 A and 0.1 A

(B) 0.4 A and 0.2 A

(C) 0.1 A and 0.2 A

(D) 0.2 A and 0.4 A

Q153. JEE Main 2015 (Online) 10th April Morning Slot

In an unbiased n-p junction electrons diffuse from n - region to p - region because :

Open solution

(A) holes in p - region attract them

(B) electrons travel across the junction due to potential difference

(C) electron concentration in n - region is more as compared to that in p - region

(D) only electrons move from n to p region and not the vice - versa

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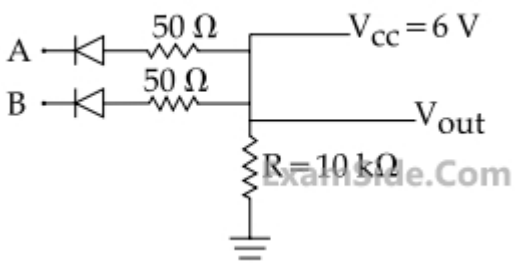
Q154. JEE Main 2014 (Online) 19th April Morning Slot



Open solution

Given : A and B are input terminals.
Logic 1 = > 5 V
Logic 0 = < 1 V
Which logic gate operation, the following circuit does ?

(A) AND Gate **(B)** OR Gate **(C)** XOR Gate **(D)** NOR Gate

Q155. JEE Main 2014 (Online) 12th April Morning Slot

For LED's to emit light in visible region of electromagnetic light, it should have energy band gap in the range of :

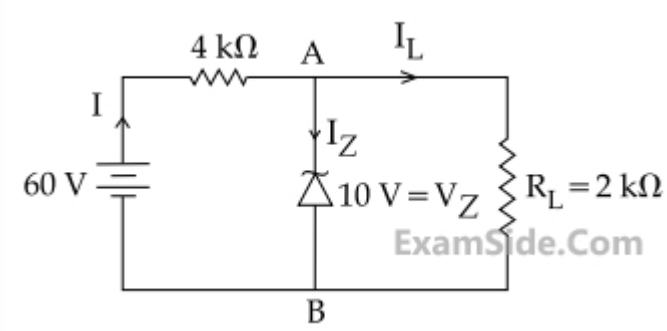
(A) 0.1 eV to 0.4 eV **(B)** 0.5 eV to 0.8 eV
(C) 0.9 eV to 1.6 eV **(D)** 1.7 eV to 3.0 eV

Open solution

Q156. JEE Main 2014 (Online) 11th April Morning Slot

A Zener diode is connected to a battery and a load as shown below :

The currents I, I_Z and I_L are respectively



Open solution

(A) 15 mA, 5 mA, 10 mA **(B)** 15 mA, 7.5 mA, 7.5 mA
(C) 12.5 mA, 5 mA, 7.5 mA **(D)** 12.5 mA, 7.5 mA, 5 mA

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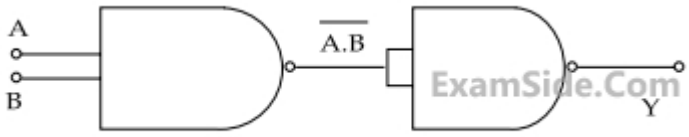
Q157. JEE Main 2014 (Online) 9th April Morning Slot

Identify the gate and match A, B, Y in bracket to check.

Open solution



(A) AND (A= 1, B = 1, Y = 1)

(B) OR (A = 1, B = 1, Y = 0)

(C) NOT (A = 1, B = 1, Y = 1)

(D) XOR (A = 0, B = 0, Y = 0)

Q158. JEE Main 2014 (Offline)

The current voltage relation of diode is given by $\mathrm{I} = \left(e^{1000V/T} - 1\right)mA$, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring $\pm 0.01\,V$ while measuring the current of $5\,mA$ at $300\,K$, what will be the error in the value of current in mA?

Open solution

(A) $0.2\,mA$

(B) $0.02\,mA$

(C) $0.5\,mA$

(D) $0.05\,mA$

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Q159. JEE Main 2014 (Offline)

The forward biased diode connection is:

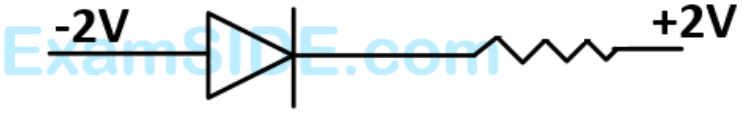
Open solution

(A)

(B)

(C)

(D)



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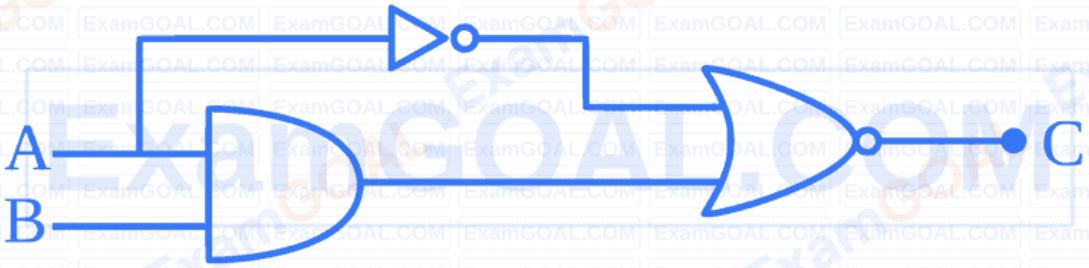
Q160. JEE Main 2013 (Online) 25th April Morning Slot

Which of the following circuits correctly represents the following truth table?

Open solution

A	B	C
0	0	0
0	1	0
1	0	1
1	1	0

(A)



(B)

(C)

(D)

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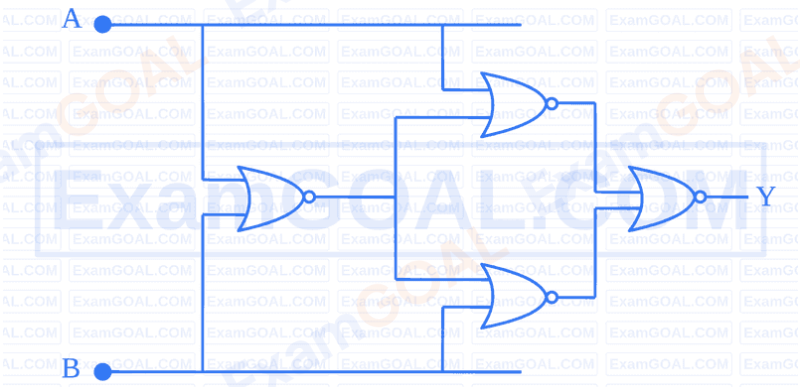
Q161. JEE Main 2013 (Online) 23th April Morning Slot

A system of four gates is set up as shown. The 'truth table' corresponding to this system is :

Open solution



(A)

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

(C)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(D)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

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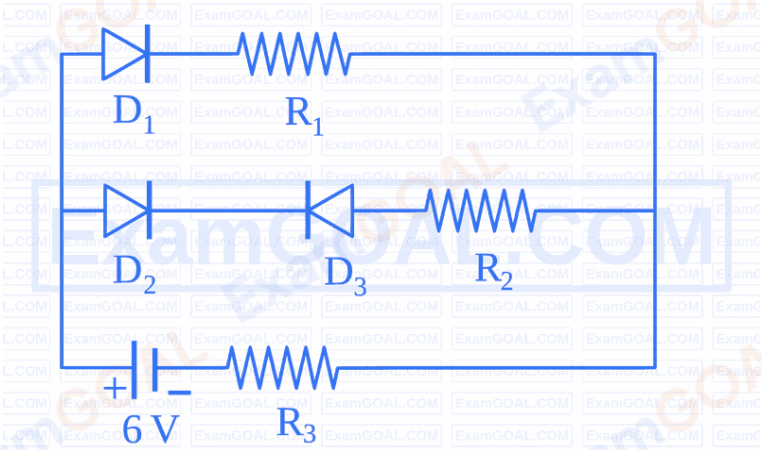
Q162. JEE Main 2013 (Online) 22th April Morning Slot

Open solution

Figure shows a circuit in which three identical diodes are used. Each diode has forward resistance of 20Ω and infinite backward resistance. Resistors $R_1 = R_2 = R_3 = 50\Omega$. Battery voltage is 6 V. The current through R_3 is :

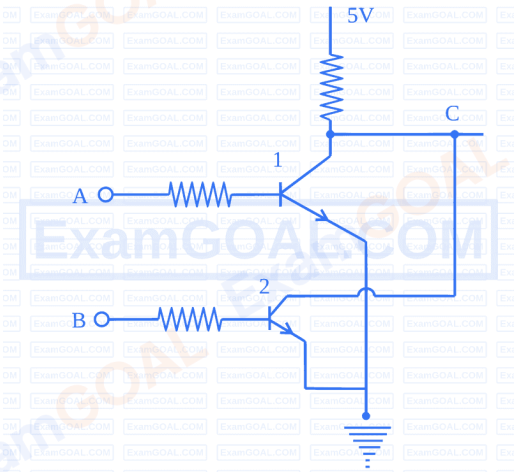


(A) 50 mA (B) 100 mA (C) 60 mA (D) 25 mA

Q163. JEE Main 2013 (Online) 9th April Morning Slot

Open solution

Consider two npn transistors as shown in the figure. If 0 Volts corresponds to false and 5 Volts correspond to true, then the output at C corresponds to :



(A) A NAND B (B) A OR B (C) A AND B (D) A NOR B

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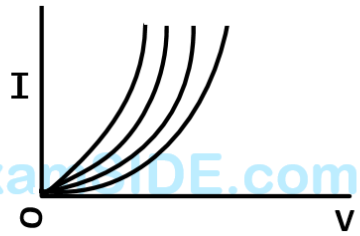
Q164. JEE Main 2013 (Offline)

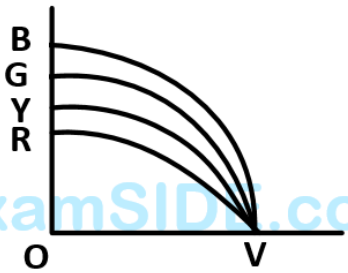
Open solution

The I-V characteristic of an LED is

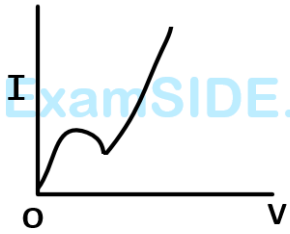
(A)



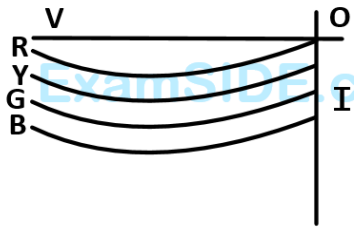
(B)



(C)



(D)



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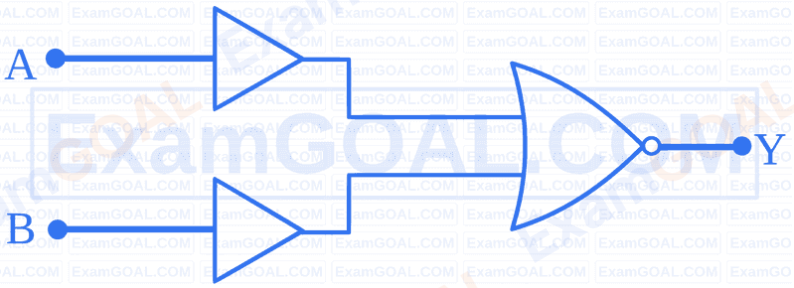
Q165. AIEEE 2012 (Online) 26th May

The figure shows a combination of two NOT gates and a NOR gate.



Open solution

The combination is equivalent to a

(A) NAND gate (B) NOR gate (C) AND gate (D) OR gate

Q166. AIEEE 2012 (Online) 19th May

Which one of the following is the Boolean expression for NOR gate?

(A) $Y = \overline{A + B}$ (B) $Y = \overline{A \cdot B}$ (C) $Y = A \cdot B$ (D) $Y = \bar{A}$

Open solution

Q167. AIEEE 2012 (Online) 12th May

This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement 1: A pure semiconductor has negative temperature coefficient of resistance.
Statement 2: On raising the temperature, more charge carriers are released into the conduction band.

(A) Statement 1 is false, Statement 2 is true.

(B) Statement 1 is true, Statement 2 is false.

(C) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.

Open solution

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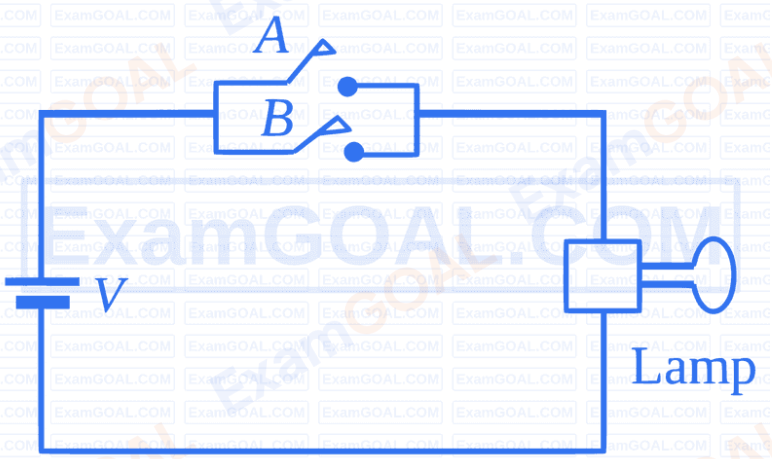
Q168. AIEEE 2012 (Online) 7th May

Open solution

Which logic gate with inputs A and B performs the same operation as that performed by the following circuit?



(A) NAND gate (B) OR gate (C) NOR gate (D) AND gate

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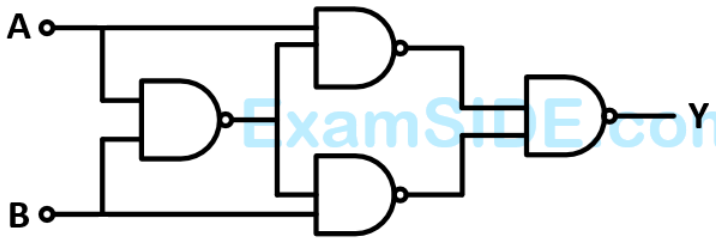
Q169. AIEEE 2012

Open solution

Truth table for system of four $NAND$ gates as shown in figure is:



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(D)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	1

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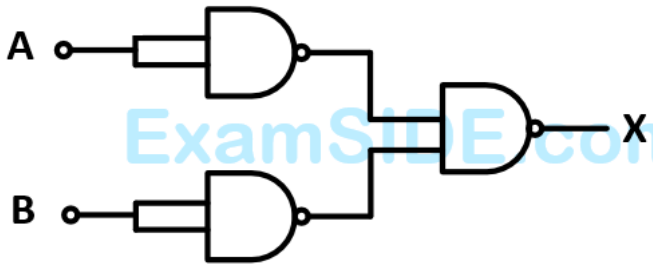
Q170. AIEEE 2010

The combination of gates shown below yields



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Open solution

(A) OR gate (B) NOT gate (C) XOR gate (D) NAND gate

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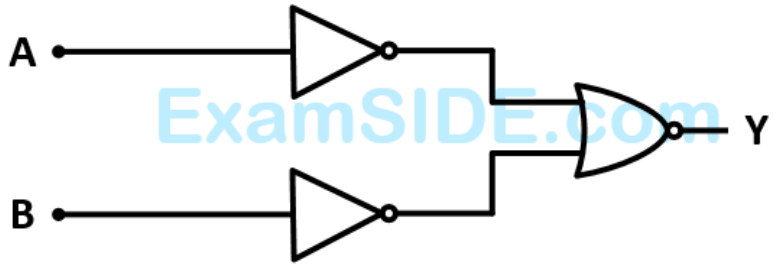
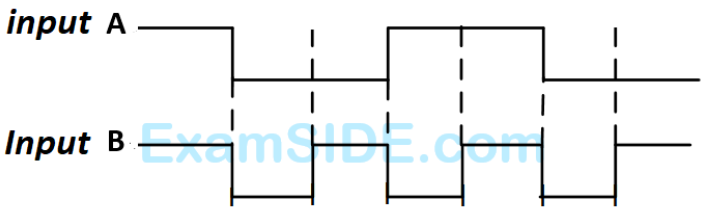
Q171. AIEEE 2009

Open solution

The logic circuit shown below has the input waveforms $'A'$ and $'B'$ as shown. Pick out the correct output waveform.





Output is

(A)

(B)

(C)

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(D)

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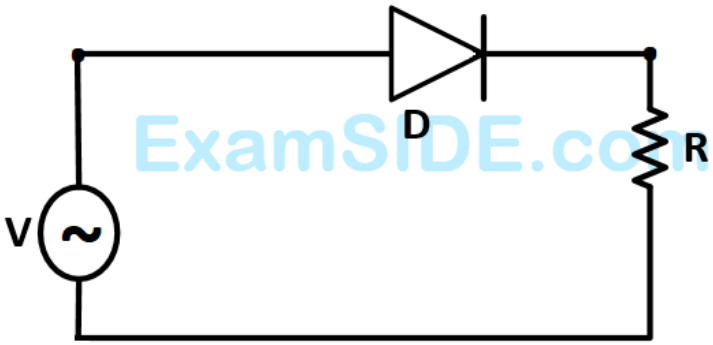
Q172. AIEEE 2009

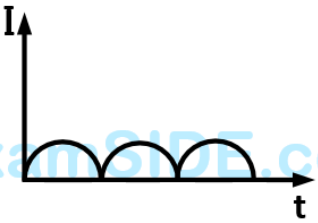
A p-n junction (D) shown in the figure can act as a rectifier. An alternating current source (V) is connected in the circuit.

Open solution

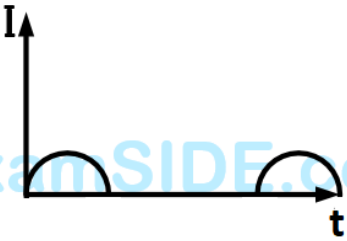


The current (I) in the resistor (R) can be shown by :

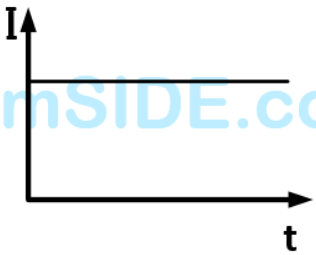
(A)



(B)



(C)



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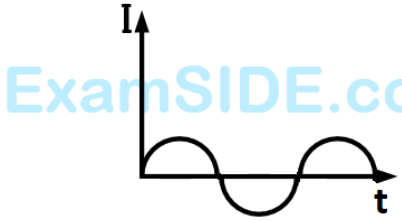
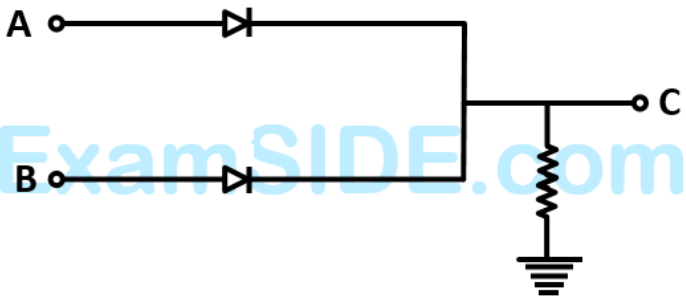
(D)



Q173. AIEEE 2008

In the circuit below, A and B represent two inputs and C represents the output.

Open solution



The circuit represents

(A) NOR gate (B) AND gate (C) NAND gate (D) OR gate

Q174. AIEEE 2007

Carbon, silicon and germanium have four valence electrons each. At room temperature which one of the following statements is most appropriate ?

Open solution

(A) The number of free electrons for conduction is significant only in Si and Ge but small in C.

(B) The number of free conduction electrons is significant in C but small in Si and Ge.

(C) The number of free conduction electrons is negligibly small in all the three.

(D) The number of free electrons for conduction is significant in all the three

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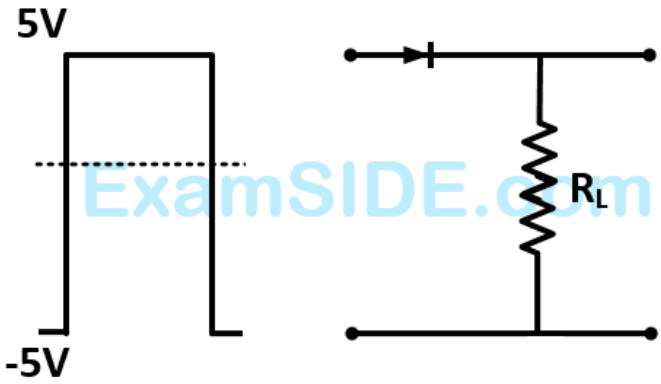
Q175. AIEEE 2007

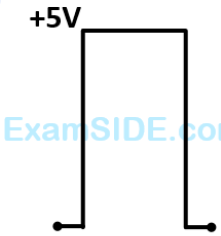
Open solution

If in a p-n junction diode, a square input signal of $10\ V$ is applied as shown

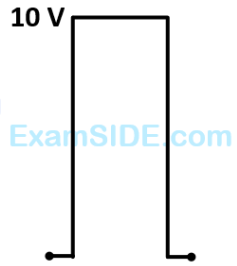


Then the output signal across R_L will be

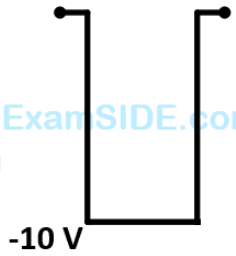
(A)



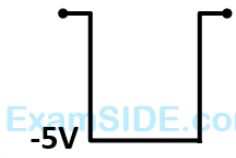
(B)



(C)



(D)



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Q176. AIEEE 2006

A solid which is not transparent to visible light and whose conductivity increases with temperature is formed by

(A) Ionic bonding
(B) Covalent bonding
(C) Vander Waals bonding
(D) Metallic bonding

Open solution

Q177. AIEEE 2006

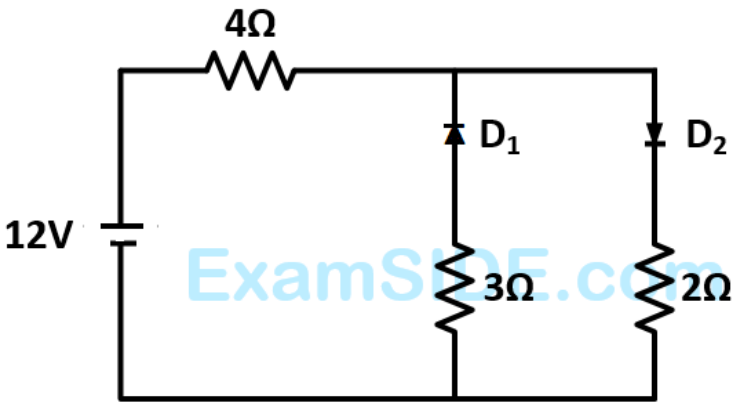
If the ratio of the concentration of electrons to that of holes in a semiconductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities?

(A) $\frac{5}{8}$ (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) $\frac{4}{7}$

Open solution

Q178. AIEEE 2006

The circuit has two oppositively connected ideal diodes in parallel. What is the current following in the circuit?



(A) $1.71\ A$ (B) $2.00\ A$ (C) $2.31\ A$ (D) $1.33\ A$

Open solution

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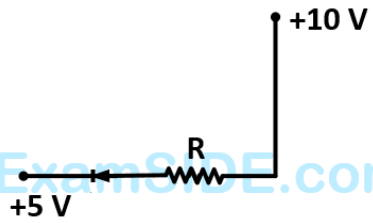
Q179. AIEEE 2006

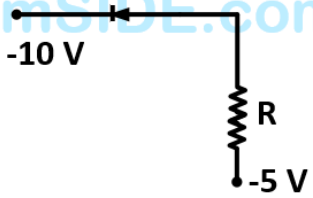
In the following, which one of the diodes is reverse biased?

Open solution

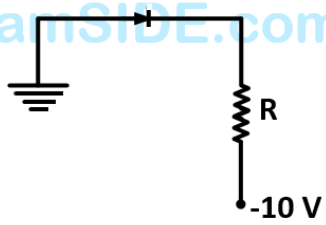
(A)



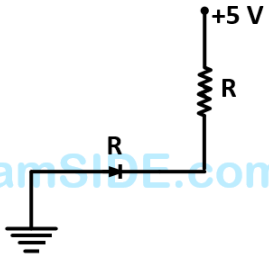
(B)



(C)



(D)



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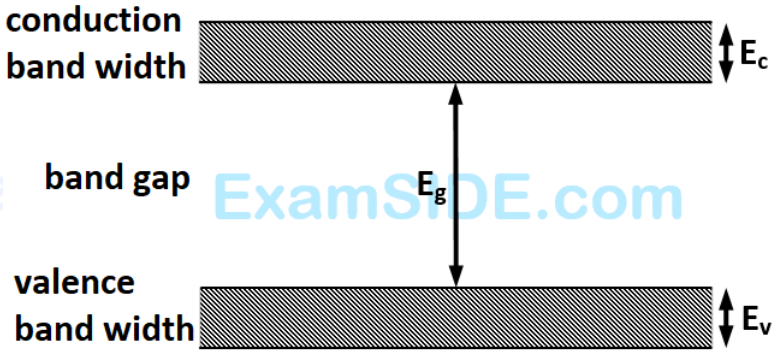
Q180. AIEEE 2006

If the lattice constant of this semiconductor is decreased, then which of the following is correct?



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Open solution

(A) All E_c, E_g, E_v increase

(B) E_c and E_v increase, but E_g decreases

(C) E_c and E_v decrease, but E_g increases

(D) All E_c, E_g, E_v decrease

Q181. AIEEE 2005

The electrical conductivity of a semiconductor increases when electromagnetic radiation of wavelength shorter than $2480\ nm$ is incident on it. The band gap in (eV) for the semiconductor is

Open solution

(A) $2.5\ eV$ (B) $1.1\ eV$ (C) $0.7\ eV$ (D) $0.5\ eV$

Q182. AIEEE 2005

In a full wave rectifier circuit operating from $50\ Hz$ mains frequency, the fundamental frequency in the ripple would be

Open solution

(A) $25\ Hz$ (B) $50\ Hz$ (C) $70.7\ Hz$ (D) $100\ Hz$

Q183. AIEEE 2004

A piece of copper and another of germanium are cooled from room temperature to $77K$, the resistance of

Open solution

(A) copper increases and germanium decreases

(B) each of them decreases

(C) each of them increases

(D) copper decreases and germanium increases

Q184. AIEEE 2004

The manifestation of band structure in solids is due to

Open solution

(A) Bohr's correspondence principle

(B) Pauli's exclusion principle

(C) Heisenberg's uncertainty principle

(D) Boltzmann's law

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Q185. AIEEE 2004

When p-n junction diode is forward biased then

(A) both the depletion region and barrier height are reduced

(B) the depletion region is widened and barrier height is reduced

(C) the depletion region is reduced and barrier height is increased

(D) both the depletion region and barrier height are increased

Open solution

Q186. AIEEE 2003

A strip of copper and another of germanium are cooled from room temperature to $80K$. The resistance of

(A) each of these decreases

(B) copper strip increases and that of germanium decreases

(C) copper strip decreases and that of germanium increases

(D) each of these increases

Open solution

Q187. AIEEE 2003

The difference in the variation of resistance with temperature in a metal and a semiconductor arises essentially due to the difference in the

(A) crystal structure

(B) variation of the number of charge carriers with temperature

(C) type of bonding

(D) variation of scattering mechanism with temperature

Open solution

Q188. AIEEE 2003

In the middle of the depletion layer of a reverse- biased p-n junction, the

(A) electric field is zero

(B) potential is maximum

(C) electric field is maximum

(D) potential is zero

Open solution

Q189. AIEEE 2002

At absolute zero, Si acts as

(A) non-metal

(B) metal

(C) insulator

(D) none of these

Open solution

Q190. AIEEE 2002

By increasing the temperature, the specific resistance of a conductor and a semiconductor

(A) increases for both

(B) decreases for both

(C) increases, decreases

(D) decreases, increases

Open solution

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Q191. AIEEE 2002

The energy band gap is maximum in

(A) metals (B) superconductors (C) insulator (D) semiconductor

Open solution

Numerical 27 questions

Q192. JEE Main 2026 (Online) 4th April Evening Shift

A diode has Zener voltage of 10 V and maximum power dissipation of 0.5 W , then the minimum resistance to be used in series with this diode for safety when it is connected to a 25 V power supply is ____ Ω.

Open solution

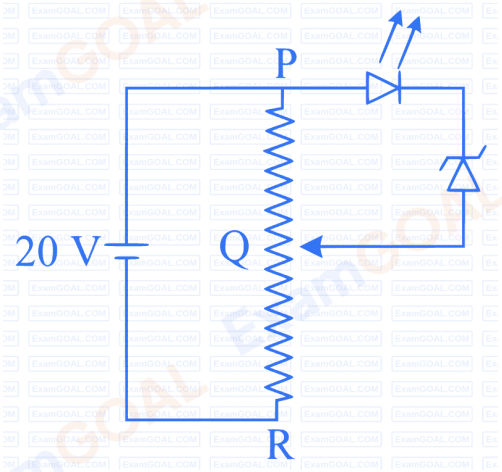
Q193. JEE Main 2026 (Online) 24th January Morning Shift

A voltage regulating circuit consisting of Zener diode, having break-down voltage of 10 V and maximum power dissipation of 0.4 W , is operated at 15 V . The approximate value of protective resistance in this circuit is ____ Ω.

Open solution

Q194. JEE Main 2024 (Online) 8th April Evening Shift

A potential divider circuit is connected with a dc source of $20\ \mathrm{V}$, a light emitting diode of glow in voltage $1.8\ \mathrm{V}$ and a zener diode of breakdown voltage of $3.2\ \mathrm{V}$. The length (PR) of the resistive wire is $20\ \mathrm{cm}$. The minimum length of PQ to just glow the LED is _____ cm.



Open solution

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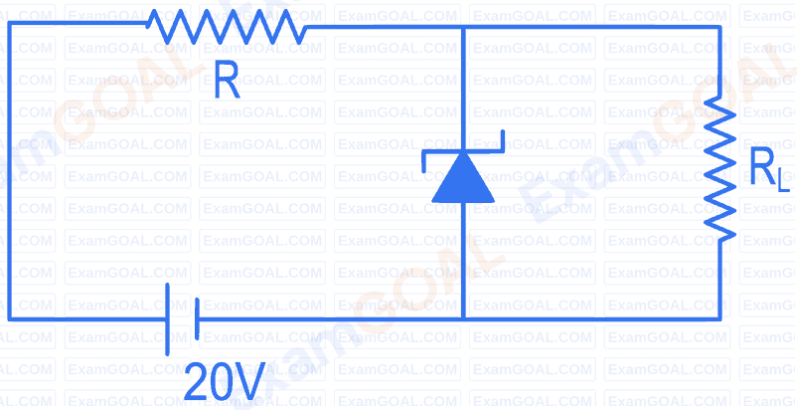
Q195. JEE Main 2022 (Online) 29th July Evening Shift

A $8\ \mathrm{V}$ Zener diode along with a series resistance R is connected across a $20\ \mathrm{V}$ supply (as shown in the figure). If the maximum Zener current is $25\ \mathrm{mA}$, then the minimum value of R will be _____ Ω.



Open solution

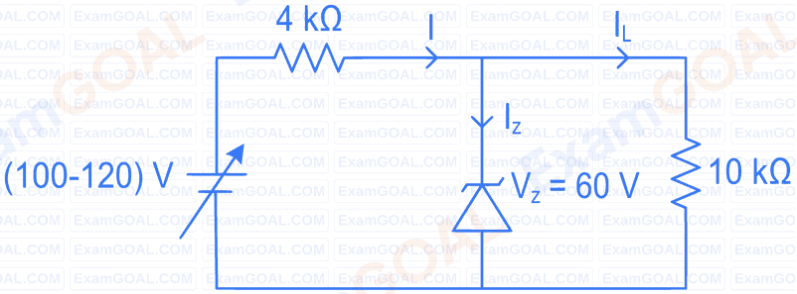
Q196. JEE Main 2022 (Online) 29th July Morning Shift

If the potential barrier across a p-n junction is $0.6\ \mathrm{V}$. Then the electric field intensity, in the depletion region having the width of $6 \times 10^{-6}\ \mathrm{m}$, will be _____ $\times 10^5\ \mathrm{N/C}$.

Open solution

Q197. JEE Main 2022 (Online) 26th July Morning Shift

In the circuit shown below, maximum zener diode current will be _____ mA.



Open solution

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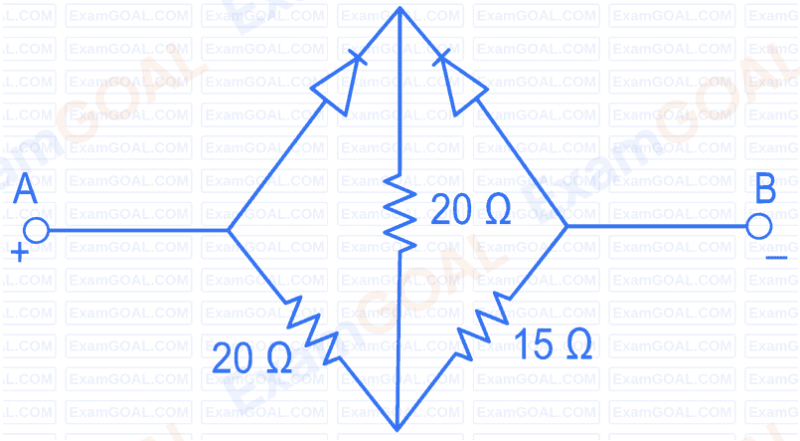
Q198. JEE Main 2022 (Online) 25th July Evening Shift

Two ideal diodes are connected in the network as shown in figure. The equivalent resistance between A and B is _____ Ω.

Open solution



Q199. JEE Main 2022 (Online) 25th July Morning Shift

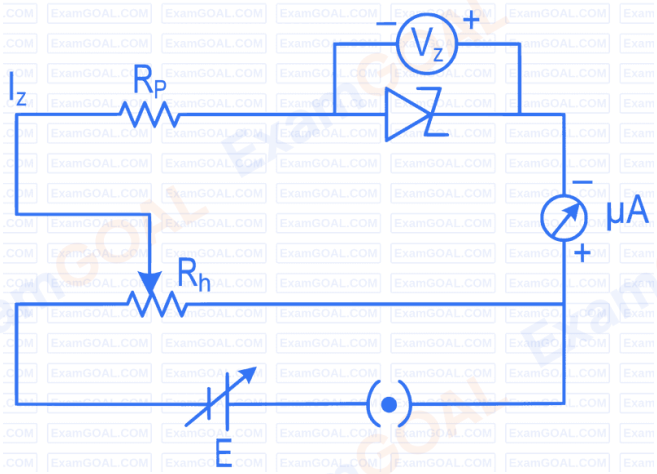
The energy band gap of semiconducting material to produce violet (wavelength = 4000$\mathring{A}$) LED is _____ eV. (Round off to the nearest integer).

Open solution

Q200. JEE Main 2022 (Online) 30th June Morning Shift

The circuit diagram used to study the characteristic curve of a zener diode is connected to variable power supply (0 − 15 V) as shown in figure. A zener diode with maximum potential V_z = 10 V and maximum power dissipation of 0.4 W is connected across a potential divider arrangement. The value of resistance R_P connected in series with the zener diode to protect it from the damage is _____ Ω.

Open solution



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Q201. JEE Main 2022 (Online) 29th June Evening Shift

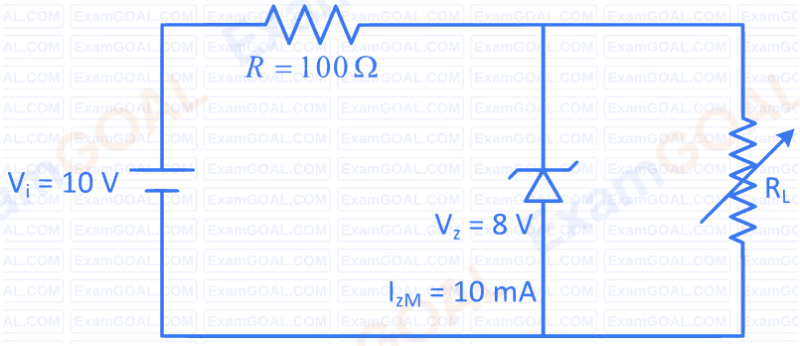
A potential barrier of 0.4 V exists across a p-n junction. An electron enters the junction from the n-side with a speed of 6.0×10^5 ms^{-1}. The speed with which electron enters the p side will be $\frac{x}{3} \times 10^5$ ms^{-1} the value of x is _____.

(Given mass of electron = 9×10^{-31} kg, charge on electron = 1.6×10^{-19} C.)

Open solution

Q202. JEE Main 2022 (Online) 28th June Evening Shift

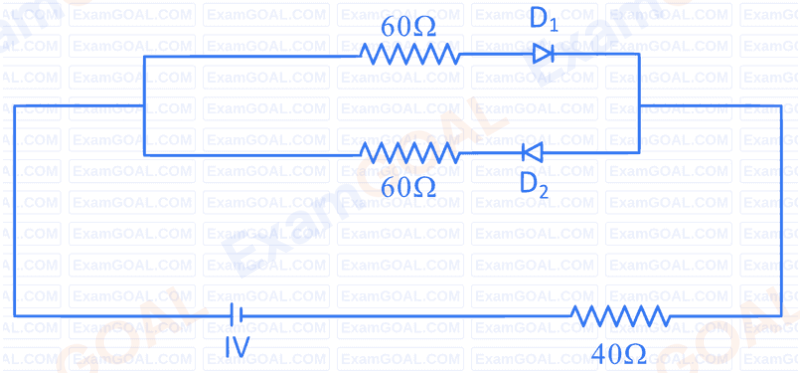
A zener of breakdown voltage V_Z = 8 V and maximum zener current, I_{ZM} = 10 mA is subjected to an input voltage V_i = 10 V with series resistance R = 100 Ω. In the given circuit R_L represents the variable load resistance. The ratio of maximum and minimum value of R_L is _____.



Open solution

Q203. JEE Main 2022 (Online) 27th June Evening Shift

The cut-off voltage of the diodes (shown in figure) in forward bias is 0.6 V. The current through the resistor of 40 Ω is _____ mA.



Open solution

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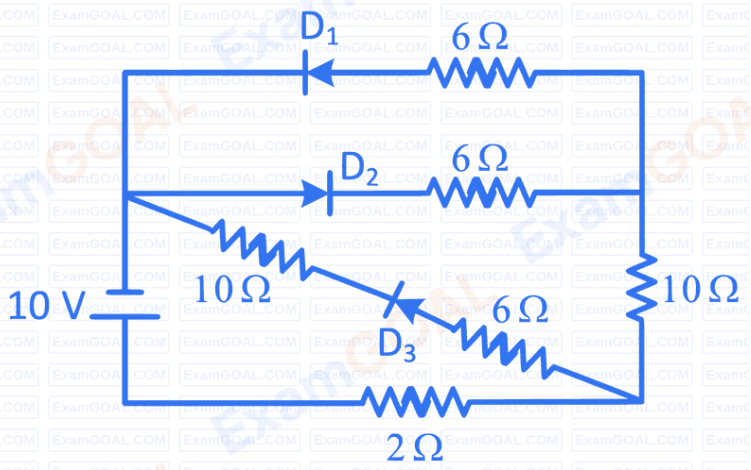
Q204. JEE Main 2022 (Online) 26th June Morning Shift

As per the given circuit, the value of current through the battery will be _____ A.

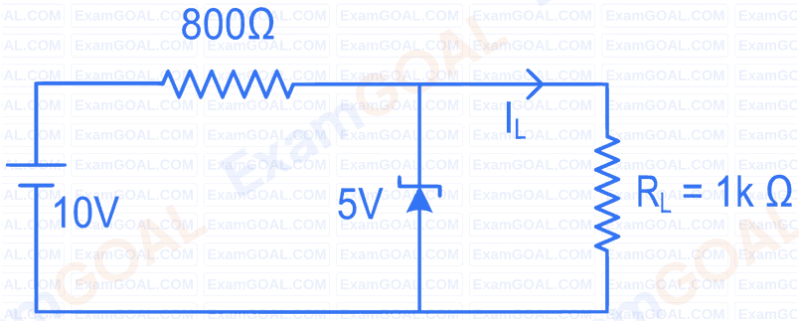
Open solution



Q205. JEE Main 2022 (Online) 24th June Evening Shift

In the given circuit, the value of current I_L will be _____ mA. (When R_L = 1kΩ)

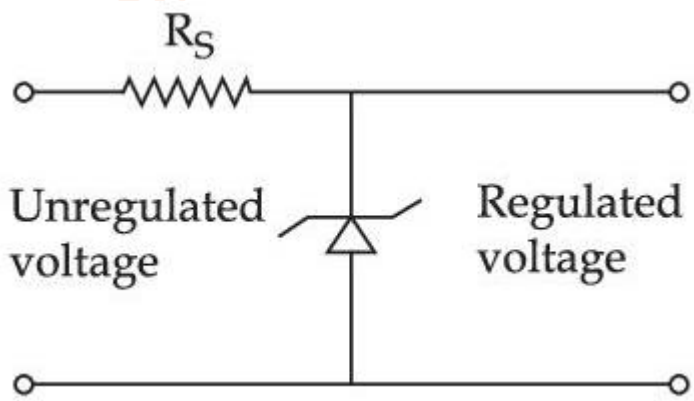
Open solution



Q206. JEE Main 2021 (Online) 27th August Evening Shift

A zener diode of power rating 2W is to be used as a voltage regulator. If the zener diode has a breakdown of 10 V and it has to regulate voltage fluctuated between 6 V and 14 V, the value of R_S for safe operation should be _____ Ω.

Open solution



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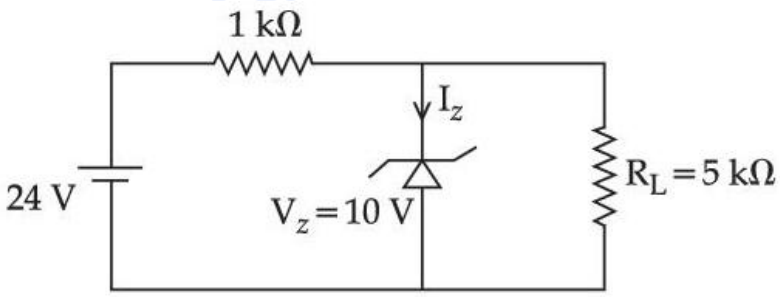
Q207. JEE Main 2021 (Online) 26th August Evening Shift

For the given circuit, the power across Zener diode is ..... mW.

Open solution



Q208. JEE Main 2021 (Online) 25th July Evening Shift

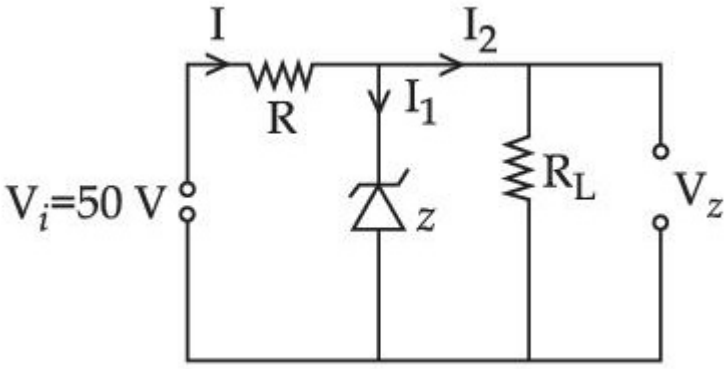
In a semiconductor, the number density of intrinsic charge carries at 27°C is $1.5 \times 10^{16}/m^3$. If the semiconductor is doped with impurity atom, the hole density increases to $4.5 \times 10^{22}/m^3$. The electron density in the doped semiconductor is _____ $\times 10^9/m^3$.

Open solution

Q209. JEE Main 2021 (Online) 22th July Evening Shift

In a given circuit diagram, a 5 V zener diode along with a series resistance is connected across a 50 V power supply. The minimum value of the resistance required, if the maximum zener current is 90 mA will be _____ Ω.

Open solution



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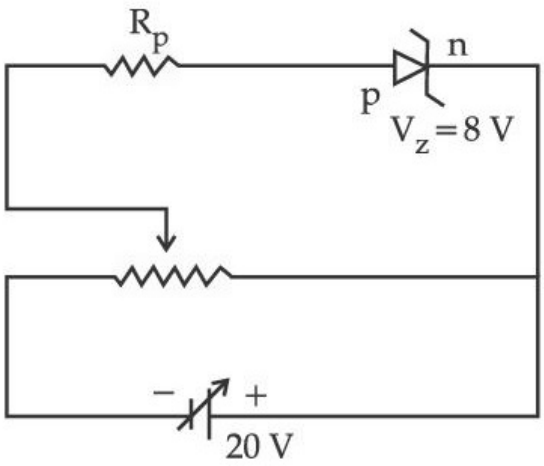
Q210. JEE Main 2021 (Online) 20th July Evening Shift

A zener diode having zener voltage 8 V and power dissipation rating of 0.5 W is connected across a potential divider arranged with maximum potential drop across zener diode is as shown in the diagram. The value of protective resistance R_p is ..... Ω.

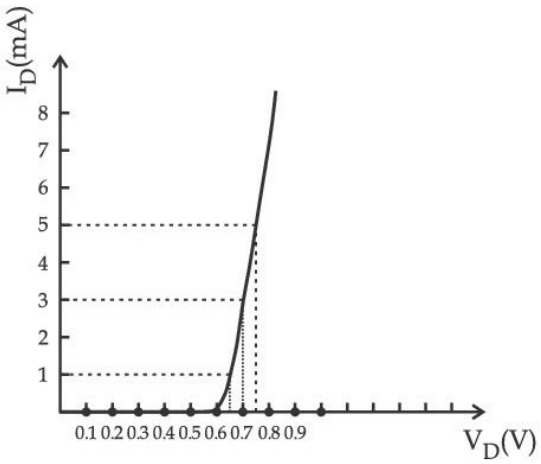
Open solution



Q211. JEE Main 2021 (Online) 20th July Evening Shift

For the forward biased diode characteristics shown in the figure, the dynamic resistance at I_D = 3 mA will be _____ Ω.

Open solution



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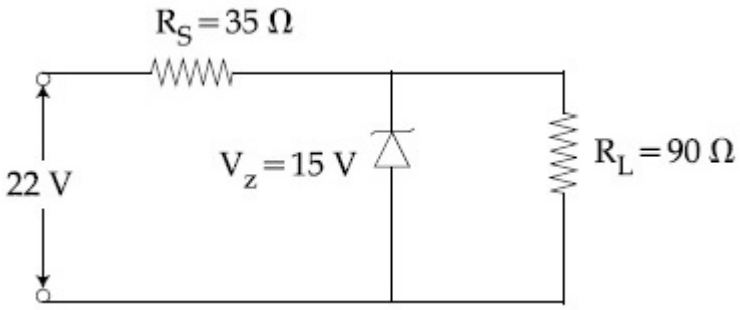
Q212. JEE Main 2021 (Online) 16th March Morning Shift

The value of power dissipated across the zener diode (V_z = 15V) connected in the circuit as shown in the figure is x $\times$ 10^{-1} watt.

Open solution



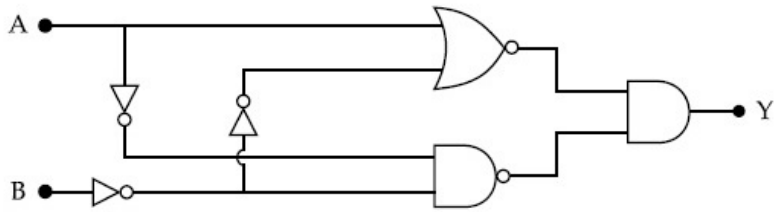
The value of x, to the nearest integer, is _____.

Q213. JEE Main 2021 (Online) 16th March Morning Shift

In the logic circuit shown in the figure, if input A and B are 0 and 1 respectively, the output at Y would be 'x'.

Open solution

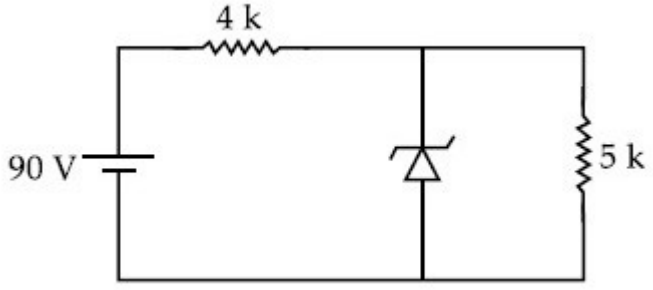
The value of x is _____.



Q214. JEE Main 2021 (Online) 26th February Evening Shift

The zener diode has a V_z = 30V. The current passing through the diode for the following circuit is _____ mA.

Open solution



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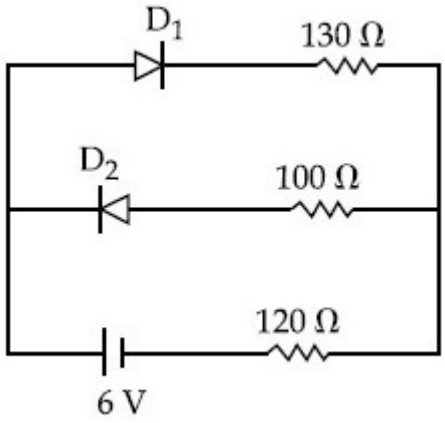
Q215. JEE Main 2021 (Online) 26th February Morning Shift

The circuit contains two diodes each with a forward resistance of 50Ω and with infinite reverse resistance. If the battery voltage is 6V, the current through the 120Ω resistance is _____ mA.

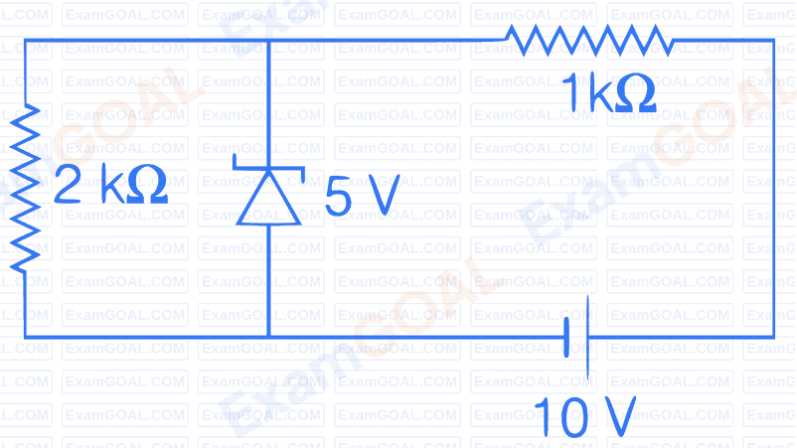
Open solution



Q216. JEE Main 2021 (Online) 24th February Morning Shift

In connection with the circuit drawn below, the value of current flowing through 2 kΩ resistor is _____ $\times 10^{-4}$ A.

Open solution



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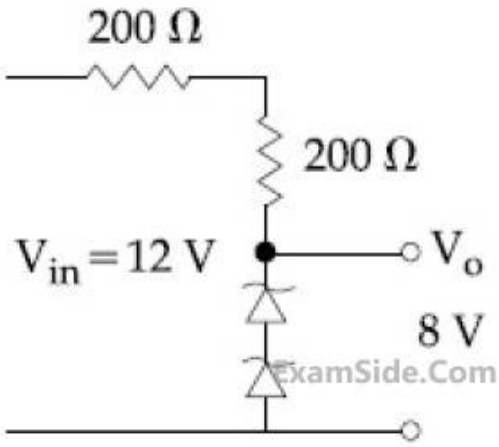
Q217. JEE Main 2020 (Online) 9th January Evening Slot

The circuit shown below is working as a 8 V dc regulated voltage source. When 12 V is used as input, the power dissipated (in mW) in each diode is; (considering both zener diodes are identical) _____.

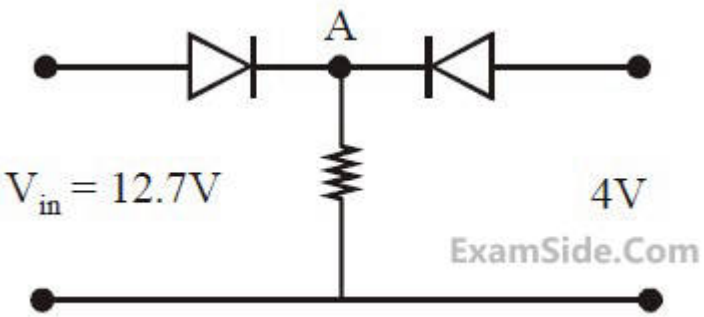
Open solution



Q218. JEE Main 2020 (Online) 9th January Morning Slot

Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in Volts) at point A is _____.

Open solution



Out of syllabus 44 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q219. JEE Main 2023 (Online) 12th April Morning Shift · **MCQ (Single Correct Answer)**

In an n-p-n common emitter (CE) transistor the collector current changes from 5 mA to 16 mA for the change in base current from 100 $\mu\mathrm{A}$ and 200 $\mu\mathrm{A}$, respectively. The current gain of transistor is _____.

Open solution

(A) 210 (B) 0.9 (C) 9 (D) 110

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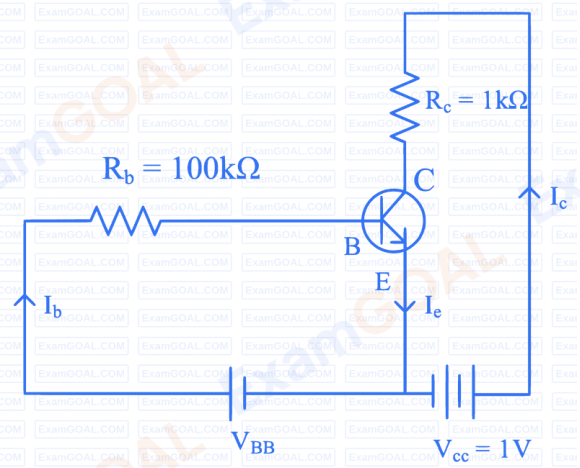
Q220. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Open solution

For a given transistor amplifier circuit in CE configuration $V_{CC} = 1\ V, R_C = 1\ k\Omega, R_b = 100\ k\Omega$ and $\beta = 100$. Value of base current I_b is



(A) $I_b = 100\ \mu A$

(B) $I_b = 0.1\ \mu A$

(C) $I_b = 1.0\ \mu A$

(D) $I_b = 10\ \mu A$

Q221. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I: In a typical transistor, all three regions emitter, base and collector have same doping level.

Statement II: In a transistor, collector is the thickest and base is the thinnest segment.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) **Statement I** is correct but **Statement II** is incorrect

(B) Both **Statement I** and **Statement II** are incorrect

(C) **Statement I** is incorrect but **Statement II** is correct

(D) Both **Statement I** and **Statement II** are correct

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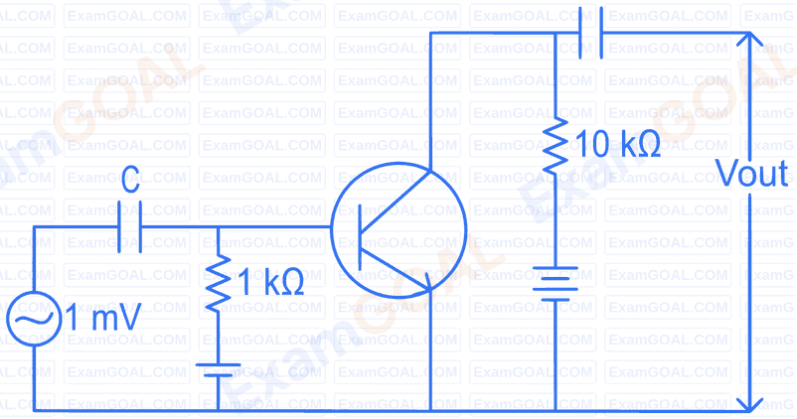
Q222. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

An n.p.n transistor with current gain $\beta = 100$ in common emitter configuration is shown in figure. The output voltage of the amplifier will be



Open solution

(A) 0.1 V (B) 1.0 V (C) 10 V (D) 100 V

Q223. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

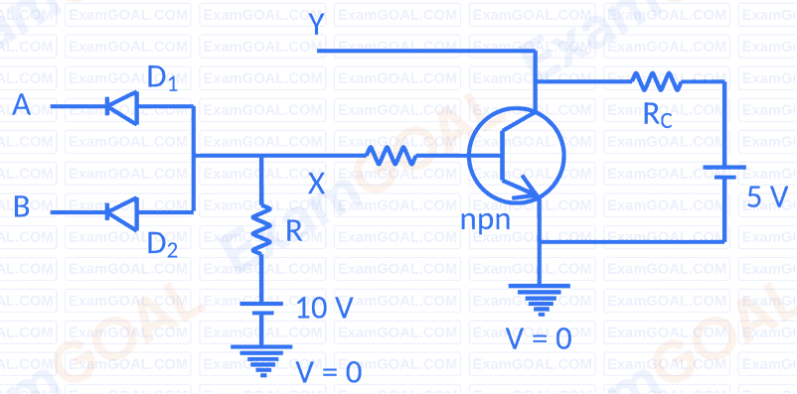
For a constant collector-emitter voltage of $8\ \text{V}$, the collector current of a transistor reached to the value of $6\ \text{mA}$ from $4\ \text{mA}$, whereas base current changed from $20\ \mu\text{A}$ to $25\ \mu\text{A}$ value. If transistor is in active state, small signal current gain (current amplification factor) will be :

Open solution

(A) 240 (B) 400 (C) 0.0025 (D) 200

Q224. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

In the following circuit, the correct relation between output (Y) and inputs A and B will be :



Open solution

(A) Y = AB (B) Y = A + B (C) $Y = \overline{AB}$ (D) $Y = \overline{A + B}$

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Q225. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : n-p-n transistor permits more current than a p-n-p transistor.

Reason R : Electrons have greater mobility as a charge carrier.

Choose the correct answer from the options given below :

(A) Both A and R are true, and R is correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q226. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Open solution

For a transistor to act as a switch, it must be operated in

(A) Active region.

(B) Saturation state only.

(C) Cut-off state only.

(D) Saturation and cut-off state.

Q227. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

Open solution

The positive feedback is required by an amplifier to act an oscillator. The feedback here means :

(A) External input is necessary to sustain ac signal in output.

(B) A portion of the output power is returned back to the input.

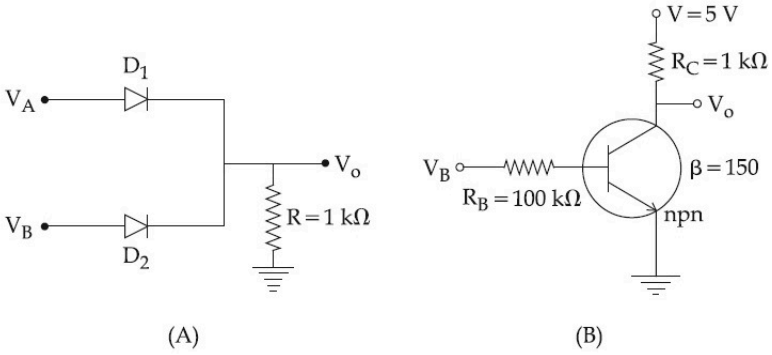
(C) Feedback can be achieved by LR network.

(D) The base-collector junction must be forward biased.

Q228. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

Open solution

If V_A and V_B are the input voltages (either 5V or 0V) and V_0 is the output voltage then the two gates represented in the following circuit (A) and (B) are :-



(A) AND and OR Gate

(B) OR and NOT Gate

(C) NAND and NOR Gate

(D) AND and NOT Gate

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Q229. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

For a transistor α and β are given as $\alpha = \frac{I_C}{I_E}$ and $\beta = \frac{I_C}{I_B}$. Then the correct relation between α and β will be :

(A) $\alpha = \frac{1-\beta}{\beta}$

(B) $\beta = \frac{\alpha}{1-\alpha}$

(C) $\alpha\beta = 1$

(D) $\alpha = \frac{\beta}{1-\beta}$

Open solution

Q230. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

For a transistor in CE mode to be used as an amplifier, it must be operated in :

(A) Both cut-off and Saturation

(B) Saturation region only

(C) Cut-off region only

(D) The active region only

Open solution

Q231. JEE Main 2021 (Online) 18th March Evening Shift · MCQ (Single Correct Answer)

The correct relation between α (ratio of collector current to emitter current) and β (ratio of collector current to base current) of a transistor is :

(A) $\beta = \frac{\alpha}{1+\alpha}$

(B) $\alpha = \frac{\beta}{1-\alpha}$

(C) $\alpha = \frac{\beta}{1+\beta}$

(D) $\beta = \frac{1}{1-\alpha}$

Open solution

Q232. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

Given below are two statements :

Statement I : PN junction diodes can be used to function as transistor, simply by connecting two diodes, back to back, which acts as the base terminal.

Statement II : In the study of transistor, the amplification factor β indicates ratio of the collector current to the base current.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Statement I is true but Statement II is false

Open solution

Q233. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

If an emitter current is changed by 4 mA, the collector current changes by 3.5 mA. The value of β will be :

(A) 0.875 (B) 0.5 (C) 3.5 (D) 7

Open solution

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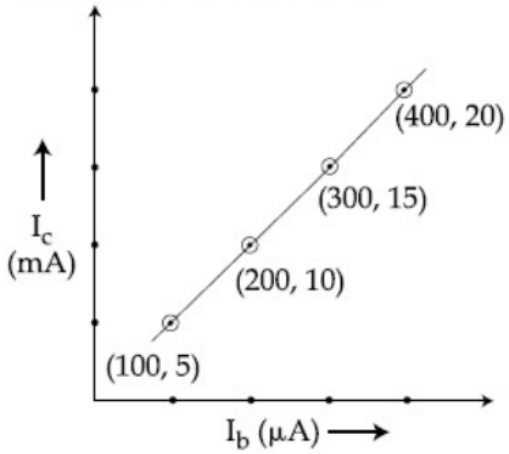
Q234. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

The transfer characteristic curve of a transistor, having input and output resistance 100 Ω and 100 kΩ respectively is shown in the figure. The voltage and power gain, are respectively:



Open solution

(A) $5 \times 10^4, 5 \times 10^5$

(B) $5 \times 10^4, 5 \times 10^6$

(C) $5 \times 10^4, 2.5 \times 10^6$

(D) $2.5 \times 10^4, 2.5 \times 10^6$

Q235. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

An n-p-n transistor operates as a common emitter amplifier, with a power gain of 60 dB. The input circuit resistance is 100Ω and the output load resistance is 10 kΩ. The common emitter current gain β is :

Open solution

(A) 10^4

(B) 10^2

(C) 6×10^2

(D) 60

Q236. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

An NPN transistor is used in common emitter configuration as an amplifier with 1 kΩ load resistance. Signal voltage of 10 mV is applied across the base-emitter. This produces a 3 mA change in the collector current and 15μA change in the base current of the amplifier. The input resistance and voltage gain are :

Open solution

(A) 0.67 kΩ, 200

(B) 0.33 kΩ, 1.5

(C) 0.67 kΩ, 300

(D) 0.33 kΩ, 300

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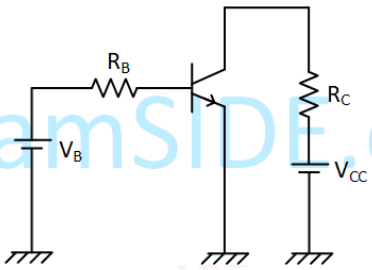
Q237. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

A common emitter amplifier circuit, built using an npn transistor, is shown in the figure. Its dc current gain is 250, R_C = 1kΩ and V_{CC} = 10 V. What is the minimum base current for V_{CE} to reach saturation ?

Open solution

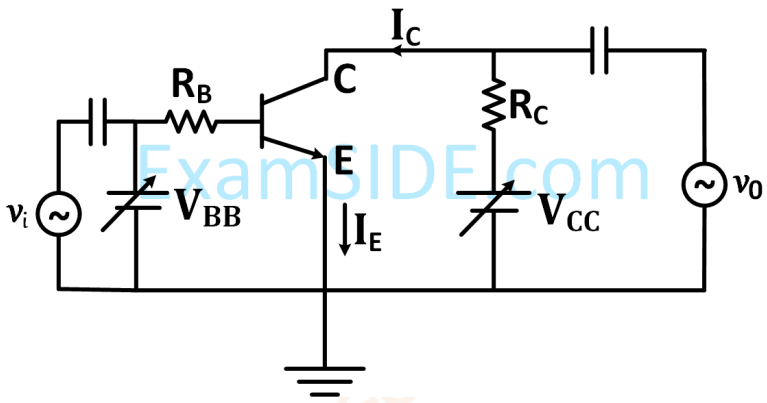


(A) 100 μA (B) 7 μA (C) 10 μA (D) 40 μA

Q238. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

In the figure, given that V_{BB} supply can vary from 0 to 5.0 V, V_{CC} = 5V, β_{dc} = 200, R_B = 100 kΩ, R_C = 1 kΩ and V_{BE} = 1.0 V. The minimum base current and the input voltage at which the transistor will go to saturation, will be respectively :

Open solution



(A) 20 μA and 2.8 V (B) 25 μA and 2.8 V (C) 20 μA and 3.5 V (D) 25 μA and 3.5 V

Q239. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

In a common emitter configuration with suitable bias, it is given that R_L is the load resistance and R_{BE} is small signal dynamic resistance (input side). Then, voltage gain, current gain and power gain are given, respectively, by :

Open solution

β is current gain, $\mathrm{I}_B, \mathrm{I}_C$ and I_E are respectively base, collector and emitter currents.

(A) $\beta\frac{R_L}{R_{BE}}, \frac{\Delta \mathrm{I}_C}{\Delta \mathrm{I}_B}, \beta^2\frac{R_L}{R_{BE}}$

(B) $\beta\frac{R_L}{R_{BE}}, \frac{\Delta \mathrm{I}_E}{\Delta \mathrm{I}_B}, \beta^2\frac{R_L}{R_{BE}}$

(C) $\beta^2\frac{R_L}{R_{BE}}, \frac{\Delta \mathrm{I}_C}{\Delta \mathrm{I}_E}, \beta^2\frac{R_L}{R_{BE}}$

(D) $\beta^2\frac{R_L}{R_{BE}}, \frac{\Delta \mathrm{I}_C}{\Delta \mathrm{I}_B}, \beta\frac{R_L}{R_{BE}}$

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Q240. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The current gain of a common emitter amplifier is 69. If the emitter current is 7.0 mA, collector current is :

(A) 9.6 mA (B) 6.9 mA (C) 0.69 mA (D) 69 mA

Open solution

Q241. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

In a common emitter amplifier circuit using an n-p-n transistor, the phase difference between the input and the output voltages will be:

(A) 180° (B) 45° (C) 90° (D) 135°

Open solution

Q242. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The ratio (R) of output resistance r_0, and the input resistance ri in measurements of input and output characteristics of a transistor is typically in the range:

(A) $R \sim 10^2 - 10^3$ (B) $R \sim 1 - 10$ (C) $R \sim 0.1 - 0.01$ (D) $R \sim 0.1 - 1.0$

Open solution

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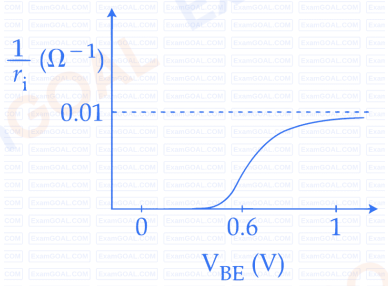
Q243. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

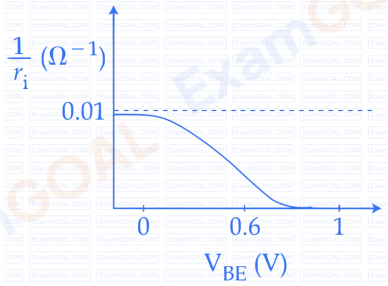
A realistic graph depicting the variation of the reciprocal of input resistance in an input characteristics measurement in a common emitter transistor configuration is :

Open solution

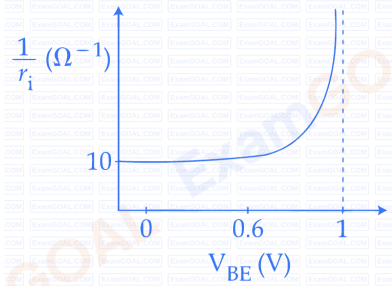
(A)



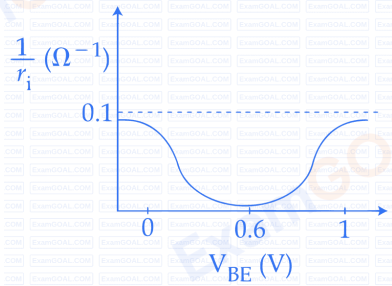
(B)



(C)



(D)



Q244. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

An unknown transistor needs to be identified as a *npn* or *pnp* type. A multimeter, with + ve and − ve terminals, is used to measure resistance between different terminals of transistor. If terminal 2 is the base of the transistor then which of the following is correct for a pnp transistor ?

Open solution

(A) + ve terminal 1, −ve terminal 2, resistance high

(B) + ve terminal 2, −ve terminal 1, resistance high

(C) + ve terminal 3, −ve terminal 2, resistance high

(D) + ve terminal 2, −ve terminal 3, resistance low

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Q245. JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

An n-p-n transistor has three leads A, B and C. Connecting B and C by moist fingers, A to the positive lead of an ammeter, and C to the negative lead of the ammeter, one finds large deflection. Then, A, B and C refer respectively to :

Open solution

(A) Emitter, base and collector
(B) Base, emitter and collector
(C) Base, collector and emitter
(D) Collector, emitter and base.

Q246. AIEEE 2008 · MCQ (Single Correct Answer)

A working transistor with its three legs marked P, Q and R is tested using a multi-meter. No conduction is found between P and Q. By connecting the common (negative) terminal of the multi-meter to R and the other (positive) terminal to P or Q, some resistance is seen on the multi-meter. Which of the following is true for the transistor?

Open solution

(A) It is an npn transistor with R as base
(B) It is an pnp transistor with R as collector
(C) It is an pnp transistor with R as emitter
(D) It is an npn transistor with R as collector

Q247. AIEEE 2006 · MCQ (Single Correct Answer)

In a common base mode of a transistor, the collector current is $5.488\ mA$ for an emitter current of $5.60mA$. The value of the base current amplification factor (β) will be

Open solution

(A) 49 (B) 50 (C) 51 (D) 48

Q248. AIEEE 2005 · MCQ (Single Correct Answer)

In a common base amplifier, the phase difference between the input signal voltage and output voltage is

Open solution

(A) π (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) 0

Q249. AIEEE 2004 · MCQ (Single Correct Answer)

When npn transistor is used as an amplifer

Open solution

(A) electrons move from collector to base
(B) holes move from emitter to base
(C) electrons move from base to collector
(D) holes move from base to emitter

Q250. AIEEE 2004 · MCQ (Single Correct Answer)

For a transistor amplifier in common emitter configuration for load impedance of $1k\,\Omega$ ($h_{fe} = 50$ and $h_{oe} = 25\,\mu A/V$) the current gain is

Open solution

(A) -24.8 (B) -15.7 (C) -5.2 (D) -48.78

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Q251. AIEEE 2002 · MCQ (Single Correct Answer)

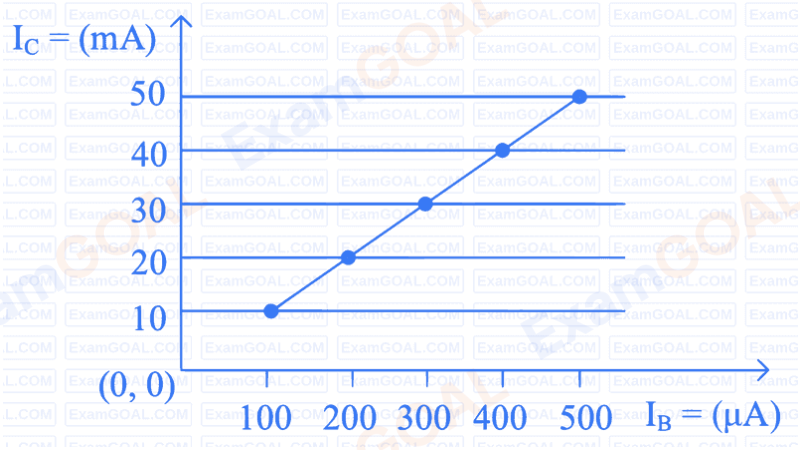
The part of a transistor which is most heavily doped to produce large number of majority carriers is

(A) emmiter
(B) base
(C) collector
(D) can be any of the above three

Open solution

Q252. JEE Main 2023 (Online) 13th April Morning Shift · Numerical

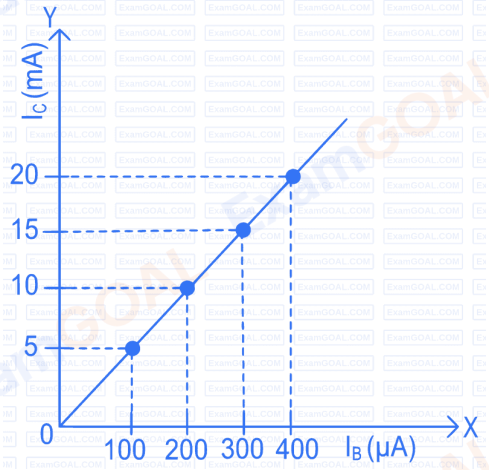
From the given transfer characteristic of a transistor in CE configuration, the value of power gain of this configuration is 10^x, for $R_B = 10\,k\Omega$, and $R_C = 1\,k\Omega$. The value of x is _____.



Open solution

Q253. JEE Main 2022 (Online) 26th July Evening Shift · Numerical

The typical transfer characteristics of a transistor in CE configuration is shown in figure. A load resistor of $2\,k\Omega$ is connected in the collector branch of the circuit used. The input resistance of the transistor is $0.50\,k\Omega$. The voltage gain of the transistor is _____.



Open solution

Q254. JEE Main 2022 (Online) 29th June Morning Shift · Numerical

A transistor is used in an amplifier circuit in common emitter mode. If the base current changes by 100 μA, it brings a change of 10 mA in collector current. If the load resistance is 2 kΩ and input resistance is 1 kΩ, the value of power gain is x $\times$ 10^4. The value of x is _____.

Open solution

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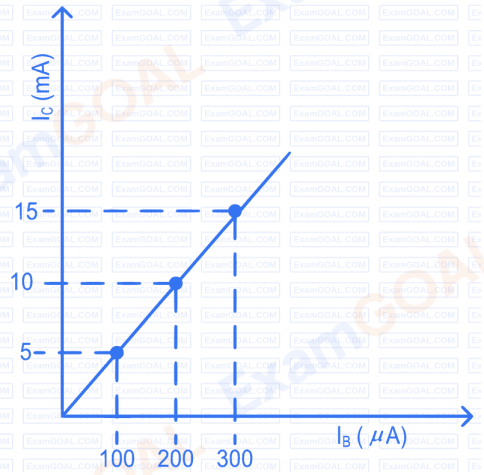
Q255. JEE Main 2022 (Online) 25th June Evening Shift · **Numerical**

In an experiment of CE configuration of n-p-n transistor, the transfer characteristics are observed as given in figure.



If the input resistance is 200 Ω and output resistance is 60 Ω, the voltage gain in this experiment will be _____.

Open solution

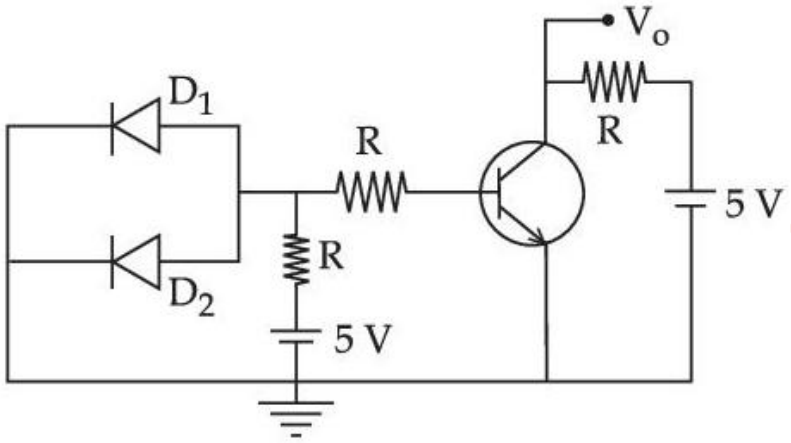
Q256. JEE Main 2022 (Online) 24th June Morning Shift · **Numerical**

A transistor is used in common-emitter mode in an amplifier circuit. When a signal of 10 mV is added to the base-emitter voltage, the base current changes by 10 μA and the collector current changes by 1.5 mA. The load resistance is 5 kΩ. The voltage gain of the transistor will be _____.

Open solution

Q257. JEE Main 2021 (Online) 27th August Morning Shift · **Numerical**

A circuit is arranged as shown in figure. The output voltage V_0 is equal to ..... V.



Open solution

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Q258. JEE Main 2021 (Online) 27th July Morning Shift · **Numerical**

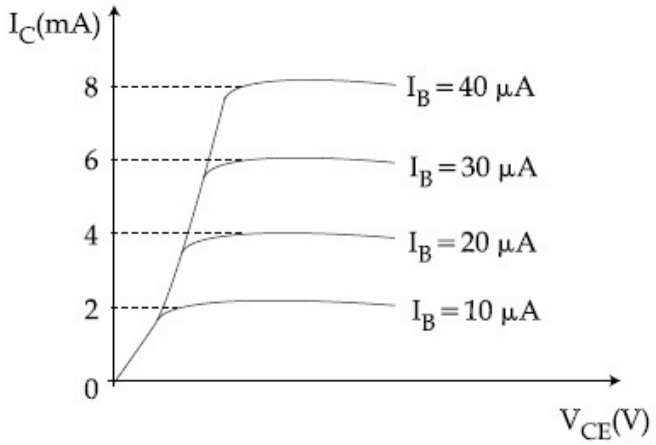
A transistor is connected in common emitter circuit configuration, the collector supply voltage is 10 V and the voltage drop across a resistor of 1000 Ω in the collector circuit is 0.6 V. If the current gain factor (β) is 24, then the base current is _____ μA. (Round off to the Nearest Integer)

Open solution

Q259. JEE Main 2021 (Online) 18th March Evening Shift · **Numerical**

The typical output characteristics curve for a transistor working in the common-emitter configuration is shown in the figure.

Open solution



The estimated current gain from the figure is _____.

Q260. JEE Main 2021 (Online) 18th March Morning Shift · **Numerical**

An npn transistor operates as a common emitter amplifier with a power gain of 10^6. The input circuit resistance is 100Ω and the output load resistance is 10 KΩ. The common emitter current gain 'β' will be _____. (Round off to the Nearest Integer).

Open solution

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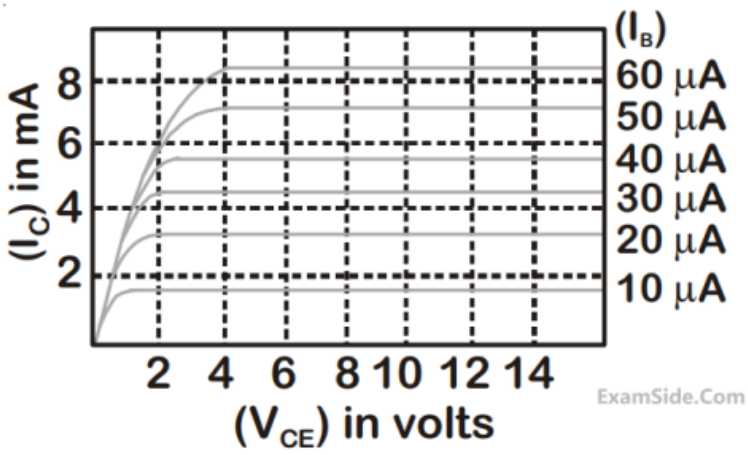
Q261. JEE Main 2020 (Online) 6th September Evening Slot · **Numerical**

The output characteristics of a transistor is shown in the figure.
When V_{CE} is 10V and I_C = 4.0 mA, then value of β_{ac} is _____.

Open solution



Q262. JEE Main 2016 (Offline) · **MCQ (More than One Correct Answer)**

For a common emitter configuration, if α and β have their usual meanings, the incorrect relationship between α and β is :

Open solution

(A) $\alpha = \frac{\beta}{1+\beta}$

(B) $\alpha = \frac{\beta^2}{1+\beta^2}$

(C) $\frac{1}{\alpha} = \frac{1}{\beta} + 1$

(D) $\alpha = \frac{\beta}{1-\beta^2}$

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **B**	4. **B**	5. **C**
6. **C**	7. **A**	8. **A**	9. **D**	10. **B**
11. **A**	12. **B**	13. **A**	14. **A**	15. **C**
16. **A**	17. **D**	18. **C**	19. **D**	20. **A**
21. **C**	22. **C**	23. **A**	24. **C**	25. **A**
26. **A**	27. **C**	28. **C**	29. **D**	30. **D**
31. **B**	32. **A**	33. **C**	34. **B**	35. **D**
36. **D**	37. *Scan QR*	38. **D**	39. **A**	40. **D**
41. **B**	42. **D**	43. **D**	44. **B**	45. **A**
46. **B**	47. **D**	48. **A**	49. **B**	50. **C**
51. **D**	52. **D**	53. **C**	54. **A**	55. **B**
56. **B**	57. **C**	58. **A**	59. **D**	60. **D**
61. **B**	62. **D**	63. **B**	64. **B**	65. **B**
66. **C**	67. **B**	68. **B**	69. **A**	70. **D**
71. **A**	72. **D**	73. **C**	74. **A**	75. **D**
76. **D**	77. **D**	78. **C**	79. **D**	80. **C**
81. **D**	82. **D**	83. **B**	84. **A**	85. **A**
86. **D**	87. **B**	88. **B**	89. **B**	90. **A**
91. **D**	92. **C**	93. **D**	94. **C**	95. **A**
96. **D**	97. **C**	98. **A**	99. **B**	100. **B**
101. **D**	102. **A**	103. **A**	104. **C**	105. **A**
106. **A**	107. **B**	108. **A**	109. **D**	110. **D**
111. **C**	112. **D**	113. **C**	114. **D**	115. **D**
116. **C**	117. **A**	118. **C**	119. **C**	120. **D**
121. **D**	122. **D**	123. **B**	124. **D**	125. **C**
126. **A**	127. **B**	128. **A**	129. **C**	130. **A**
131. **D**	132. **D**	133. **D**	134. **D**	135. **A**
136. **D**	137. **B**	138. **D**	139. **C**	140. **D**
141. **D**	142. **C**	143. **A**	144. **B**	145. **C**
146. **B**	147. **D**	148. **A**	149. **A**	150. **A**

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151. **C**	152. **B**	153. **C**	154. **A**	155. **D**
156. **D**	157. **A**	158. **A**	159. **A**	160. **A**
161. **A**	162. **A**	163. **D**	164. **A**	165. **C**
166. **A**	167. **D**	168. **B**	169. **A**	170. **A**
171. **D**	172. **B**	173. **D**	174. **A**	175. **A**
176. **B**	177. **C**	178. **B**	179. **D**	180. **B**
181. **D**	182. **D**	183. **D**	184. **B**	185. **A**
186. **C**	187. **B**	188. **C**	189. **C**	190. **C**
191. **C**				

Numerical

192. **300**	193. **125**	194. **5**	195. **480**	196. **1**
197. **9**	198. **25**	199. **3**	200. **125**	201. **14**
202. **2**	203. **4**	204. **1**	205. **5**	206. **20**
207. **120**	208. **5**	209. **500**	210. **192**	211. **25**
212. **5**	213. **0**	214. **9**	215. **20**	216. **25**
217. **40**	218. **12**			

Out of syllabus

219. **D**	220. **D**	221. **C**	222. **B**	223. **B**
224. **C**	225. **A**	226. **D**	227. **B**	228. **B**
229. **B**	230. **D**	231. **C**	232. **B**	233. **D**
234. **C**	235. **B**	236. **C**	237. **D**	238. **D**
239. **A**	240. **B**	241. **A**	242. **A**	243. **C**
244. **B**	245. **A**	246. **A**	247. **A**	248. **D**
249. **C**	250. **D**	251. **A**	252. **3**	253. **200**
254. **2**	255. **15**	256. **750**	257. **5**	258. **25**
259. **200**	260. **100**	261. **150**	262. **B, D**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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